

University of Ioannina
Department of Physics



Scientific Review

Dark Energy and Modified Gravity: A Review of Cosmological Models and Observational Probes

Apostolos Bisas

Supervisor: Leandros Perivolaropoulos

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Abstract

This work presents a comprehensive review of the theoretical framework of modern cosmology, anchored in the principles of General Relativity. Beginning with a rigorous derivation of the Einstein Field Equations and the FLRW metric, the standard Λ CDM model is established through the Friedmann equations, describing the dynamics of an expanding, homogeneous, and isotropic universe. The review systematically explores the thermal history of the early universe, the physics of primordial fluctuations, and the hierarchical structure formation predicted by the Cold Dark Matter paradigm, contextualizing these phenomena with the theory of cosmic inflation as a solution to the initial condition problems. A central focus is the critical examination of dark energy, contrasting the successes and theoretical shortcomings of the cosmological constant with dynamical alternatives that include a scalar field approach. This exploration includes a detailed analysis of modified gravity theories—such as Brans-Dicke, $f(R)$ gravity with its screening mechanisms, 4D Einstein-Gauss-Bonnet gravity, and the generalized Horndeski framework—and scalar field models of quintessence, whose dynamics are investigated using phase-space analysis and the Statefinder diagnostic. Throughout the review, theoretical concepts are consistently linked to their observational signatures, with in-depth discussions on how cosmological parameters are constrained by key probes: Type Ia Supernovae, the Cosmic Microwave Background, and Baryon Acoustic Oscillations.

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Finally, I wish to extend my deepest thanks to my beloved family and close friends. Their unconditional support, constant encouragement and belief in me were indispensable throughout the composition of this scientific review. They not only helped with my basic needs but also provided the psychological boost that made progress possible. This work marks just the beginning of a long scientific road and I am grateful to have them by my side.

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1 Introduction

1.1 Einstein's Equation

In 1915, Albert Einstein proposed [1] the field equation that changed our understanding of gravitation by linking the geometry of spacetime to the distribution of matter and energy. His fundamental insight was to describe gravity not as a force, but as a result of spacetime curvature caused by energy and momentum. The central equation is:

$$G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$$

where:

- $G_{\mu\nu}$ is the Einstein tensor:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2} R g_{\mu\nu}$$

where $R_{\mu\nu}$ is the Ricci curvature tensor, R the Ricci scalar (the trace of the Ricci tensor), and $g_{\mu\nu}$ the metric tensor(cf. Derivation of the Einstein Tensor from the Metric and conservation of energy/pomementum in the appendix).

- $T_{\mu\nu}$ is the energy-momentum tensor:

$$T_{\mu\nu} = (\rho + p)u_\mu u_\nu + p g_{\mu\nu} \quad (1)$$

where ρ is the energy density and p is the pressure (initially $\rho + \frac{p}{c^2}$ but we simplify with $c=1$).

- G the gravitational constant and c the speed of light

We will propose an analytic derivation of this equation in the appendix (cf. Derivation of the Einstein's Field Equation from the Einstein's - Hilbert's Action)

Einstein later introduced the cosmological constant Λ to his field equations in an attempt to allow for a static universe, which was the prevailing belief at the time. The modified field equations take the form:

$$G_{\mu\nu} + \Lambda g_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu} \quad (2)$$

Here, $\Lambda g_{\mu\nu}$ acts as an effective energy-momentum contribution from the vacuum itself, often interpreted today as a form of dark energy.

1.2 Geodesic Equation

In general relativity, geodesics represent the natural trajectories of free-falling particles—paths determined entirely by the geometry of spacetime in the absence of non-gravitational forces. They generalize the concept of straight lines to curved spacetime, as geodesics locally extremize the spacetime interval (or proper time for massive particles). The geodesic equation ensures that a particle's velocity vector is parallel transported along its own path, meaning the particle follows the curvature of spacetime.

Everything mentioned above is in agreement with the **Equivalence Principle**: in any local freely falling (inertial) frame, the laws of physics reduce to those of special relativity, and gravitational effects can be "transformed away." This implies that gravity is not treated as a force, but as the effect of spacetime curvature encoded in the metric $g_{\mu\nu}$.

It is convenient to work with the quadratic action:

$$S = \frac{1}{2} \int_{\lambda_1}^{\lambda_2} g_{\mu\nu}(x(\lambda)) \dot{x}^\mu(\lambda) \dot{x}^\nu(\lambda) d\lambda$$

where over-dots denote derivatives with respect to an arbitrary parameter λ (i.e. curve parametrisation) that's used to describe the position $x^\mu(\lambda)$ along the worldline of a particle or a general path in spacetime.

By treating the coordinates $x^\mu(\lambda)$ as generalized coordinates of a Lagrangian:

$$L = \frac{1}{2} g_{\mu\nu}(x) \dot{x}^\mu \dot{x}^\nu$$

The Euler-Langrange equation (cf. Derivation of the Euler-Langrange Equation in the appendix) for this Langrangian gives us:

$$\frac{d}{d\lambda} \left(\frac{\partial L}{\partial \dot{x}^\rho} \right) - \frac{\partial L}{\partial x^\rho} = 0 \quad (3)$$

By computing the two terms separately:

$$\frac{\partial L}{\partial \dot{x}^\rho} = g_{\rho\nu} \dot{x}^\nu \quad \Rightarrow \quad \frac{d}{d\lambda} (g_{\rho\nu} \dot{x}^\nu) = g_{\rho\nu} \ddot{x}^\nu + \partial_\sigma g_{\rho\nu} \dot{x}^\sigma \dot{x}^\nu$$

$$\frac{\partial L}{\partial x^\rho} = \frac{1}{2} \partial_\rho g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu$$

Insert these into the Euler–Lagrange equation (3):

$$g_{\rho\nu} \ddot{x}^\nu + \partial_\sigma g_{\rho\nu} \dot{x}^\sigma \dot{x}^\nu - \frac{1}{2} \partial_\rho g_{\mu\nu} \dot{x}^\mu \dot{x}^\nu = 0$$

Contract with the inverse metric $g^{\rho\lambda}$ and rearrange:

$$\ddot{x}^\lambda + \Gamma_{\mu\nu}^\lambda \dot{x}^\mu \dot{x}^\nu = 0 \quad (4)$$

describes how a free particle moves under the influence of spacetime curvature. The Christoffel symbols $\Gamma_{\mu\nu}^\lambda$ act as effective gravitational "forces" arising from the geometry, not from a potential. This equation is the general relativistic analogue of Newton's second law, where instead of acceleration being caused by a force $\vec{F} = m\vec{a}$, the motion results from the curvature of spacetime itself guiding the particle along its geodesic path.

1.3 The Schwarzschild Metric: A Static, Spherically Symmetric Vacuum

One of the first and most important exact solutions to Einstein's field equation was discovered by Karl Schwarzschild in 1916, just months after Einstein published his theory. The **Schwarzschild solution** [2] describes the geometry of spacetime outside a non-rotating, uncharged, spherically symmetric body of mass M . It is a vacuum solution, meaning it is valid only in the region where the energy-momentum tensor is zero ($T_{\mu\nu} = 0$).

The derivation of the metric relies on several key simplifying assumptions that reflect the physical situation:

- **Static:** The spacetime geometry is unchanging in time. This implies that all metric components are independent of the time coordinate t .
- **Spherically Symmetric:** The geometry is isotropic around a central point, meaning it looks the same in all radial directions. This allows the use of spherical coordinates (r, θ, ϕ) where the metric components depend only on the radial coordinate r .
- **Asymptotically Flat:** Far from the central mass (as $r \rightarrow \infty$), the gravitational field must weaken, and spacetime should approach the flat Minkowski spacetime of special relativity.

These conditions are powerful enough to single out a unique solution. This is formally stated by Birkhoff's theorem, which proves that the assumption of a static spacetime is actually redundant.

Birkhoff's Theorem. Any spherically symmetric solution of the vacuum Einstein field equations is necessarily static and asymptotically flat [3, 2]. This means the external gravitational field of a spherical body must be described by the Schwarzschild metric.

Under these conditions, the vacuum Einstein equations ($R_{\mu\nu} = 0$) can be solved to find the line element. The detailed derivation is presented in the appendix (cf. Derivation of the Schwarzschild Metric). The resulting Schwarzschild metric is:

$$ds^2 = - \left(1 - \frac{2GM}{c^2 r} \right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r} \right)^{-1} dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad (5)$$

This metric reveals a critical feature of general relativity. The metric components become singular at a specific radius known as the **Schwarzschild radius**, r_s :

$$r_s = \frac{2GM}{c^2} \quad (6)$$

This radius defines the **event horizon**, a boundary in spacetime from which nothing, not even light, can escape the gravitational pull of the central mass. For any object with a physical radius smaller than its Schwarzschild radius, it is classified as a black hole. The singularity at $r = r_s$ is a coordinate singularity, which can be removed by a different choice of coordinates, while the true physical singularity lies at $r = 0$.

1.4 The Newtonian Limit and the Poisson Equation in General Relativity

A crucial test for any new theory of gravity is that it must reproduce the successful predictions of its predecessor, Newtonian gravity, in the appropriate domain. For General Relativity, this is known as the Newtonian limit, which applies under three conditions:

1. **Weak gravitational fields:** Spacetime is nearly flat, allowing the metric $g_{\mu\nu}$ to be treated as a small perturbation $h_{\mu\nu}$ from the Minkowski metric $\eta_{\mu\nu}$, i.e., $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ where $|h_{\mu\nu}| \ll 1$.

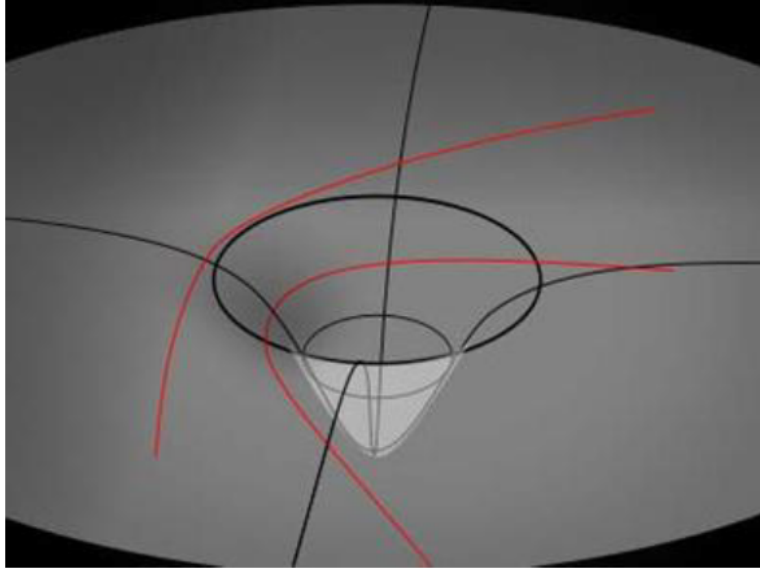


Figure 1: An embedding diagram (adapted from Perivolaropoulos, L., Antoniou, I. (2023), Introduction to General Relativity and Cosmology, edition Kallipos) illustrating the curvature of two-dimensional space around a massive, spherically symmetric object, as predicted by the Schwarzschild solution. The central depression represents the gravitational "well" created by the object's mass, warping the fabric of spacetime. The red lines on the surface represent **geodesics**—the paths that free-falling objects follow.

2. **Slow-moving sources:** The matter generating the gravitational field is moving at speeds much less than the speed of light ($v \ll c$).
3. **Static or slowly-varying fields:** The field itself is not changing rapidly with time, meaning time derivatives of the metric perturbation are negligible compared to spatial derivatives.

Under these assumptions, the full Einstein Field Equations, $G_{\mu\nu} = \frac{8\pi G}{c^4} T_{\mu\nu}$, can be simplified dramatically. The component that corresponds to the Newtonian gravitational potential, Φ , is the time-time component of the metric perturbation, h_{00} , through the relation $\Phi \approx \frac{1}{2}c^2 h_{00}$.

The derivation (shown in Appendix A.5) yields a relativistic version of the Poisson equation for the gravitational potential:

$$\nabla^2 \Phi = 4\pi G (\rho + 3p) \quad (7)$$

where ∇^2 is the Laplacian operator.

Comparing this to the classical Poisson equation, $\nabla^2 \Phi = 4\pi G \rho$, reveals a profound difference. In General Relativity, the source of gravity is not just mass-energy (ρ), but also pressure (p). The term $\rho + 3p$ (initially $3p/c^2$ but we have considered $c=1$) shows that **pressure gravitates**. This is a purely relativistic effect. While the pressure contribution is utterly negligible for ordinary objects like planets or even the Sun, it becomes critically important in the study of highly dense, relativistic objects such as neutron stars. For a neutron star, the immense internal pressure required to prevent gravitational collapse contributes significantly to the star's own gravitational field, a key factor in determining its maximum possible mass (known as the Tolman-Oppenheimer-Volkoff limit).

1.5 Precession of the Perihelion

One of the most celebrated early successes of General Relativity was its ability to explain the anomalous precession of the perihelion of Mercury's orbit. In Newtonian gravity, an isolated planet orbiting a perfectly spherical star would trace out a perfect ellipse that remains fixed in space. However, the observed orbit of Mercury slowly rotates; its perihelion—the point of closest approach to the Sun—advances by a small amount with each orbit. While most of this precession is due to the gravitational tugs of other planets, a residual amount of $42.56'' \pm 0.94''$ per century [4] remained unexplained by Newtonian mechanics.

From Geodesics to Orbits In the framework of General Relativity, the problem is redefined: a planet's orbit is not a path dictated by a force, but a **geodesic**—the straightest possible path—through the curved spacetime created by the Sun's mass. The geometry of this spacetime is described by the **Schwarzschild metric** (5). Therefore, to find the orbit of a planet, we must solve the geodesic equations for a massive particle in this

metric. A direct solution of the geodesic equations can be cumbersome; a more elegant approach is to exploit the symmetries of the spacetime.

Conserved Quantities and the Effective Potential The Schwarzschild metric is independent of time t and the azimuthal angle ϕ . These symmetries imply the existence of conserved quantities along the geodesic, which correspond to energy and angular momentum. For a particle of mass m moving in the equatorial plane ($\theta = \pi/2$), these conserved quantities per unit mass are:

- **Conserved Energy:** $E = c^2 \left(1 - \frac{2GM}{c^2 r}\right) \frac{dt}{d\tau}$
- **Conserved Angular Momentum:** $L = r^2 \frac{d\phi}{d\tau}$

These conservation laws allow us to reduce the four coupled geodesic equations to a single radial equation of motion, which can be expressed in a form reminiscent of classical mechanics:

$$\frac{1}{2} \left(\frac{dr}{d\tau} \right)^2 + V_{\text{eff}}(r) = \frac{E^2 - c^2}{2c^2}$$

Here, $V_{\text{eff}}(r)$ is the relativistic effective potential:

$$V_{\text{eff}}(r) = \underbrace{-\frac{GM}{r}}_{\text{Newtonian Gravity}} + \underbrace{\frac{L^2}{2r^2}}_{\text{Centrifugal Barrier}} - \underbrace{\frac{GML^2}{c^2 r^3}}_{\text{GR Correction}}$$

The first two terms are identical to the effective potential in Newtonian mechanics. The third term, an attractive inverse-cube potential, is a uniquely relativistic correction. This term breaks the perfect symmetry of the Newtonian potential, which is what prevents the orbits from being perfectly closed ellipses and causes them to precess.

The Relativistic Orbit Equation (Binet's Equation) To find the shape of the orbit, $r(\phi)$, it is convenient to use the variable $u = 1/r$ and derive an equation for $u(\phi)$, known as Binet's equation. In Newtonian gravity, this equation is a simple harmonic oscillator:

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{L^2} \quad (\text{Newtonian})$$

The solution is $u(\phi) = \frac{GM}{L^2} (1 + e \cos \phi)$, which describes a perfect, non-precessing ellipse. However, when derived from the relativistic effective potential, the equation becomes:

$$\frac{d^2 u}{d\phi^2} + u = \frac{GM}{L^2} + \frac{3GM}{c^2} u^2 \quad (\text{General Relativity})$$

The additional term, $\frac{3GM}{c^2} u^2$, which arises directly from the $1/r^3$ term in the potential, acts as a small perturbation. It ensures that after one full radial oscillation (from perihelion to aphelion and back), the azimuthal angle ϕ will have advanced by slightly more than 2π . This "extra" angle is the perihelion precession.

The Precession Formula and its Significance Solving this perturbed equation (as detailed in the appendix, cf. Derivation of the Perihelion Precession Formula) yields the advance in the perihelion angle, $\delta\phi$, for each orbit:

$$\delta\phi = \phi - 2\pi = 2\pi \left(\frac{3G^2 M^2}{c^2 L^2} \right) = \frac{6\pi G^2 M^2}{c^2 L^2}$$

This is the precession per orbit. The term L represents the conserved angular momentum per unit mass. To express this in terms of orbital parameters, we use the semi-latus rectum of the orbit, which is defined as $p = L^2/(GM)$. Substituting $L^2 = p(GM)$ into our result yields:

$$\delta\phi = \frac{6\pi G^2 M^2}{c^2 (pGM)} = \frac{6\pi GM}{pc^2}$$

This is the standard result for the precession of the perihelion per orbit, where p is the semi-latus rectum of the elliptical orbit.

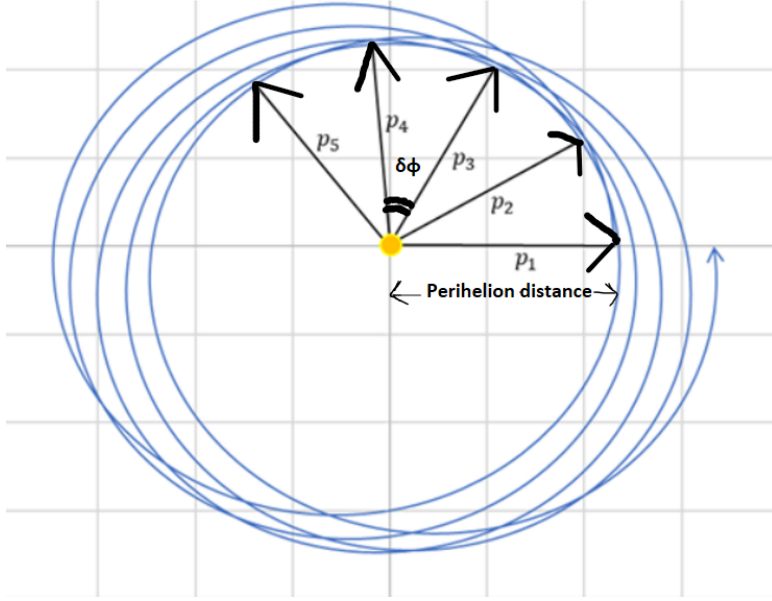


Figure 2: **Visualization of Perihelion Precession.** This diagram illustrates the gradual rotation of an elliptical orbit, a phenomenon known as the precession of the perihelion, as predicted by Einstein’s General Theory of Relativity. The vectors p_1, p_2, p_3, p_4 , and p_5 show the orientation of the orbit at the beginning of five successive orbital periods. The slow rotation of these vectors in the direction of the orbit visually represents the perihelion advance. The angle between each successive vector and the previous one is the precession per orbit ($\delta\phi$), which is greatly exaggerated here for illustrative purposes. For Mercury, this advance is only $42.56'' \pm 0.94''$ per century [4]. Adapted from: <https://www.general-relativity.net/2020/03/precession-of-perihelion-of-mercury.html>

1.6 The FLRW Metric

The choice of the FLRW (Friedmann–Lemaître–Robertson–Walker) metric is rooted in the cosmological principle, which states that the universe is spatially homogeneous and isotropic on large scales (in the order of 100 Mpc).

These two symmetries severely constrain the possible form of the spatial metric. This leads us to adopt a spherically symmetric coordinate system for the spatial part of the metric. A general 3-dimensional spatial metric compatible with these symmetries can then be written as:

$$ds_3^2 = A(r)dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2) \quad (8)$$

where $A(r)$ encodes the radial dependence of the spatial curvature. This form ensures that the geometry respects isotropy around any point (via the angular part) and homogeneity (via the radial function $A(r)$).

We finally calculate (cf. Derivation of the Radial Function $A(r)$ in a symmetric 3D Space in the appendix):

$$A(r) = \frac{1}{1 - Kr^2} \quad \text{for a real constant } K$$

We now verify that this form is consistent with the requirement of $A(r)=1$ when $K=0$ which matches with an Euclidean flat space.

To express the space metric in a more unified form, we define the dimensionless coordinate $\chi = \sqrt{|K|}r$, which leads to a spatial line element of the form

$$ds_3^2 = \frac{1}{1 - Kr^2} dr^2 + r^2 d\Omega^2 = \begin{cases} d\chi^2 + \frac{1}{K} \sin^2(\sqrt{K}\chi) d\Omega^2, & K > 0 \\ d\chi^2 + \chi^2 d\Omega^2, & K = 0 \\ d\chi^2 + \frac{1}{|K|} \sinh^2(\sqrt{|K|}\chi) d\Omega^2, & K < 0 \end{cases}$$

where $d\Omega^2 = d\theta^2 + \sin^2 \theta d\phi^2$ is the metric on the unit 2-sphere.

To simplify notation, we normalize the curvature by setting $k = \frac{K}{|K|} \in \{+1, 0, -1\}$:

$$ds_3^2 = d\chi^2 + S_k^2(\chi) d\Omega^2,$$

with

$$S_k(\chi) = \begin{cases} \sin \chi, & k = +1 \\ \chi, & k = 0 \\ \sinh \chi, & k = -1 \end{cases}.$$

This includes all three spatial curvature cases—positive 3 (spherical), zero (flat), and negative 4 (hyperbolic).

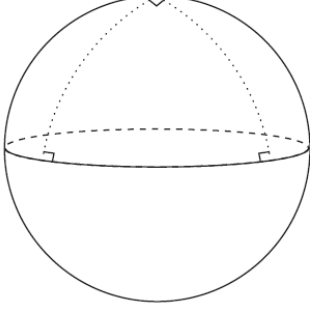


Figure 3: Representation of a positively curved surface ($k > 0$). The triangle drawn on the spherical geometry clearly shows that the total of its internal angles exceeds 180° .

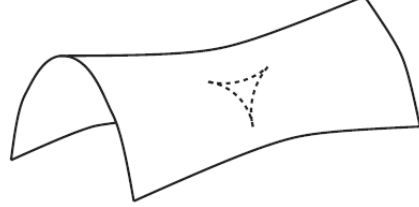


Figure 4: Illustration of a negatively curved surface (hyperbolic geometry, $k < 0$). The triangle drawn on that geometry clearly shows that the total of its internal angles falls short of 180° .

To obtain the full spacetime geometry, we now introduce a global cosmic time coordinate t , which is the proper time measured by comoving observers (those at rest with respect to the large-scale structure of the universe).

The expansion of space is encoded in the scale factor $a(t)$, a time-dependent function that multiplies the entire spatial part of the metric and leads to the full FLRW metric:

$$ds^2 = -dt^2 + a^2(t) [d\chi^2 + S_k^2(\chi) d\Omega^2], \quad (9)$$

We adopt throughout this work the $(-+++)$ signature convention for the spacetime metric.

1.7 Friedmann Equations and Fluid Equation

The dynamical evolution of a homogeneous and isotropic universe is governed by Einstein's field equations applied to the FLRW metric. The **first Friedmann equation** arises from the tt (time-time) component of Einstein's equation (2) combined with the FLRW metric (9) which gives¹:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} + \frac{\Lambda}{3} \quad (10)$$

The **fluid equation**, which expresses energy conservation, is derived from the vanishing of the covariant divergence of the energy-momentum tensor,

$$\nabla^\mu T_{\mu\nu} = 0,$$

and takes the final form:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0 \quad (11)$$

Finally, combining the first Friedmann equation (10) with the fluid equation (11) allows us to derive the **second Friedmann equation**, which describes the acceleration of the universe:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) + \frac{\Lambda}{3}. \quad (12)$$

This equation describes the acceleration (or deceleration) of the cosmic scale factor and shows how it is influenced by the energy density ρ , pressure p , and cosmological constant Λ .

¹The complete derivations of these three fundamental equations from Einstein's field equations are presented in the appendix (cf. Derivation of Friedmann and Fluid Equations from Einstein's Equations)

1.8 Cosmological Redshift

Consider a photon emitted (radially) from a distant galaxy at time $t = t_E$ and received by an observer on Earth at time $t = t_R$. Along the photon path, $ds^2 = 0$ and $d\theta = d\phi = 0$, so the FLRW metric (9) for a flat universe implies:

$$c^2 dt^2 = a^2(t) d\chi^2 \quad \Rightarrow \quad \frac{c dt}{a(t)} = d\chi$$

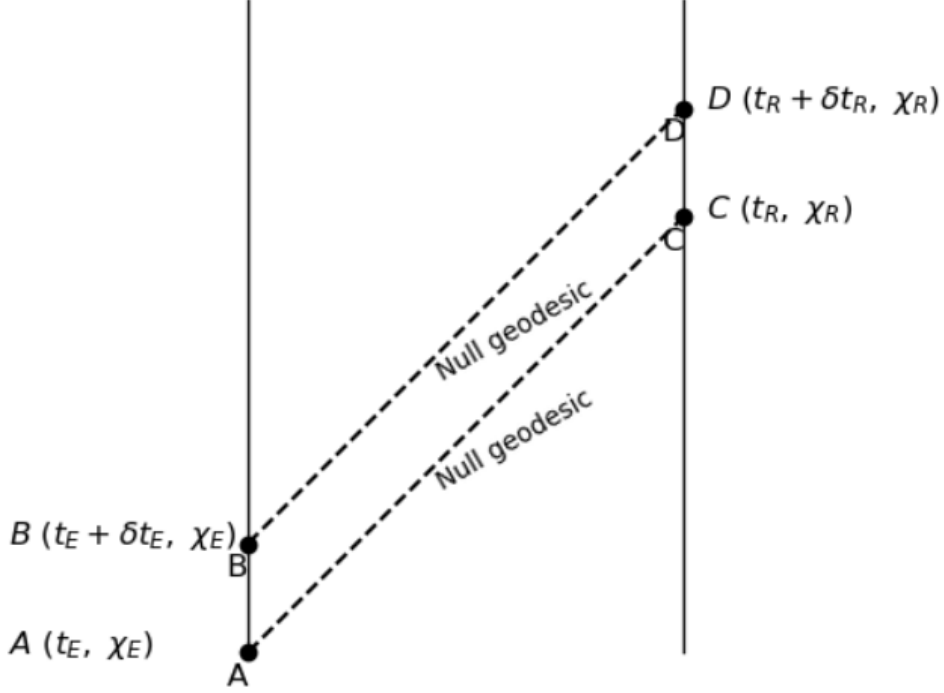


Figure 5: **Redshift diagram in proper time.** Points A and B represent the emission of two photons at χ_E and times $t_E, t_E + \delta t_E$. Points C and D are their reception at χ_R and times $t_R, t_R + \delta t_R$. Dashed lines are null geodesics.

For an incoming photon moving toward us, the comoving coordinate χ decreases as time increases, so:

$$\frac{c dt}{a(t)} = -d\chi \quad \Rightarrow \quad \int_{t_E}^{t_R} \frac{c dt}{a(t)} = - \int_{\chi_E}^0 d\chi = \int_0^{\chi_E} d\chi = \chi_E$$

Now consider a second photon emitted at $t_E + \delta t_E$ and received at $t_R + \delta t_R$. Applying the same reasoning:

$$\int_{t_E + \delta t_E}^{t_R + \delta t_R} \frac{c dt}{a(t)} = \chi_E.$$

By equating both expressions, we obtain:

$$\int_{t_E}^{t_R} \frac{c dt}{a(t)} = \int_{t_E + \delta t_E}^{t_R + \delta t_R} \frac{c dt}{a(t)} \quad \Rightarrow \quad \int_{t_R}^{t_R + \delta t_R} \frac{c dt}{a(t)} = \int_{t_E}^{t_E + \delta t_E} \frac{c dt}{a(t)}.$$

If $\delta t_E, \delta t_R$ are small, we can approximate $a(t)$ as constant in each interval:

$$\frac{c \delta t_R}{a(t_R)} = \frac{c \delta t_E}{a(t_E)} \quad \Rightarrow \quad \frac{\delta t_R}{\delta t_E} = \frac{a(t_R)}{a(t_E)}.$$

From the definition of the frequency of light that is inversely proportional to the period, we have for the redshift:

$$\frac{\nu_E}{\nu_R} = \frac{\delta t_R}{\delta t_E} = \frac{a(t_R)}{a(t_E)} = \frac{\lambda_R}{\lambda_E} = 1 + z \quad (13)$$

In modern cosmology, we denote the emission time t_E as a generic time t , and the reception time t_R as today t_0 , so:

$$1 + z = \frac{a(t_0)}{a(t)}.$$

If we normalize the scale factor such that $a(t_0) = 1$, this simplifies to:

$$a(t) = \frac{1}{1 + z}.$$

This relation is fundamental in connecting observations of redshift with the expansion history of the universe.

1.9 Hubble's Law and Hubble's Parameter

At large scales, galaxies are observed to recede from each other at a rate proportional to their separation. This is expressed by **Hubble's law** (cf. Derivation of Hubble's law in the appendix) :

$$v = H_0 d$$

where:

- $v = c \cdot z$ is the recession velocity of a galaxy
- d its proper distance
- H_0 is the present-day value of the Hubble parameter, defined as $H(t) = \dot{a}(t)/a(t)$. It is measured to be $H_0 = 67.4 \pm 0.5 \text{ km} \cdot \text{s}^{-1} \cdot \text{Mpc}^{-1}$ (Planck 2018 [5]) or $H_0 = 73.24 \pm 1.74 \text{ km} \cdot \text{s}^{-1} \cdot \text{Mpc}^{-1}$ (SHOES collaboration, Riess et al. 2018 [6]).

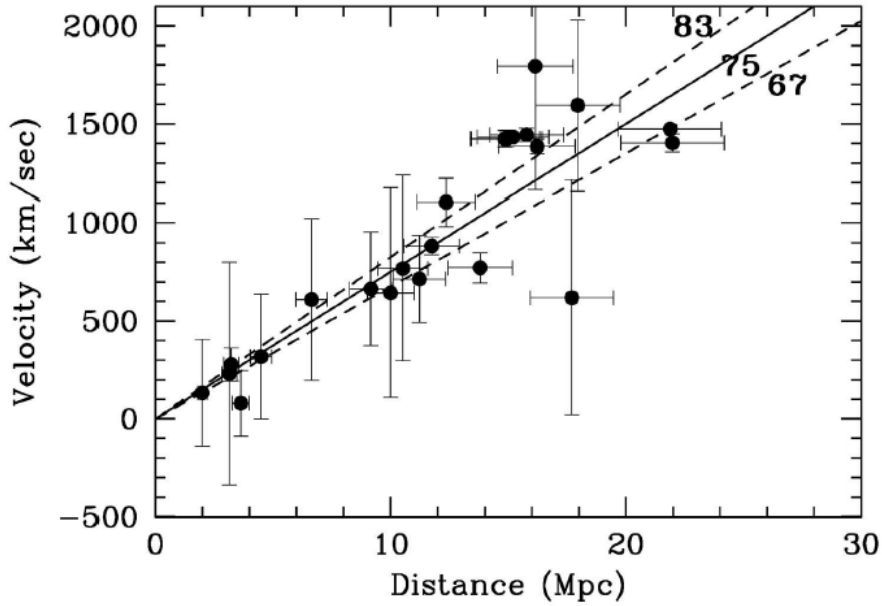


Figure 6: Hubble diagram showing galaxy velocities versus distances (in Mpc). Distances are derived using Cepheid variable stars as standard candles. The solid and dashed lines correspond to Hubble constants $H_0 = 67, 75, 83 \text{ km/s/Mpc}$. Adapted from Freedman et al. (2001), ApJ, 553, 47 [7].

The linear relation between redshift and distance was first established by **Edwin Hubble** in 1929. By analyzing the redshifts and estimating the distances to galaxies using Cepheid variable stars², Hubble discovered that distant galaxies appeared to be moving away from us at velocities proportional to their distance. This provided the first empirical evidence that the universe is expanding, fundamentally shifting cosmology away from the static models that had previously dominated.

²Cepheid variable stars are pulsating stars whose luminosity varies periodically due to oscillations in their outer layers. Their key property is a well-defined period–luminosity relation: the longer the pulsation period, the brighter the luminosity. This makes them reliable standard candles for measuring extragalactic distances.

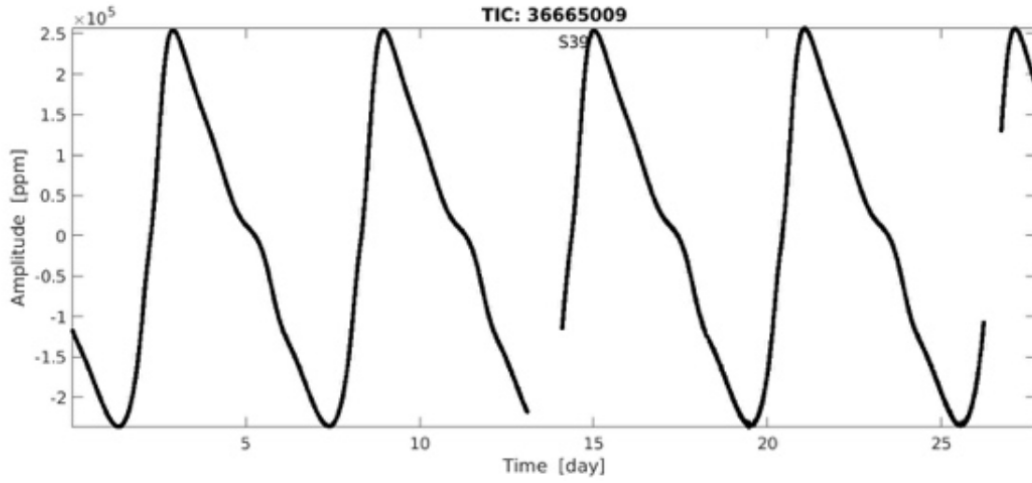


Figure 7: TESS light curve of the classical Cepheid variable RV Sco (TESS magnitude 6.01), observed during Sector 39 from May 27 to June 24, 2021. The light curve clearly shows the periodic pulsations characteristic of Cepheids. Gaps and faded regions are due to TESS’s regular data downlink interruptions. Adapted from Guzik et al. (2021) [8].

1.10 Cosmological Parameters and the Dimensionless Friedmann Equation

To interpret the Friedmann equations in terms of measurable quantities, it is convenient to introduce several cosmological parameters.

1.10.1 Critical Density and Spatial Curvature

The **critical density** $\rho_c(t)$ is the energy density required to make the universe spatially flat at a given time t . It is defined as:

$$\rho_c(t) := \frac{3H(t)^2}{8\pi G}$$

At the present epoch t_0 , this becomes:

$$\rho_c(t_0) := \frac{3H_0^2}{8\pi G}$$

To express this in astrophysical units, we define the dimensionless reduced Hubble parameter h by:

$$H_0 = 100 h \text{ km} \cdot \text{s}^{-1} \cdot \text{Mpc}^{-1}.$$

Inserting into the formula, we get:

$$\rho_c(t_0) = 1.88 h^2 \times 10^{-26} \text{ kg} \cdot \text{m}^{-3} \quad \text{or} \quad \rho_c(t_0) = 2.78 h^2 \times 10^{11} M_\odot \cdot (h^{-1} \text{ Mpc})^{-3}$$

Using the Planck 2018 [5] value $h = 0.674 \pm 0.005$ and by calculating the propagation of incertitude, we numerically estimate:

$$\rho_c(t_0) = (8.53 \pm 0.13) \times 10^{-27} \text{ kg} \cdot \text{m}^{-3}.$$

This value serves as a reference point: if the actual total energy density $\rho(t_0) > \rho_c(t_0)$, the universe is spatially **closed** ($k = +1$); if $\rho(t_0) < \rho_c(t_0)$, it is **open** ($k = -1$); and if $\rho(t_0) = \rho_c(t_0)$, it is **flat** ($k = 0$).

1.10.2 Dimensionless Density Parameters

We now define the following dimensionless parameters by normalizing each energy component by the critical density:

- Matter density parameter: $\Omega_m(t) = \frac{\rho_m(t)}{\rho_c(t)}$
- Radiation density parameter:

$$\Omega_r(t) = \frac{\rho_r(t)}{\rho_c(t)}$$
- Cosmological constant (dark energy): $\Omega_\Lambda = \frac{\Lambda}{3H_0^2}$
- Curvature density parameter: $\Omega_k(t) = -\frac{k}{a(t)^2 H(t)^2}$

1.10.3 Present-Day Values

Precise cosmological observations(Planck 2018[5]) suggest that the total energy density of the universe today is extremely close to the critical density $\rho_c(t_0)$. These observations constrain the total density parameter to be :

$$\Omega_{\text{tot}} = \frac{\rho_{\text{tot}}}{\rho_c} = 1.0007 \pm 0.0019$$

which is consistent with $\Omega_{\text{tot}} = 1$ within measurement uncertainty.

At time $t = t_0$, by normalizing the scale factor such that $a(t_0) = 1$, the Friedmann equation (10) becomes:

$$1 = \frac{8\pi G}{3H_0^2} \rho(t_0) - \frac{k}{H_0^2} + \frac{\Lambda}{3H_0^2}.$$

Thus, by using the definitions of the dimensionless density parameters 1.10.2, we find:

$$1 = \Omega_{m,0} + \Omega_{r,0} + \Omega_{\Lambda,0} + \Omega_{k,0}.$$

where the subscript 0 indicates evaluation today. The observed values (from Planck 2018 [5]) are approximately:

- $\Omega_{m,0} = 0.315 \pm 0.007$
- $\Omega_{k,0} = 0.001 \pm 0.002$
- $\Omega_{r,0} \approx 9 \times 10^{-5}$ (negligible today)
- $\Omega_{\Lambda,0} = 1 - \Omega_{m,0} - \Omega_{r,0} - \Omega_{k,0} = 0.684 \pm 0.007$

1.10.4 Dimensionless Friedmann Equation

We try to rewrite the first Friedmann equation (10) at any time t in terms of the dimensionless parameters:

$$\left(\frac{H(t)}{H_0}\right)^2 = \Omega_{r,0} \left(\frac{a_0}{a(t)}\right)^4 + \Omega_{m,0} \left(\frac{a_0}{a(t)}\right)^3 + \Omega_{k,0} \left(\frac{a_0}{a(t)}\right)^2 + \Omega_{\Lambda,0}.$$

Setting $a_0 = 1$ (by convention), this simplifies to:

$$\left(\frac{H(t)}{H_0}\right)^2 = \Omega_{r,0} a^{-4} + \Omega_{m,0} a^{-3} + \Omega_{k,0} a^{-2} + \Omega_{\Lambda,0}$$

1.11 Constant and Dynamical Equation of State

In cosmology, a convenient way to describe the relation between pressure and energy density of a component of the universe is through the **equation of state parameter** w , defined as:

$$w = \frac{p}{\rho} \tag{14}$$

This relation characterizes how the energy density evolves with the expansion of the universe. For a constant equation of state, combining this with the fluid equation (11) and substituting $p = w\rho$, we get:

$$\dot{\rho} + 3\frac{\dot{a}}{a}\rho(1+w) = 0 \tag{15}$$

This is a separable differential equation, and its solution(cf. Derivation of $\rho(a) \propto a^{-3(1+w)}$ in the appendix) shows the scaling behavior of the energy density:

$$\rho(a) \propto a^{-3(1+w)} \tag{16}$$

This relation tells us how different components of the universe dilute as it expands:

- For radiation: $w = 1/3 \Rightarrow \rho_r \propto a^{-4}$
- For matter: $w = 0 \Rightarrow \rho_m \propto a^{-3}$
- For a cosmological constant: $w = -1 \Rightarrow \rho_{\Lambda} = \text{constant} \propto a^0$

For more general forms of dark energy, $w = w(t)$ or $w = w(a)$ can vary dynamically with time or scale factor. In such cases, we can express the solution(cf. Derivation of the $\rho(z) = \rho_0 \exp\left[3 \int_0^z \frac{1+w(z')}{1+z'} dz'\right]$) in terms of the redshift z :

$$\rho(z) = \rho_0 \exp\left[3 \int_0^z \frac{1+w(z')}{1+z'} dz'\right] \tag{17}$$

1.12 Solutions of Friedmann's Equations

1.12.1 Matter-Dominated Universe ($w = 0$)

In this case, the universe is filled with pressureless dust ($p = 0$) and from the equation (16) we have that $\rho(a) \propto a^{-3}$. Inserting into the first Friedmann equation (10) and setting $k=0$:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0 a^{-3}.$$

Separate and integrate:

$$\int a^{1/2} da = \sqrt{\frac{8\pi G}{3}\rho_0} \int dt \Rightarrow a(t) \propto t^{2/3}.$$

1.12.2 Radiation-Dominated Universe ($w = 1/3$)

The radiation energy density evolves as $\rho(a) \propto a^{-4}$ by replacing $w=1/3$ in (16). Then the Friedmann equation (10) with $k=0$ becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_0 a^{-4}$$

Separate and integrate:

$$\int a da = \sqrt{\frac{8\pi G}{3}\rho_0} \int dt \Rightarrow a(t) \propto t^{1/2}$$

1.12.3 Cosmological Constant-Dominated Universe ($w = -1$) - de Sitter Model

In this case, the energy density is due entirely to a cosmological constant Λ , corresponding to a vacuum energy with equation of state $w = -1$, so that $p = -\rho$. From equation (16), we already found that $\rho(a) \propto a^0$ and by inserting this into the first Friedmann equation (10), and setting $k = 0$, we obtain:

$$\left(\frac{\dot{a}}{a}\right)^2 = \frac{8\pi G}{3}\rho_\Lambda = \frac{\Lambda}{3}.$$

This is a separable equation with constant right-hand side. Solving:

$$\frac{\dot{a}}{a} = H_\Lambda = \sqrt{\frac{\Lambda}{3}} \Rightarrow \frac{da}{a} = H_\Lambda dt \Rightarrow a(t) \propto e^{H_\Lambda t}.$$

This solution describes an exponentially expanding universe dominated by dark energy.

1.12.4 Flat Universe with Matter and Cosmological Constant (Λ CDM Model)

We consider a flat universe ($k = 0$) dominated by both matter and a cosmological constant. We define the scale factor a_m at which the energy densities of matter and cosmological constant are equal:

$$a_m = \left(\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}}\right)^{1/3}$$

Using this, the Friedmann equation (10) becomes:

$$\left(\frac{\dot{a}}{a}\right)^2 = H_0^2 (\Omega_{m,0} a^{-3} + \Omega_{\Lambda,0}) = H_0^2 \Omega_{\Lambda,0} \left(\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} a^{-3} + 1\right).$$

So the final result (cf. Derivation of the Scaling Factor $a(t)$ for Flat Universe with Matter and Cosmological Constant in the appendix) is :

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m,0}}} \ln \left(y^{3/2} + \sqrt{1+y^3} \right) \quad \text{where} \quad y = \frac{a(t)}{a_m}$$

In order to understand how this expression works for different time periods, we can distinguish the following cases:

- Early-Time Behavior: $a \ll a_m \Rightarrow y \ll 1$:

$$a(t) \approx a_m \left(\frac{3}{2} \sqrt{1-\Omega_{m,0}} H_0 t \right)^{2/3}.$$

- Late-Time Behavior: $a \gg a_m \Rightarrow y \gg 1$:

$$a(t) \approx a_m \exp\left(\sqrt{1 - \Omega_{m,0}} H_0 t\right).$$

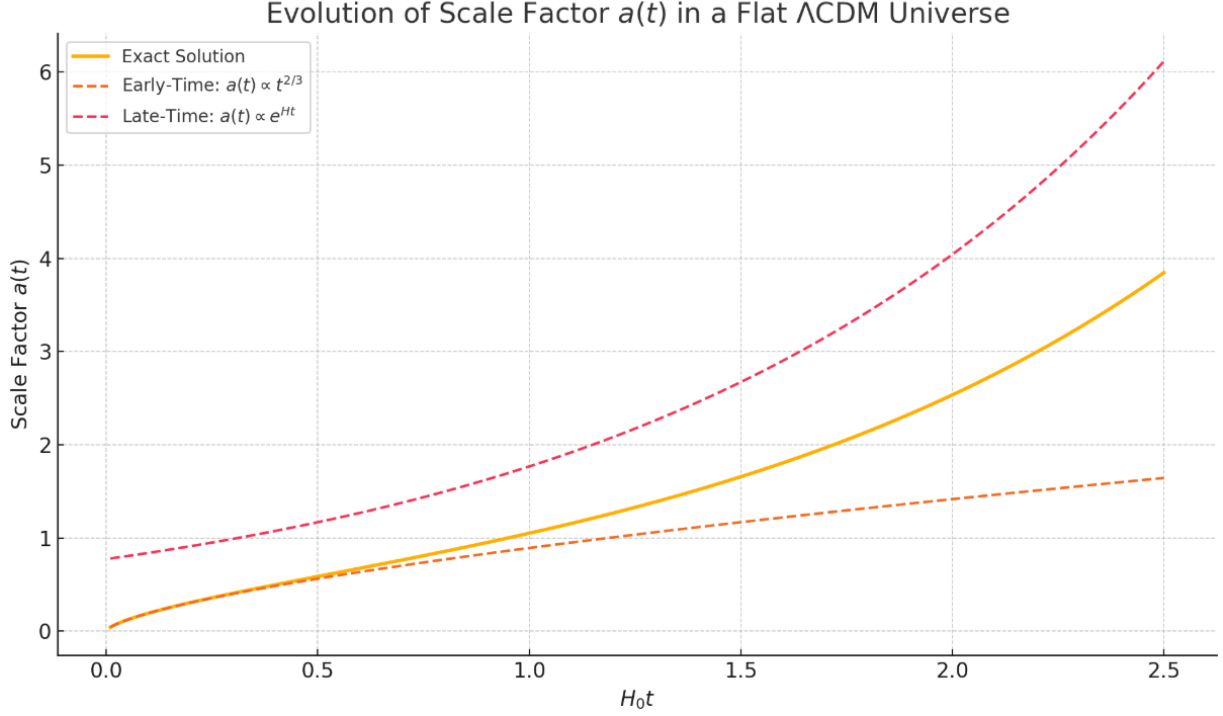


Figure 8: Evolution of the scale factor $a(t)$ in a flat Λ CDM universe. The solid curve shows the exact numerical solution of the Friedmann equation with present-day values $\Omega_{m,0} = 0.315$ and $\Omega_{\Lambda,0} = 0.684$. The dashed curves show analytic approximations valid in two asymptotic regimes: early-time behavior $a(t) \propto t^{2/3}$ (matter-dominated), and late-time behavior $a(t) \propto e^{Ht}$ (dark energy-dominated).

1.13 Cosmological Distances

In an expanding universe, distance is not a unique concept. We distinguish between several types of distances depending on the method of measurement. In this section, we focus on those that are relevant to observations: *physical*, *angular diameter*, and *luminosity* distances.

1.13.1 Theoretically Defined Distances

While not directly measurable, theoretically defined distances serve as the basis for interpreting observational data. Two important examples are the metric distance and the comoving distance.

Metric Distance In the FLRW metric (9), the metric distance corresponds to the radial factor that multiplies the angular part $d\Omega^2$. We define it as:

$$d_m = S_k(\chi)$$

In the case of a flat universe where $k = 0$ and $S_k(\chi) = \chi$, this simplifies to: $d_m = \chi$.

Comoving Distance The comoving distance $d_{co}(z)$ measures the spatial separation of two objects at rest with respect to the Hubble flow. It is obtained by considering light-like paths $ds^2 = 0$ and gives:

$$d_{co}(z) = \int_{t_e}^{t_0} \frac{dt'}{a(t')} = \int_0^z \frac{dz'}{H(z')} = \frac{1}{a_0 H_0} \int_0^z \frac{dz'}{E(z')} \quad (18)$$

where

$$E(z) \equiv \frac{H(z)}{H_0}$$

is the normalized Hubble parameter. For convenience, we often set $a_0 = 1$, which renders distances in units of c/H_0 .

1.13.2 Physical Distance

The physical (or proper) distance $d_p(z)$ is the instantaneous separation between two events at the same cosmic time. In terms of comoving distance:

$$d_p(z) = a(t) d_{co}(z) \quad (19)$$

where $a(t)$ is the scale factor at the moment of observation.

1.13.3 Angular Diameter Distance

Consider a source located at comoving coordinate χ , subtending an angle $\delta\theta$ in the observer's sky. If the object has a known proper size D , the angular diameter distance is defined as:

$$d_A = \frac{D}{\delta\theta}$$

From the FLRW metric, the transverse proper size is given by:

$$D = a(t_1) S_k(\chi) \delta\theta = \frac{d_m}{1+z} \delta\theta$$

Thus, we find:

$$d_A = \frac{d_m}{1+z}$$

This quantity is central to analyses involving standard rulers, such as baryon acoustic oscillations and CMB anisotropies.

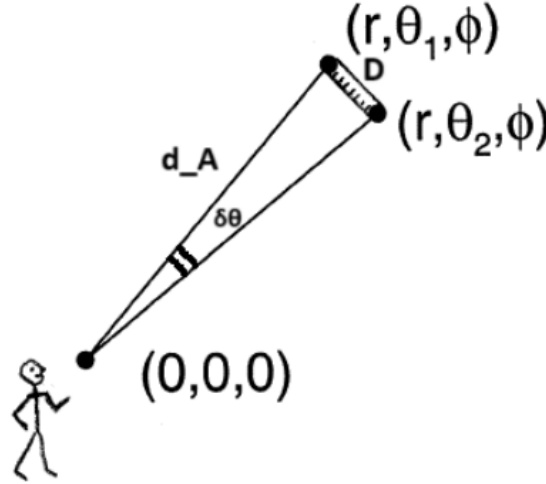


Figure 9: A standard ruler of known size D subtends an angle $\delta\theta = \theta_1 - \theta_2$ in the sky. The angular diameter distance d_A links the physical size to the observed angle. Adapted from Ryden's *Introduction to Cosmology* [9].

1.13.4 Luminosity Distance

Let a source of intrinsic luminosity L lie at comoving distance χ . In a static Euclidean universe, the flux F received would satisfy:

$$F = \frac{L}{4\pi\chi^2} \quad (20)$$

However, in an expanding universe, cosmological effects modify the observed flux due to:

- The increase in the surface area of the light sphere: $4\pi d_m^2$
- The redshift of photon energy and arrival rate: together contributing a factor $(1+z)^{-2}$

Thus, the flux observed is:

$$F = \frac{L}{4\pi d_m^2 (1+z)^2}$$

We define the luminosity distance d_L as the quantity that restores the inverse-square law:

$$d_L = \sqrt{\frac{L}{4\pi F}} = d_m(1+z) \quad (21)$$

This definition is crucial for the study of standard candles such as Type Ia supernovae.

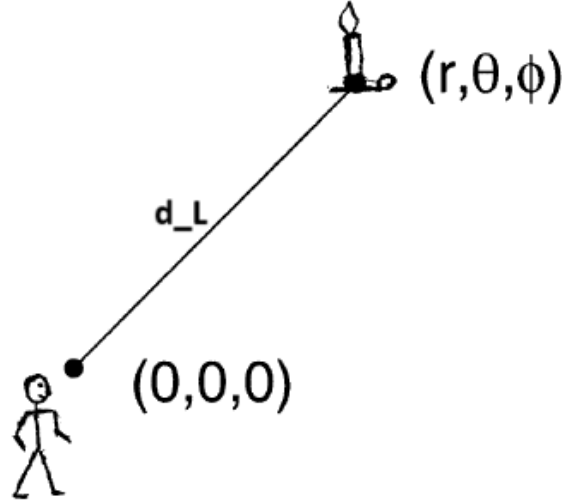


Figure 10: An observer measures the flux F from a source of known luminosity L . The luminosity distance d_L links the source's intrinsic properties to the observed signal. Adapted from Ryden's *Introduction to Cosmology* [9].

1.13.5 Distance Modulus and Magnitude Relationships

To express brightness on a logarithmic scale, astronomers define the apparent magnitude m as:

$$m = -2.5 \log_{10} \left(\frac{F}{F_{\text{ref}}} \right)$$

The absolute magnitude M is the apparent magnitude the source would have at 10 parsecs:

$$M = -2.5 \log_{10} \left(\frac{F_{10 \text{ pc}}}{F_{\text{ref}}} \right) \quad (22)$$

The difference is called the *distance modulus* μ , and relates directly to the luminosity distance:

$$\mu = m - M = 5 \log_{10} \left(\frac{d_L}{10 \text{ pc}} \right) \quad (23)$$

When d_L is expressed in megaparsecs (Mpc), this becomes:

$$\mu = 5 \log_{10} d_L + 25$$

This compact form is widely used in observational cosmology.

1.14 Theoretical Background for Dark Energy

In the standard model of cosmology, the late-time acceleration of the universe cannot be explained by matter or radiation alone, since both components contribute positive pressure or none at all. This leads to a decelerating expansion, as seen from the acceleration equation (12):

For matter:

$$p_m = 0 \quad \Rightarrow \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \rho_m < 0$$

For radiation:

$$p_r = \frac{1}{3} \rho_r \quad \Rightarrow \quad \frac{\ddot{a}}{a} = -\frac{4\pi G}{3} \left(\rho_r + 3 \cdot \frac{1}{3} \rho_r \right) = -\frac{8\pi G}{3} \rho_r < 0$$

Thus, to explain the observed accelerated expansion ($\ddot{a} > 0$), an additional energy component is required, known as **Dark Energy**:

- A positive energy density: $\rho_X > 0$
- A negative pressure: $p_X = w \rho_X$ (with $w < -\frac{1}{3}$)

Including this component in the acceleration equation (12) gives:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho_m + \rho_X + 3p_X) = -\frac{4\pi G}{3}[\rho_m + \rho_X(1 + 3w)]$$

To achieve accelerated expansion ($\ddot{a} > 0$), the following condition must be satisfied:

$$w < -\frac{1}{3}$$

Assuming a spatially flat universe ($k = 0$) composed of matter and this general dark energy component, the Friedmann equation (10) expressed in terms of redshift $z = \frac{1}{a} - 1$, can be written:

$$H^2(z) = H_0^2 [\Omega_{m,0}(1+z)^3 + \Omega_X(z)]$$

The functional form of $\Omega_X(z)$ depends on the dark energy equation of state (17) :

$$\rho_X(z) = \rho_{X,0} \exp \left[3 \int_0^z \frac{1 + w(z')}{1 + z'} dz' \right]$$

Hence, by specifying a form for $w(z)$, we can fully determine the evolution of the dark energy density and its effect on the expansion of the universe.

Note: When $w = -1$, dark energy becomes equivalent to a cosmological constant, corresponding to the standard Λ CDM model. More general models with $w \neq -1$, or even time-dependent $w(z)$, are called **dynamical dark energy** models.

1.15 The Flatness Problem

We consider a flat FLRW model filled with a perfect fluid with pressure $P = w\rho$, where w is a constant. From the definition of the density parameter $\Omega(t)$ and the use of the two Friedmann's equations (10), (12), we can show(cf. Derivation of the Scaling Factor's $a(t)$ - density parameter $\Omega(t)$ relationship in the appendix) the following :

$$\frac{d\Omega}{d \ln a} = (1 + 3w)\Omega(\Omega - 1) \quad (24)$$

Let us consider a small deviation from flatness:

$$\Omega(a) = 1 + \delta(a), \quad \delta \ll 1$$

Inserting into Eq. (24) and keeping only first-order terms:

$$\frac{d\delta}{d \ln a} = (1 + 3w)\delta \quad \Rightarrow \quad \delta(a) = \delta_i \left(\frac{a}{a_i} \right)^{1+3w}$$

- If $1 + 3w > 0$ (e.g., matter or radiation), then δ grows with expansion, and $\Omega = 1$ is unstable.
- If $1 + 3w < 0$, such as during inflation or dark energy domination, then δ decays and $\Omega = 1$ becomes a solution.

Implications: This result highlights the so-called **flatness problem**: unless the universe began with an extremely fine-tuned value of Ω very close to 1, any small initial curvature would have grown rapidly over time. For the universe to still appear flat today, $\Omega(a) \approx 1$, the initial condition must have been incredibly close to 1. For more details observe the inflation's subsection 2.5.1.

1.16 Gravitational Waves: Ripples in Spacetime

A direct consequence of the dynamical nature of spacetime in General Relativity is the existence of gravitational waves. First predicted by Einstein in 1916 [1], these are propagating disturbances in the curvature of spacetime, generated by accelerating massive objects. Much like accelerating electric charges produce electromagnetic waves, accelerating masses radiate energy away in the form of gravitational waves. However, due to the weakness of the gravitational interaction, only the most extreme astrophysical events—such as the merger of black holes or neutron stars—can produce waves strong enough to be detected.

The Weak-Field Limit and Linearized Gravity To derive the wave-like behavior, we consider gravity in the **weak-field limit**, where the spacetime metric $g_{\mu\nu}$ is a small perturbation around a flat Minkowski background $\eta_{\mu\nu}$:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}, \quad |h_{\mu\nu}| \ll 1$$

Here, $h_{\mu\nu}$ is the metric perturbation. In this limit, we can linearize the Einstein field equations by keeping only terms of first order in $h_{\mu\nu}$. The Ricci tensor, to first order, is given by:

$$R_{\mu\nu} \approx \frac{1}{2} (\partial_\mu \partial^\lambda h_{\lambda\nu} + \partial_\nu \partial^\lambda h_{\lambda\mu} - \partial_\mu \partial_\nu h - \eta^{\alpha\beta} \partial_\alpha \partial_\beta h_{\mu\nu})$$

where $h = \eta^{\mu\nu} h_{\mu\nu}$ is the trace of the perturbation. The vacuum field equation $R_{\mu\nu} = 0$ becomes a complicated-looking system of partial differential equations for $h_{\mu\nu}$.

To simplify this, we introduce the **trace-reversed perturbation**, $\bar{h}_{\mu\nu}$:

$$\bar{h}_{\mu\nu} = h_{\mu\nu} - \frac{1}{2} \eta_{\mu\nu} h$$

This definition has the convenient property that its trace is the negative of the original trace, $\bar{h} = -h$. With this redefinition, the Einstein tensor becomes:

$$G_{\mu\nu} \approx \frac{1}{2} (\partial_\mu \partial^\lambda \bar{h}_{\lambda\nu} + \partial_\nu \partial^\lambda \bar{h}_{\lambda\mu} - \eta_{\mu\nu} \partial^\alpha \partial^\beta \bar{h}_{\alpha\beta} - \eta^{\alpha\beta} \partial_\alpha \partial_\beta \bar{h}_{\mu\nu})$$

Gauge freedom In field theories, gauge freedom refers to a redundancy in the mathematical description, where different configurations of the underlying potentials describe the exact same physical system. A specific coordinate choice made to eliminate this redundancy is called a **gauge**. In linearized gravity, the metric perturbation $h_{\mu\nu}$ acts as the potential, while the physical spacetime curvature is the invariant quantity. The gauge freedom corresponds to infinitesimal coordinate transformations, $x^\mu \rightarrow x'^\mu = x^\mu + \xi^\mu(x)$, which change the form of the perturbation to $h'_{\mu\nu} = h_{\mu\nu} - \partial_\mu \xi_\nu - \partial_\nu \xi_\mu$ but leave the physics unchanged.

We can simplify the equations by making a specific gauge choice. The most common is the **Lorenz gauge** condition:

$$\partial^\mu \bar{h}_{\mu\nu} = 0$$

Imposing this condition causes the first three terms in the linearized Einstein tensor to vanish, dramatically simplifying the vacuum field equations to a simple wave equation:

$$\eta^{\alpha\beta} \partial_\alpha \partial_\beta \bar{h}_{\mu\nu} = 0 \tag{25}$$

This is the equation for a wave propagating at the speed of light, confirming that gravitational disturbances travel at c .

Plane Wave Solutions and Polarizations We seek a plane wave solution to equation (25) of the form:

$$\bar{h}_{\mu\nu} = A_{\mu\nu} e^{ik_\alpha x^\alpha}$$

where $A_{\mu\nu}$ is a constant, symmetric polarization tensor and k_α is the wave four-vector. Substituting this into the wave equation gives $k_\alpha k^\alpha = 0$, which confirms that the wave is null, or light-like. The Lorenz gauge condition becomes $k^\mu A_{\mu\nu} = 0$, which means the polarization tensor is orthogonal to the direction of propagation.

There is a residual gauge freedom that allows us to impose further conditions, leading to the **Transverse-Traceless (TT) gauge**. In this gauge, the wave has only two independent, non-zero components. For a wave propagating in the z -direction, $k^\alpha = (\omega, 0, 0, \omega)$, the polarization tensor has the form:

$$h_{\mu\nu}^{TT} = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & h_+ & h_\times & 0 \\ 0 & h_\times & -h_+ & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

The two remaining components, h_+ and $h_\times$, represent the two fundamental polarizations of gravitational waves.

- The **plus polarization** (h_+) stretches and squeezes spacetime in the x and y directions.
- The **cross polarization** ($h_\times$) produces a similar effect, but rotated by 45 degrees.

The effect of a passing gravitational wave is to induce a physical strain in spacetime, causing the distance between free-falling objects to oscillate. This physical strain is precisely what is measured by ground-based detectors like the Laser Interferometer Gravitational-Wave Observatory (LIGO) and Virgo. These observatories are giant Michelson interferometers (cf. figure 11). The first detection, designated GW150914 and announced in 2016, was the result of two black holes merging over a billion light-years away [10]. The detected strain, $h = \Delta L/L$, was on the order of 10^{-21} , a displacement thousands of times smaller than the diameter of a proton over a 4-kilometer arm length. This monumental achievement required unprecedented sensitivity and confirmed the existence of both gravitational waves and binary black hole systems, heralding the new era of gravitational-wave astronomy [11].

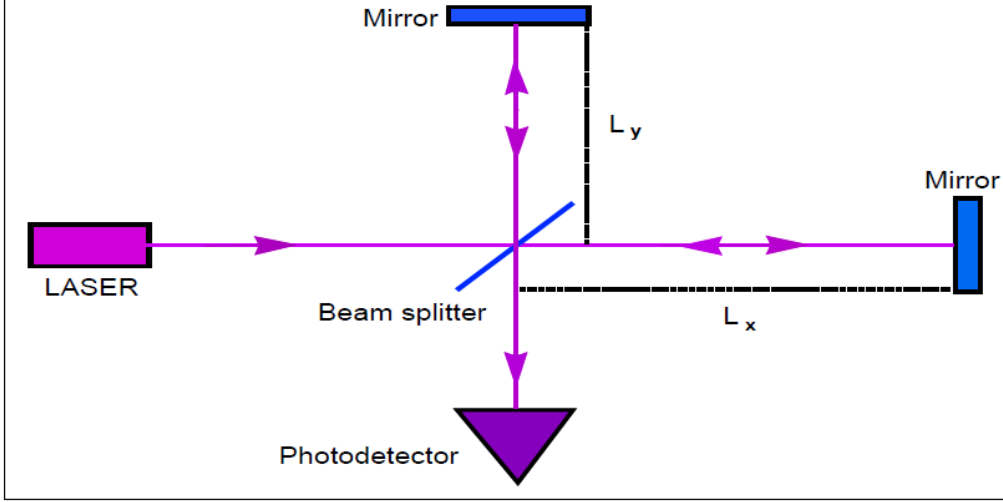


Figure 11: Schematic diagram of a Michelson interferometer, the fundamental design for gravitational wave detectors like LIGO and Virgo. A laser beam is divided by a beam splitter, sending coherent light down two perpendicular arms of lengths L_x and L_y . The beams reflect off mirrors at the end of each arm and recombine at the beam splitter. The resulting interference pattern is monitored by a photodetector. A passing gravitational wave induces a differential strain, altering the physical lengths of the arms ($\Delta L_x \neq \Delta L_y$). This change in the relative path length shifts the interference pattern, producing a detectable signal proportional to the gravitational wave amplitude. Adapted from Dirkes, A. (2019), *Gravitational Waves - A Review on the Conceptual Foundations of Gravitational Radiation*.

2 The Early Universe

In the previous chapters, we established the geometric properties of a homogeneous and isotropic universe governed by the Friedmann equations (10), (12). Now, we turn to its thermal history and the origin of the structures within it. The period between the very first moments of creation and the era of recombination is characterized by a sequence of profound physical events and phase transitions. These events not only shaped the present state of the universe but also seeded and influenced the primordial perturbations that would eventually grow into galaxies and clusters. Our considerations are based on well-understood and tested laws of particle, nuclear, and atomic physics below a few TeV, providing a robust foundation for understanding the hot, expanding universe.

2.1 The Thermal History of the Universe

The cosmic microwave background radiation is a near-perfect black body. According to **Wien's Displacement Law**, the temperature T of a black-body radiator is inversely proportional to the peak wavelength λ_{peak} of its emission spectrum: $T \propto \frac{1}{\lambda_{\text{peak}}}$. Therefore, the ratio of the temperature at some past time, $T(z)$, to the temperature today, T_0 , is inversely proportional to the ratio of their respective peak wavelengths.

$$\frac{T(z)}{T_0} = \frac{\lambda_0}{\lambda(z)} \quad (26)$$

Here, $T(z)$ is the temperature at redshift z and $\lambda(z)$ is the peak wavelength at that time. T_0 is the temperature today (at $z = 0$) and λ_0 is the peak wavelength today.

From equation (13), we know that $\frac{\lambda_0}{\lambda(z)} = \frac{a(t_0)}{a(t_e)} = 1 + z$. Substituting these into our temperature relation from equation (26):

$$\frac{T(z)}{T_0} = 1 + z$$

This shows that the temperature of the cosmic plasma was higher in the past, at higher redshifts, by a factor of $(1 + z)$. The temperature of the cosmic plasma is unambiguously related to the redshift by:

$$T(z) = T_0(1 + z)$$

This allows us to use temperature, energy, or time interchangeably to chronicle the key epochs of our universe, moving backward from the present day.

2.1.1 The Planck Era and the Limits of Physics ($t \sim 10^{-43}$ s, $E \sim 10^{19}$ GeV).

Our current understanding of physics breaks down at the Planck scale. Here, the quantum mechanical nature of spacetime itself becomes dominant, and General Relativity can no longer be trusted. This is the realm of nonperturbative quantum gravity, where concepts like cosmic singularities must be addressed by a yet-unknown theory, possibly a form of string/M-theory.

2.1.2 The Grand Unification Epoch ($t \sim 10^{-36}$ s, $E \sim 10^{16}$ GeV).

At energies slightly below the Planck scale, it is hypothesized that the strong, weak, and electromagnetic forces were unified into a single Grand Unified Theory (GUT). The phase transition associated with the breaking of this GUT symmetry is a candidate for driving a period of extraordinarily rapid, accelerated expansion known as **inflation**. Inflation provides an elegant solution to several classic cosmological puzzles (the horizon, flatness, and monopole problems) and, most importantly, provides a physical mechanism for generating the primordial density perturbations. During inflation, microscopic quantum vacuum fluctuations were stretched to astrophysical scales, becoming the classical, nearly scale-invariant, and predominantly adiabatic perturbations that seeded all future structure.

2.1.3 Electroweak Symmetry Breaking ($t \sim 10^{-10}$ s, $E \sim 100$ GeV).

As the universe continued to cool, it underwent another phase transition where the electroweak force separated into the distinct electromagnetic and weak forces we observe today. Above this energy scale, particles like the W and Z bosons were massless.

2.1.4 The Quark-Hadron Transition ($t \sim 10^{-5}$ s, $E \sim 200$ MeV).

Prior to this time, the universe was a hot soup of fundamental particles, including quarks and gluons. At this transition, quarks and gluons became confined within composite particles called hadrons, primarily protons and neutrons.

2.1.5 Neutrino Decoupling ($t \sim 1$ s, $E \sim 1$ MeV).

Around this time, the rate of weak interactions became slower than the expansion rate of the universe. As a result, neutrinos decoupled from the thermal bath of photons and electrons. From this moment on, they have traveled essentially unimpeded through the cosmos, forming a Cosmic Neutrino Background (CνB). Their temperature today is predicted to be $T_\nu \approx 1.9$ K.

2.1.6 Big Bang Nucleosynthesis (BBN) ($t \sim 3$ min, $E \sim 0.1$ MeV).

Once the universe cooled to about a tenth of an MeV, protons and neutrons could fuse to form light atomic nuclei without being immediately destroyed by high-energy photons. This process created the primordial abundances of deuterium, helium-3, helium-4, and lithium-7. The remarkable agreement between the predictions of BBN and the observed abundances of these light elements is one of the strongest pillars of the Big Bang model.

2.1.7 Matter-Radiation Equality ($t \sim 10^4$ years, $z_{eq} \sim 3400$).

For thousands of years, the energy density of the universe was dominated by relativistic particles (radiation). However, as per Eq. (16), the energy density of radiation ($w = \frac{1}{3}$, $\rho_r \propto a^{-4}$) dilutes faster than that of non-relativistic matter ($w=0$, $\rho_m \propto a^{-3}$). The condition for equality is $\rho_m(z_{eq}) = \rho_r(z_{eq})$. Substituting the scaling relations gives:

$$\rho_{m0}(1 + z_{eq})^3 = \rho_{r0}(1 + z_{eq})^4$$

Solving for the redshift factor yields:

$$1 + z_{eq} = \frac{\rho_{m0}}{\rho_{r0}}$$

We can express the present-day energy densities in terms of the critical density, ρ_{cr0} , and the density parameters, $\Omega_i = \rho_{i0}/\rho_{cr0}$:

$$1 + z_{eq} = \frac{\Omega_{m0}\rho_{cr0}}{\Omega_{r0}\rho_{cr0}} = \frac{\Omega_{m0}}{\Omega_{r0}}$$

The total radiation density parameter, Ω_{r0} , includes contributions from both photons (γ) and neutrinos (ν):

$$\Omega_{r0} = \Omega_{\gamma0} + \Omega_{\nu0} = \Omega_{\gamma0} \left(1 + N_{\text{eff}} \frac{7}{8} \left(\frac{4}{11} \right)^{4/3} \right)$$

where $N_{\text{eff}} \approx 3.046$ [12] is the effective number of relativistic neutrino species. The total radiation density parameter is $\Omega_{r0}h^2 \approx 9 \times 10^{-5}$ [5]. Substituting this into our expression for the equality redshift, we arrive at the analytical formula:

$$1 + z_{eq} = \frac{\Omega_{m0}}{\Omega_{r0}} = \frac{\Omega_{m0}h^2}{\Omega_{r0}h^2} \approx \frac{\Omega_{m0}h^2}{9 \times 10^{-5}} \approx 1.1 \times 10^4 \Omega_{m0}h^2 \quad (27)$$

This epoch is of paramount importance for structure formation. Before equality, the intense radiation pressure prevented baryonic perturbations from growing and slowed the growth of dark matter perturbations inside the horizon. After equality, the gravitational potential wells created by the dominant dark matter component could begin to grow efficiently, pulling in both dark matter and baryons.

2.1.8 Recombination and Photon Decoupling ($t \sim 380,000$ years, $z \sim 1100$).

As the temperature dropped to about 0.3 eV, it became energetically favorable for electrons and protons to combine and form neutral hydrogen atoms. This event, known as recombination, drastically reduced the number density of free electrons, rendering the universe transparent to photons. The photons, which had been constantly scattering off electrons, decoupled from the baryons and began to free-stream across the universe. These are the photons we observe today as the Cosmic Microwave Background. The CMB is therefore a snapshot of the universe at this "surface of last scattering," carrying an imprint of the density and velocity fluctuations of the photon-baryon fluid at that precise moment.

2.2 The Nature of Primordial Fluctuations

The inflationary epoch is not only responsible for smoothing the universe but is also believed to have generated the primordial density perturbations that seeded all cosmic structure. These fluctuations can be classified into two fundamental modes.

2.2.1 Adiabatic Perturbations.

An adiabatic perturbation is one in which the local equation of state of the universe remains unchanged. This implies that the number density ratios of the different particle species are constant everywhere. A local overdensity in one component is therefore matched by a corresponding overdensity in all other components, such that the entropy per particle is spatially uniform. For the key species, this relationship can be expressed as:

$$\frac{\delta n_b}{n_b} = \frac{\delta n_c}{n_c} = \frac{\delta n_\nu}{n_\nu} = \frac{\delta n_\gamma}{n_\gamma}$$

Given that the energy density of relativistic species like photons scales as $\rho_\gamma \propto n_\gamma^{4/3}$, while for non-relativistic matter it scales as $\rho_m \propto n_m$ (cf. Derivation of $\rho_m \propto n_m$ and of $\rho_\gamma \propto n_\gamma^{4/3}$ in the Appendix), the fractional energy density fluctuations are related by:

$$\delta_b = \delta_c = \frac{3}{4}\delta_\gamma = \frac{3}{4}\delta_\nu$$

where $\delta_i = \delta\rho_i/\rho_i$. This type of perturbation represents a uniform compression or expansion of all components of the cosmic fluid. It was the evolution of these modes that was first calculated in detail by Peebles & Yu [13] and which forms the basis of the standard Λ CDM model [14, 15].

2.2.2 Isocurvature Perturbations

In stark contrast to adiabatic modes, an **isocurvature perturbation** is characterized by an initially uniform total energy density, such that the net energy density fluctuation is zero: $\delta\rho_{\text{tot}} = 0$. Because the total density is unperturbed, the spatial curvature remains homogeneous, lending these modes the name “isocurvature.” The perturbation, therefore, does not arise from a total over- or under-density, but rather from spatial fluctuations in the relative composition of the universe’s components. This is fundamentally an **entropy perturbation**, as the entropy per baryon, which is proportional to the photon-to-baryon number density ratio ($S_b \propto n_\gamma/n_b$), varies from place to place (cf. Derivation of $S_b \propto n_\gamma/n_b$ in the Appendix).

A canonical example involves a compensated fluctuation between non-relativistic matter (baryons and CDM, subscript m) and relativistic radiation (photons, subscript γ). The condition $\delta\rho_{\text{tot}} = \delta\rho_m + \delta\rho_\gamma = 0$ implies that any local overdensity in matter must be exactly balanced by a corresponding underdensity in radiation. We can relate these energy density fluctuations to the number density fluctuations using the scaling relations $\rho_m \propto n_m$ and $\rho_\gamma \propto n_\gamma^{4/3}$. For small perturbations, this gives:

$$\frac{\delta\rho_m}{\rho_m} = \frac{\delta n_m}{n_m} \quad \text{and} \quad \frac{\delta\rho_\gamma}{\rho_\gamma} = \frac{4}{3} \frac{\delta n_\gamma}{n_\gamma}$$

Substituting these into the condition $\delta\rho_m = -\delta\rho_\gamma$ yields a direct relationship between the fractional number density fluctuations:

$$\rho_m \left(\frac{\delta n_m}{n_m} \right) = -\rho_\gamma \left(\frac{4}{3} \frac{\delta n_\gamma}{n_\gamma} \right)$$

This shows that the number densities must fluctuate in opposition to maintain a constant total energy density. The gauge-invariant entropy perturbation, δS , is formally defined to capture this compositional fluctuation. For a matter-radiation system, its definition is:

$$\delta S \equiv \frac{\delta n_m}{n_m} - \frac{3}{4} \frac{\delta n_\gamma}{n_\gamma}$$

Given the anti-correlation between the number density fluctuations derived above, the two terms in this definition do not cancel, leading to the condition $\delta S \neq 0$. While these modes were explored as a viable cosmological scenario in early research [16], they are not a natural prediction of the simplest single-field inflationary models, which predominantly generate adiabatic perturbations.

2.2.3 The Primordial Power Spectrum.

To describe the statistical properties of these initial fluctuations, we use the power spectrum, $P(k)$, which quantifies the variance of the fluctuation amplitude as a function of comoving wavenumber k . It is often parameterized as a simple power law:

$$P(k) = \langle |\delta_k|^2 \rangle \propto k^n$$

where δ_k is the Fourier transform of the density contrast. A particularly important case is the scale-invariant or **Harrison-Zel’dovich spectrum**, which corresponds to $n = 1$ [17]. This spectrum has the unique property that the amplitude of gravitational potential fluctuations is independent of scale, meaning that perturbations of all sizes have the same “strength” as they enter the horizon. This specific initial condition was a key assumption in the development of the Cold Dark Matter model, which successfully addressed the shortcomings of earlier theories by predicting CMB anisotropies consistent with observational limits [17, 15].

2.3 The Cold Dark Matter Model

The Cold Dark Matter (CDM) paradigm posits that the vast majority of matter in the Universe is composed of non-baryonic particles that are non-relativistic (i.e., *cold*) at the epoch of matter-radiation equality, and interact with ordinary matter and radiation only through gravity (i.e., *dark*) [18, 19]. This model, particularly in its modern incarnation as Lambda-CDM 1.12.4, has become the standard framework for cosmology due to its remarkable success in explaining the large-scale structure of the Universe, from the anisotropies in the Cosmic Microwave Background (CMB) to the distribution of galaxies and the Lyman- α forest [20].

A key feature of the CDM model is its prediction of **hierarchical**, or “bottom-up,” structure formation. Because the dark matter particles are cold, primordial density fluctuations survive down to very small mass scales. This allows small dark matter halos to collapse first under their own gravity at high redshift. These small halos then progressively merge over cosmic time to form larger and more massive structures, such as galaxies and galaxy clusters [21]. This hierarchical assembly is a cornerstone prediction of the model.

2.3.1 The CDM Transfer Function

The link between the primordial power spectrum 2.2.3, $P_{\text{prim}}(k) \propto k^n$ (with $n \approx 1$ from inflation) and the linear matter power spectrum at later times is the transfer function, $T(k)$. The late-time linear power spectrum is given by $P(k) \propto k^n T(k)^2$. The transfer function encapsulates the physics of perturbation growth. For CDM, density perturbations on scales smaller than the Hubble horizon at the time of matter-radiation equality enter a radiation-dominated era where their growth is suppressed. This physical process imprints a characteristic "bend" in the power spectrum, distinguishing it from a pure power law.

Numerical simulations in the 1980s, which established the viability of the CDM model, used an analytical fit to the transfer function. Following the work of [21], a good approximation for the CDM power spectrum $|\delta_k|^2$ at comoving wavenumber k is given by:

$$|\delta_k|^2 = \frac{Ak}{(1 + \alpha k + \beta k^{3/2} + \gamma k^2)^2} \quad (28)$$

where A is a normalization constant. For a universe with a critical matter density ($\Omega_m = 1$) and a Hubble constant $H_0 = 100h \text{ km s}^{-1} \text{ Mpc}^{-1}$, the parameters are $\alpha = 1.7(\Omega_m h^2)^{-1} \text{ Mpc}$, $\beta = 9.0(\Omega_m h^2)^{-3/2} \text{ Mpc}^{3/2}$, and $\gamma = 1.0(\Omega_m h^2)^{-2} \text{ Mpc}^2$. This form accurately captures the transition from the large-scale ($|\delta_k|^2 \propto k$) to small-scale ($|\delta_k|^2 \propto k^{-3}$) behavior of the processed power spectrum.

2.3.2 The Universal Halo Profile

A fundamental prediction arising from high-resolution N-body simulations of hierarchical clustering in a CDM universe is that collapsed dark matter halos, regardless of their mass, share a nearly "universal" density profile (cf. figure 12).

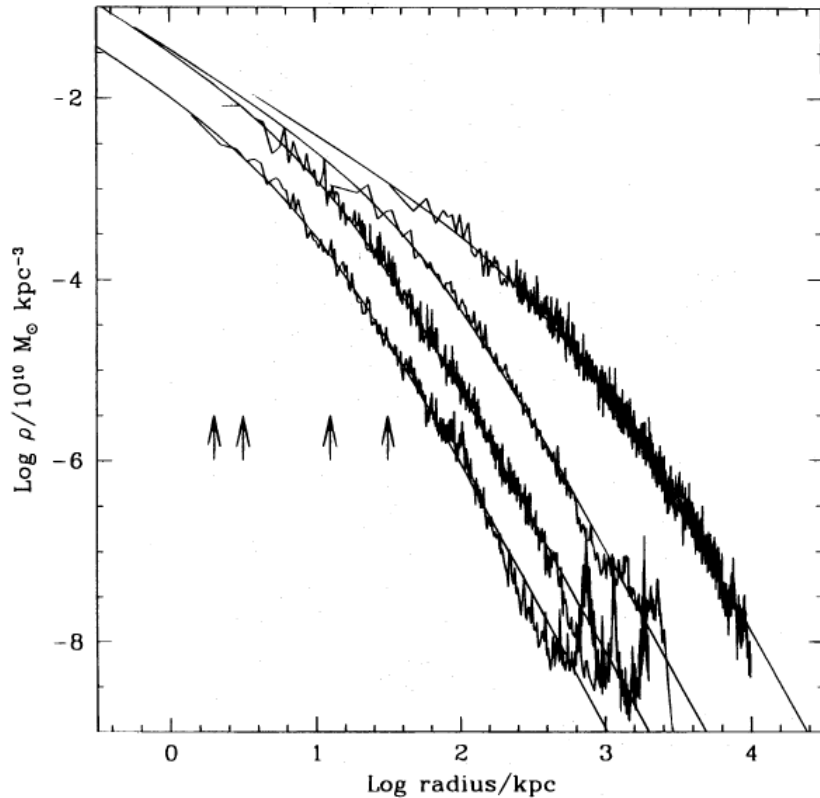


Figure 12: The universality of the CDM halo density profile, adapted from [22]. The plot displays the logarithm of the spherically-averaged density (y-axis, in units of $10^{10} M_{\odot} \text{ kpc}^{-3}$) versus the logarithm of radius (x-axis, in kpc) for four simulated dark matter halos spanning four orders of magnitude in mass. The smooth lines are fits using the NFW profile (Equation 29), which provide an excellent match to the simulation data (noisy lines) over two decades in radius. This demonstrates that halos of vastly different masses settle into a similar, universal structural form. The arrows indicate the gravitational softening length of each simulation, below which the profiles are not numerically resolved.

This profile was famously characterized by [22] and is now known as the Navarro-Frenk-White (NFW) profile.

It is given by:

$$\rho(r) = \frac{\rho_{\text{crit}} \delta_c}{(r/r_s)(1 + r/r_s)^2} \quad (29)$$

where ρ_{crit} is the critical density for closure, r_s is a characteristic scale radius, and δ_c is a dimensionless characteristic density. The profile has a characteristic asymptotic behavior: it approaches $\rho(r) \propto r^{-1}$ in the inner regions (a "cusp") and steepens to $\rho(r) \propto r^{-3}$ in the outer regions.

The shape of an individual halo's profile is determined by its concentration, $c = r_{\text{vir}}/r_s$, where r_{vir} is the virial radius³. A key prediction of the hierarchical model is a direct correlation between halo mass and concentration: lower-mass halos are more concentrated because they typically formed at earlier epochs when the mean density of the universe was higher (cf. figure 13).

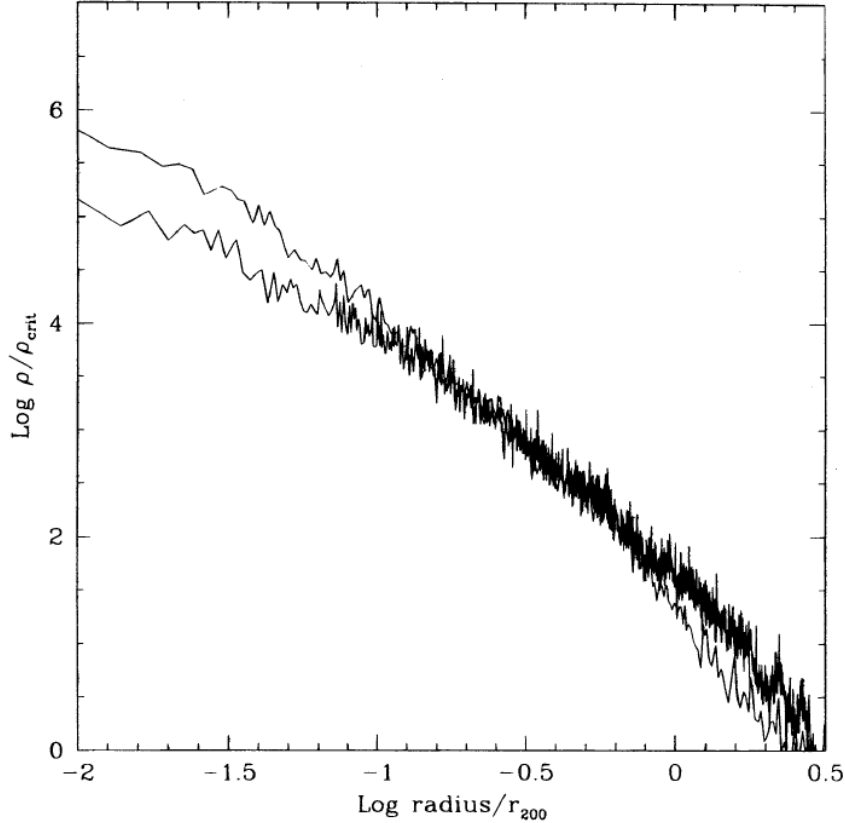


Figure 13: Visual evidence for the mass-concentration relation in CDM halos, adapted from [22]. The plot shows the density profiles of the most massive (lower curve) and least massive (upper curve) halos from Figure 12. Here, the radius (x-axis) is scaled by the virial radius (r_{200}) and the density (y-axis) is scaled by the critical density (ρ_{crit}). This scaling allows a direct comparison of the profile shapes. The less massive halo is systematically denser at small scaled radius, demonstrating that it is more centrally concentrated than the more massive halo. This supports the hierarchical formation scenario where smaller objects collapse earlier from a denser background universe.

2.3.3 Halo Substructure and Small-Scale Challenges

Another direct consequence of hierarchical formation is that dark matter halos are not smooth, monolithic structures. Instead, they are filled with a vast population of smaller subhalos that survived the merging process [21]. The predicted abundance of this substructure, combined with other observational tensions, has led to several persistent challenges to the CDM model [20]:

1. **The Cusp-Core Problem:** The cuspy inner profile ($\rho \propto r^{-1}$) predicted by Eq. 29 is in tension with observations of some galaxies, particularly dwarfs, which appear to favor a constant-density "core" [20].
2. **The Missing Satellites Problem:** Early simulations predicted that a Milky Way-sized halo should contain hundreds of subhalos massive enough to form visible galaxies, whereas only a few dozen are observed. This has evolved into the "Too Big to Fail" problem, which notes that the most massive

³The virial radius defines the boundary of a gravitationally collapsed and relaxed dark matter halo.

predicted subhalos are too dense to be consistent with the observed kinematics of the brightest Milky Way satellite galaxies [20].

3. **Excessive Clustering Power in Standard CDM:** While the Λ CDM model is highly successful on large scales, early work on galaxy clustering revealed that the original "standard" CDM model (with $\Omega_m = 1, h = 0.5$) produced excessive clustering power on scales of $\sim 30 - 40h^{-1}$ Mpc when compared to redshift surveys of IRAS galaxies [23]. This discrepancy was one of the key pieces of evidence that pushed cosmology towards the current Λ CDM model with $\Omega_m \approx 0.3$.
4. **Galaxy Downsizing:** The evolution of spheroidals⁴ presents a nuanced picture. The stellar populations of massive galaxies appear to have formed early and evolved passively, consistent with simple models. However, less massive systems show evidence of significant recent star formation, a trend known as "downsizing," which suggests a more complex assembly history than predicted by simple hierarchical models [24].

2.4 The Dynamics of Perturbations

The evolution of the metric potentials and the matter and radiation fluids is governed by the coupled Einstein and Boltzmann equations.

2.4.1 Gauge Choice and Metric Perturbations

To mathematically describe the evolution of primordial fluctuations (cf. Subsection 2.2), one must first perturb the background FLRW metric (9). The choice of coordinate system, or **gauge**, is a critical step in this process. While early, seminal works often used the synchronous gauge, this choice is known to have complications, such as the appearance of spurious, unphysical gauge modes that can obscure the interpretation of results [25].

A more physically intuitive and widely adopted choice in modern cosmology is the **Conformal Newtonian Gauge**, also known as the longitudinal gauge [25]. This gauge is particularly convenient for scalar perturbations because it is free from gauge ambiguities and offers a clear physical interpretation. The perturbed line element for a spatially flat universe is written as:

$$ds^2 = a^2(\tau) \left[-(1 + 2\Psi)d\tau^2 + (1 - 2\Phi)\delta_{ij}dx^i dx^j \right] \quad (30)$$

where τ is the conformal time, and the two scalar potentials $\Psi(\mathbf{x}, \tau)$ and $\Phi(\mathbf{x}, \tau)$ generalize the concept of the Newtonian potential to a cosmological context. Their physical interpretations are distinct and crucial [25]:

- Φ is the generalized **Newtonian gravitational potential**. In the non-relativistic limit, it governs the motion of particles and is the source of gravitational lensing and the Sachs-Wolfe effect.
- Ψ represents the **perturbation to the spatial curvature**.

One of the key advantages of this gauge is that the metric tensor is diagonal, which significantly simplifies calculations such as the geodesic equations. Furthermore, the potentials ψ and Φ are directly related to the gauge-invariant variables constructed by Bardeen, which ensures that they represent true physical degrees of freedom [25].

2.4.2 The Perturbed Einstein Equations

We start from the full Einstein field equations (2). By linearizing these equations using the perturbed metric from Eq. (30), we obtain a set of equations that link the geometry (Φ, Ψ) to the perturbed sources. The energy-momentum tensor, previously defined in Eq. (1), must now be described in terms of its perturbed components: the energy density perturbation $\delta\rho$, pressure perturbation δp , peculiar velocity field v , and anisotropic stress σ .

The key components of the perturbed Einstein equations in Fourier space become:

- **The (0-0) Component (Generalized Poisson Equation (7)):**

$$k^2\Psi + 3\mathcal{H}(\Psi' + \mathcal{H}\Phi) = 4\pi G a^2 \delta\rho_{\text{tot}} \quad (31)$$

where a prime denotes a derivative with respect to conformal time τ , and $\mathcal{H} = a'/a$ (cf. Derivation of the Generalised Poisson Equation in the Appendix).

⁴Spheroidals, or spheroidal stellar systems, are gravitationally bound collections of stars that are roughly spherical or ellipsoidal in shape. This category includes elliptical galaxies (from dwarf to giant) and the central bulges of spiral and lenticular galaxies. They are primarily defined by their dynamics, where the stars move on largely random, eccentric orbits, making the system 'pressure-supported' rather than rotationally-supported like a spiral disk. Consequently, they are typically composed of older stellar populations (often called 'red and dead') and contain little to no cold gas or ongoing star formation.

- **The (i-j) Traceless Component:** This equation provides a direct, algebraic relationship between the difference in the two potentials and the anisotropic stress σ (cf. Derivation of the Anisotropic Stress Equation in the Appendix):

$$\Phi - \Psi = \frac{8\pi G a^2}{k^2} (\rho_{\text{tot}} + p_{\text{tot}}) \sigma \quad (32)$$

The second equation shows that in a universe filled only with perfect fluids ($\sigma = 0$), the two potentials must be equal. The primary source of anisotropic stress in the early universe is the population of free-streaming relativistic particles, most notably neutrinos after they decouple from the thermal bath. As these particles travel unimpeded along geodesics, their directional distribution of momentum becomes anisotropic. On sub-horizon scales ($k \gg \mathcal{H}$) and when the anisotropic stress σ is zero ($\Phi = \Psi$) because there are no free-streaming relativistic particles dominating the dynamics, the Eq. (31) simplifies to the familiar Poisson equation (7), $k^2 \Phi \approx -4\pi G a^2 \delta \rho_{\text{tot}}$.

The Role of Neutrinos As established in foundational papers [14], [26], neutrinos are the key source of anisotropic stress in the early universe.

- **Before Decoupling ($T > 1 \text{ MeV}$):** Neutrinos are tightly coupled to the cosmic plasma through weak interactions. Frequent collisions keep their distribution function highly isotropic, meaning their pressure is the same in all directions. In this regime, $\sigma_\nu \approx 0$.
- **After Decoupling ($T < 1 \text{ MeV}$):** The weak interaction rate falls below the Hubble expansion rate, and neutrinos begin to free-stream. As they move collisionlessly from overdense to underdense regions, they create a quadrupole moment in their phase-space distribution. This quadrupole is precisely the source of the anisotropic stress, σ_ν .

This "directional dispersion" or "phase mixing" of the collisionless neutrinos leads to a non-zero σ , which in turn generates the gravitational slip $\Phi \neq \Psi$. Therefore, measuring the gravitational slip provides a direct observational window into the physics of free-streaming particles and serves as a powerful test of the standard cosmological model.

2.4.3 The Boltzmann Equation for Photons

While the Einstein equations govern the evolution of the spacetime geometry, the evolution of the distribution of particles within that geometry is described by the **Boltzmann equation**. For components that are not perfectly coupled to the cosmic fluid, such as photons near recombination or collisionless dark matter, a simple fluid description is inadequate. The Boltzmann equation describes the evolution of the particle distribution function $f(\mathbf{x}, \mathbf{p}, \tau)$ in phase space.

For photons, it is more convenient to work with the fractional temperature fluctuation, $\Theta(\mathbf{x}, \hat{n}, \tau) \equiv \frac{\delta T}{T}$, where $\hat{n}$ is the direction of photon propagation. The linearized Boltzmann equation(cf. Derivation of the Linearized Boltzmann Equation in the Appendix) in the Conformal Newtonian Gauge (30) takes the form:

$$\frac{\partial \Theta}{\partial \tau} + ik\mu\Theta + \frac{\partial \Psi}{\partial \tau} - ik\mu\Phi = C[\Theta] \quad (33)$$

where $\mu = \hat{k} \cdot \hat{n}$ is the cosine of the angle between the wavevector and the direction of propagation. Let us analyze each term:

- $\frac{\partial \Theta}{\partial \tau} + ik\mu\Theta$: This is the Liouville operator, describing the free-streaming of photons. An anisotropy pattern with wavenumber k simply propagates through space.
- $\frac{\partial \Psi}{\partial \tau}$: This is the **Integrated Sachs-Wolfe (ISW) effect**. A time-varying spatial curvature perturbation causes a change in the photon energy. This effect is active whenever the gravitational potentials are evolving, such as during the transition from radiation to matter domination.
- $-ik\mu\Phi$: This term represents the gravitational redshift/blueshift as photons stream through the potential Φ . Photons falling into a potential well ($\Phi < 0$) are blueshifted, and those climbing out are redshifted.
- $C[\Theta]$: This is the **collision term**, which accounts for interactions. For photons in the early universe, the dominant interaction is Thomson scattering with free electrons. The collision term takes the form:

$$C[\Theta] = -\dot{\kappa} (\Theta - \Theta_0 - \mu v_b)$$

where $\dot{\kappa} = n_e \sigma_T a$ is the differential optical depth, Θ_0 is the monopole (isotropic part) of the photon temperature fluctuation, and v_b is the peculiar velocity of the baryons. This term drives the photon distribution towards isotropy in the rest frame of the baryons, with a small Doppler shift contribution (μv_b).

This single equation elegantly encapsulates the key physics of CMB anisotropies. The gravitational potentials (Φ, Ψ) act as source terms, driving the evolution of the temperature fluctuations, while the collision term governs the coupling between photons and baryons. To solve this equation, it is standard practice to expand the angular dependence in Legendre polynomials, $\Theta(\mu) = \sum_l (-i)^l (2l+1) \Theta_l P_l(\mu)$, which transforms the single partial differential equation into an infinite hierarchy of coupled ordinary differential equations for the multipole moments Θ_l (cf. subsection 4.2.2 for more details). The tight-coupling approximation used by Hu & Sugiyama [26] is a method to analytically solve this hierarchy in the limit of very frequent scattering ($\bar{\kappa} \rightarrow \infty$).

2.5 Inflation

The standard Hot Big Bang model, while remarkably successful in describing the universe from the era of Big Bang Nucleosynthesis (BBN) onwards, faces several profound problems when extrapolated to its earliest moments. These issues are not inconsistencies within the theory itself, but rather problems of initial conditions that appear to require an extreme degree of fine-tuning for the universe to have evolved into its currently observed state. The theory of **cosmic inflation** was proposed as a dynamical mechanism to resolve these puzzles by positing a period of extraordinarily rapid, accelerated expansion in the very early universe [27, 28].

2.5.1 The Initial Condition Puzzles

The Flatness Problem 1.15 The first Friedmann equation, when written in terms of the total density parameter $\Omega(t) \equiv \rho_{\text{tot}}(t)/\rho_c(t)$, establishes a direct link between the universe's energy content and its spatial geometry [29]:

$$|\Omega(t) - 1| = \frac{|k|}{a^2 H^2} \quad (34)$$

Observations today, particularly from the Cosmic Microwave Background (CMB), indicate that our universe is spatially flat to a high degree of precision, with $\Omega_0 \approx 1$, [5]. However, in a standard decelerating universe (dominated by matter or radiation), the term $aH = \dot{a}$ decreases with time. Consequently, the deviation of Ω from unity must have grown dramatically over cosmic history [30]. For Ω_0 to be of order unity today, its value at the Planck time ($t_p \sim 10^{-43}$ s) must have been fine-tuned to be extraordinarily close to 1 [30]:

$$|\Omega(t_p) - 1| \leq 10^{-60}$$

The standard Big Bang model offers no explanation for this exquisite fine-tuning, which is known as the **flatness problem**.

The Horizon Problem The CMB exhibits a nearly uniform temperature of $T_0 \approx 2.725$ K across the entire sky, with anisotropies of only one part in 10^5 . This thermal equilibrium implies that all regions of the last scattering surface were in causal contact. However, a detailed mathematical analysis within the standard decelerating cosmology reveals a severe contradiction. The maximum distance that information, limited by the speed of light, could have traveled from the beginning of the universe ($t = 0$) until a given time t defines the physical particle horizon, $d_p(t)$, defined in the Equation (19). The key to the problem is to compare the size of a causally connected region at the time of decoupling (t_{dec} , corresponding to $z_{\text{dec}} \approx 1100$) with the size of our observable universe at that time. The comoving radius of a causally connected patch at decoupling is the comoving particle horizon, $\chi_p(t_{\text{dec}})$. The comoving radius of the surface we observe today as the CMB sky is the comoving distance light has traveled from decoupling to us, $\chi_{ls} = \int_{t_{\text{dec}}}^{t_0} \frac{c dt'}{a(t')}$.

The angular size that a single causally connected patch from the decoupling era subtends on our sky today is given by the ratio of its comoving size to the comoving distance to it:

$$\theta_H \approx \frac{\chi_p(t_{\text{dec}})}{\chi_{ls}} = \frac{\int_0^{t_{\text{dec}}} dt'/a(t')}{\int_{t_{\text{dec}}}^{t_0} dt'/a(t')} = \frac{\eta_{\text{dec}}}{\eta_0 - \eta_{\text{dec}}}$$

where η is the conformal time. To evaluate this, we must integrate through the radiation- and matter-dominated eras. A detailed calculation, accounting for the expansion history with parameters constrained by modern observations, yields $\eta_{\text{dec}} \approx 280$ Mpc and $\eta_0 \approx 14000$ Mpc [29]. Using these values, we find the angular size of the horizon at decoupling:

$$\theta_H \approx \frac{280 \text{ Mpc}}{14000 \text{ Mpc} - 280 \text{ Mpc}} \approx 0.02 \text{ radians} \approx 1.15^\circ$$

This implies that regions on the CMB sky separated by more than this angle were causally disconnected at the time of decoupling. Our sky is composed of thousands of such causally independent patches. Why, then, do they all share the same temperature to a precision of 10^{-5} ? This is the essence of the **horizon problem**.

The Monopole Problem. Grand Unified Theories (GUTs), which unify the strong and electroweak forces at energies around 10^{16} GeV, predict the copious production of magnetic monopoles during the GUT phase transition in the early universe [30]. If produced, their density would scale with the expansion like non-relativistic matter. Their predicted abundance would be so high that they would dominate the energy density of the universe today by many orders of magnitude, a result in stark contradiction with observations [31]. The standard model provides no mechanism to dilute this catastrophic overproduction of relics.

2.5.2 The Inflationary Solution: A Period of Accelerated Expansion

All these problems can be resolved by positing that the universe underwent a brief, intense period of accelerated expansion, $\ddot{a} > 0$, prior to the hot, radiation-dominated era [27]. From the second Friedmann equation (12), it is clear that acceleration is possible only if the universe is dominated by an energy component with a sufficiently negative pressure, specifically :

$$p < -\frac{\rho}{3}$$

In the limit where the pressure is maximally negative, $p \approx -\rho$, the energy density ρ remains nearly constant according to the fluid equation $\dot{\rho} + 3H(\rho + p) = 0$. The Friedmann equation $H^2 = 8\pi G\rho/3$ then implies that the Hubble parameter H is also nearly constant. This leads to an almost exponential expansion of the scale factor:

$$a(t) \propto e^{Ht}$$

This quasi-exponential expansion provides a powerful solution to the initial condition problems.

- **Solution to the Flatness Problem:** During inflation, the term (aH) in Eq. (34) grows enormously (exponentially). Consequently, any initial deviation of Ω from 1 is driven rapidly towards zero, $|\Omega - 1| \propto e^{-2Ht}$. Inflation does not require the universe to start precisely flat, but rather predicts that it will become flat as a result of the accelerated expansion.
- **Solution to the Horizon Problem:** Inflation provides a mechanism to generate a vast, homogeneous universe from a single, tiny, causally connected patch. Before inflation began, this patch was smaller than the Hubble radius at that time, allowing it to achieve thermal equilibrium. Inflation then stretched this smooth patch by an enormous factor, making it large enough to encompass the entire observable universe today. All points on our last scattering surface thus originated from this single, causally connected pre-inflationary region.
- **Solution to the Monopole Problem:** If magnetic monopoles were produced before the onset of inflation, the subsequent exponential expansion would dilute their number density by a factor of e^{3N} to an unobservably low level today, as their number would be conserved while the volume of the universe increased by a colossal factor.

To solve the horizon and flatness problems, the inflationary epoch must last for a sufficient **number of e-folds**, $N \equiv \ln(a_f/a_i)$, where a_i and a_f are the scale factors at the beginning and end of inflation, respectively. The required duration is typically estimated to be at least $N \approx 60 - 70$ [29, 30].

2.5.3 Realizing Inflation: The Inflaton Field

The required condition of negative pressure can be realized by the potential energy of a scalar field, dubbed the **inflation**, ϕ . For a homogeneous scalar field, the energy density and pressure are given by :

$$\rho_\phi = \frac{1}{2}\dot{\phi}^2 + V(\phi) \tag{35}$$

$$p_\phi = \frac{1}{2}\dot{\phi}^2 - V(\phi) \tag{36}$$

The equation of state parameter is $w_\phi = p_\phi/\rho_\phi$. If the dynamics of the field are such that its kinetic energy is much smaller than its potential energy, a condition known as the **slow-roll approximation**, then :

$$\frac{1}{2}\dot{\phi}^2 \ll V(\phi) \quad \Rightarrow \quad p_\phi \approx -V(\phi) \approx -\rho_\phi$$

This yields an equation of state $w_\phi \approx -1$, which drives the required accelerated expansion. The dynamics of the field are governed by the Klein-Gordon equation in an expanding background (cf. Derivation of the Klein-Gordon Equation for a Homogeneous Scalar Field in the Appendix) :

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0 \tag{37}$$

where a prime denotes a derivative with respect to ϕ . This equation describes the evolution of the inflaton field, where the expansion of the universe acts as a friction term ($3H\dot{\phi}$) on the field's motion within its potential $V(\phi)$.

The slow-roll condition is satisfied if the potential $V(\phi)$ is sufficiently flat. This is formalized by the **slow-roll conditions**, which require that the friction term $3H\dot{\phi}$ dominates the acceleration term $\ddot{\phi}$, and that the kinetic energy remains subdominant. These conditions can be expressed in terms of the potential via two dimensionless **slow-roll parameters**:

$$\epsilon_V \equiv \frac{m_{Pl}^2}{2} \left(\frac{V'}{V} \right)^2 \ll 1 \quad (38)$$

$$|\eta_V| \equiv m_{Pl}^2 \left| \frac{V''}{V} \right| \ll 1 \quad (39)$$

where $m_{Pl} = (8\pi G)^{-1/2}$ is the reduced Planck mass. Under these conditions, the equations of motion simplify to the slow-roll equations (cf. Derivation of the Slow-Roll Equations of Motion in the appendix):

$$H^2 \approx \frac{V(\phi)}{3m_{Pl}^2}$$

$$3H\dot{\phi} \approx -V'(\phi)$$

Inflation continues as long as these conditions are met. It ends when the potential becomes too steep, causing the parameters to approach unity. The field then begins to oscillate around the minimum of its potential, decaying into standard model particles and reheating the universe, thus initiating the Hot Big Bang era.

2.5.4 The Origin of Structure: Quantum Fluctuations

Inflation not only smooths the universe but also provides a causal mechanism for generating the primordial density perturbations that seeded the formation of galaxies and large-scale structure. The source of these perturbations is the quantum vacuum fluctuations of the inflaton field itself.

During inflation, pairs of virtual particles are constantly being created and annihilated in the vacuum, as dictated by the Heisenberg uncertainty principle. On sub-Hubble scales ($k > aH$), these fluctuations behave as they would in flat spacetime. However, as the universe expands exponentially, the physical wavelength of these fluctuations, $\lambda \propto a$, is stretched. When a fluctuation's wavelength becomes larger than the Hubble radius, $\lambda > H^{-1}$, it "freezes out" because it is no longer in causal contact. The amplitude of the inflaton field fluctuation, $\delta\phi$, at the moment of horizon crossing ($k = aH$) is given by (cf. Derivation of Quantum Fluctuations in de Sitter Space):

$$\langle (\delta\phi)^2 \rangle^{1/2} \approx \frac{H}{2\pi}$$

Because the Hubble parameter H is nearly constant during inflation, fluctuations of different wavelengths all freeze out with nearly the same amplitude. These frozen fluctuations of the scalar field correspond to slight time delays in the end of inflation in different regions of space, leading to small variations in the final energy density from one region to another. These energy density fluctuations, in turn, source perturbations in the spacetime metric, creating the primordial gravitational potentials that seed structure formation.

2.5.5 Predictions of Inflation

The inflationary paradigm makes several key, testable predictions about the nature of these primordial perturbations.

Scalar Perturbations. The primary prediction concerns the statistical properties of the scalar (density) perturbations. These are predicted to [30]:

- **Adiabatic:** All particle species (photons, baryons, dark matter) have their number density ratios held constant, corresponding to a uniform compression of the primordial fluid.
- **Gaussian:** The fluctuations follow a Gaussian random distribution, meaning their statistical properties are fully characterized by the two-point correlation function, or its Fourier transform, the power spectrum.
- **Nearly Scale-Invariant:** The power spectrum of the primordial curvature perturbation, ζ , is predicted to be a near power-law in wavenumber k , $P_\zeta(k) \propto k^{n_s-1}$. The amplitude of the spectrum is given by (cf. Derivation of the Scalar Perturbation Power Spectrum in the Appendix):

$$\Delta_\zeta^2(k) = \frac{k^3 P_\zeta(k)}{2\pi^2} = \frac{1}{24\pi^2 m_{Pl}^4} \frac{V}{\epsilon_V} \Big|_{k=aH}$$

The **spectral index** n_s , defined by $n_s - 1 \equiv d \ln \Delta_{\mathcal{R}}^2 / d \ln k$, is predicted to be very close to 1 but slightly red-tilted (i.e., less power on small scales), with deviations from scale-invariance given by the slow-roll parameters. To find the spectral index $n_s - 1 \equiv d \ln \Delta_{\mathcal{R}}^2 / d \ln k$, we note that $d \ln k = d \ln(aH) \approx H dt$.

$$n_s - 1 = \frac{1}{H} \frac{d}{dt} \ln \left(\frac{V}{\epsilon_V} \right) = \frac{1}{H} \left(\frac{V' \dot{\phi}}{V} - \frac{\epsilon'_V \dot{\phi}}{\epsilon_V} \right)$$

Using the slow-roll relations $\dot{\phi} \approx -V'/(3H)$ and the definitions of ϵ_V and η_V , this simplifies to:

$$n_s - 1 = 2\eta_V - 6\epsilon_V$$

Since the slow-roll parameters must be small, this naturally leads to $n_s \approx 1$.

Tensor Perturbations. Inflation also predicts the generation of a stochastic background of gravitational waves, or tensor perturbations. These are also generated from quantum vacuum fluctuations and are stretched to super-horizon scales. Their power spectrum is also nearly scale-invariant, with an amplitude given by (cf. Derivation of the Tensor Perturbation Power Spectrum in the Appendix):

$$\Delta_h^2(k) = \frac{2V}{3\pi^2 m_{Pl}^4} \Big|_{k=aH}$$

The relative amplitude of the tensor and scalar perturbations is a key prediction, typically characterized by the tensor-to-scalar ratio, r :

$$r \equiv \frac{\Delta_h^2}{\Delta_\zeta^2} = 16\epsilon_V$$

This relation, known as **the inflationary consistency relation**, provides a powerful observational test of the inflationary paradigm. A detection of primordial B-mode polarization in the CMB would provide a measurement of r , which would in turn determine the energy scale of inflation, $V^{1/4}$, and test the entire framework.

The predictions of a nearly flat universe ($\Omega_0 \approx 1$) with nearly scale-invariant, adiabatic, and Gaussian primordial perturbations are the cornerstones of the inflationary model and have been spectacularly confirmed by CMB experiments like COBE, WMAP, and Planck [32, 33, 5].

3 Modified Gravity

While the Λ CDM model 1.12.4 provides an excellent description of a wide range of cosmological observations, its reliance on a cosmological constant with a finely-tuned value has motivated the exploration of alternatives. One major avenue of research involves modifying the laws of gravity on cosmological scales. Instead of introducing a dark energy component, these models propose that the observed cosmic acceleration arises from a deviation of gravity from General Relativity (GR).

3.1 The Brans-Dicke Theory

One of an influential early attempt to modify GR is the scalar-tensor theory developed by Carl Brans and Robert Dicke in 1961 [34]. Their work was deeply motivated by **Mach's principle**, a concept suggesting that the inertia of a physical body is not an intrinsic property but is determined by the distribution of all other matter in the universe. Brans and Dicke argued that GR does not fully incorporate this principle, as spacetime can exist (e.g., the Minkowski vacuum) in the complete absence of matter.

3.1.1 The Scalar Field and the Brans-Dicke Action

The central idea of the Brans-Dicke theory is to promote the gravitational constant G to a dynamical field. Instead of being a fundamental constant of nature, the strength of gravity is determined by the value of a scalar field, $\phi(x^\mu)$, that permeates spacetime. The effective gravitational coupling is inversely proportional to this field:

$$G_{\text{eff}} \propto \frac{1}{\phi}$$

Since ϕ is a dynamical field, it is sourced by the matter content of the universe. This feature provides a mechanism to realize Mach's principle: the local inertial properties of spacetime (related to G_{eff}) are influenced by the global distribution of matter.

The theory is defined by the following action, which generalizes the Einstein-Hilbert action to include the scalar field ϕ and its dynamics:

$$S = \int d^4x \sqrt{-g} \left[\phi R - \frac{\omega}{\phi} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + \mathcal{L}_m \right] \quad (40)$$

where:

- R is the Ricci scalar, as in GR.
- $\mathcal{L}_m$ is the Lagrangian density for matter, which is assumed not to couple directly to ϕ .
- ω is a new, dimensionless free parameter known as the **Brans-Dicke parameter**. It controls the strength of the coupling between the scalar field and gravity.

The second term in the action represents the kinetic energy of the scalar field. In the limit where $\omega \rightarrow \infty$, this kinetic term dominates unless $\partial_\mu \phi = 0$. This forces ϕ to be constant, and the theory smoothly reduces to General Relativity with an effective gravitational constant $G = 1/\phi$. Thus, observational constraints on ω serve as a direct test of deviations from GR.

3.1.2 Field Equations

Varying the action (40) with respect to the metric tensor $g_{\mu\nu}$ and the scalar field ϕ yields the field equations for the theory.

Gravitational Field Equation Varying with respect to $g^{\mu\nu}$ gives the modified Einstein field equations (cf. Derivation of the Brans-Dicke Field Equations in the Appendix):

$$G_{\mu\nu} = \frac{8\pi}{\phi} T_{\mu\nu} + \frac{\omega}{\phi^2} \left(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} \partial_\lambda \phi \partial^\lambda \phi \right) + \frac{1}{\phi} (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)) \quad (41)$$

The right-hand side contains three terms:

1. The standard energy-momentum tensor of matter, but with the gravitational coupling set by $1/\phi$.
2. The energy-momentum tensor associated with the scalar field ϕ itself.
3. A term describing the direct interaction between the scalar field and the curvature of spacetime.

Scalar Field Equation Varying the action with respect to ϕ gives a wave-like equation for the scalar field (cf. Derivation of the Brans-Dicke Scalar Equations in the Appendix):

$$g^{\mu\nu} \nabla_\mu \nabla_\nu \phi = \frac{8\pi}{3 + 2\omega} T \quad (42)$$

where $T = g^{\mu\nu} T_{\mu\nu}$ is the trace of the energy-momentum tensor. This equation explicitly shows that matter acts as the source for the scalar field. For standard matter and radiation, $T \neq 0$, causing ϕ (and thus G_{eff}) to evolve dynamically.

3.1.3 The Weak-Field Approximation

To understand the observational consequences of the Brans-Dicke theory, we first examine its predictions in the weak-field limit, where the spacetime metric $g_{\mu\nu}$ and the scalar field ϕ are small perturbations of a background flat spacetime and a constant background scalar field ϕ_0 . We write:

$$g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$$

$$\phi = \phi_0 + \varphi$$

where $\eta_{\mu\nu}$ is the Minkowski metric, and $|h_{\mu\nu}| \ll 1$ and $|\varphi| \ll \phi_0$. In this limit, the field equations can be linearized. The wave equation for the scalar field (42) becomes:

$$\eta^{\mu\nu} \partial_\mu \partial_\nu \phi = \frac{8\pi}{3 + 2\omega} T$$

This equation shows that the perturbation φ is sourced by the trace of the energy-momentum tensor. For a static, localized source of mass M , the solution (cf. Derivation of the Scalar Field Solution in the Weak-Field Limit in the Appendix) is:

$$\varphi(r) = -\frac{2M}{r(3 + 2\omega)}$$

The linearized gravitational field equation (41) can also be solved. For a static point mass M , the relevant components of the metric perturbation are found to be:

$$h_{00} = \frac{2M}{r\phi_0} \left(\frac{2+\omega}{3+2\omega} \right)$$

$$h_{ij} = \delta_{ij} \frac{2M}{r\phi_0} \left(\frac{1+\omega}{3+2\omega} \right)$$

The effective gravitational constant in this limit is identified from the h_{00} component, which corresponds to the Newtonian potential. We define $G_{\text{eff}} = \frac{1}{\phi_0} \left(\frac{2(2+\omega)}{3+2\omega} \right)$. This allows us to write the metric components in the familiar post-Newtonian form:

$$g_{00} \approx -1 + \frac{2G_{\text{eff}}M}{r}$$

$$g_{ij} \approx \left(1 + \gamma \frac{2G_{\text{eff}}M}{r} \right) \delta_{ij}$$

where γ is the post-Newtonian parameter that measures the amount of spatial curvature produced by a unit rest mass. In Brans-Dicke theory, it is given by:

$$\gamma = \frac{1+\omega}{2+\omega}$$

In the limit $\omega \rightarrow \infty$, $\gamma \rightarrow 1$, and the theory recovers the weak-field predictions of General Relativity.

3.1.4 Static Spherically Symmetric Solution and Perihelion Precession

For strong-field tests, such as the perihelion precession of Mercury, the weak-field approximation is insufficient. An exact vacuum solution for a static, spherically symmetric gravitational source is required. This solution, expressed in isotropic coordinates, is given by:

$$ds^2 = -e^{\nu(r)} dt^2 + e^{\lambda(r)} (dr^2 + r^2 d\Omega^2)$$

where $e^{\nu(r)}$ and $e^{\lambda(r)}$ are more complex functions of r , the mass of the source M , and the Brans-Dicke parameter ω . We can try to solve this with parametrisation and we end up deriving a solution (cf. Derivation of the Static Spherically Symmetric Vacuum Solution in Brans-Dicke Theory in the Appendix) of the form:

$$\phi(r) = \phi_0 \left(\frac{1 - K/r}{1 + K/r} \right)^{-\frac{1+\omega}{3+2\omega}}$$

This solution allows for the calculation of the advance of the perihelion of an orbiting body. The predicted rate of precession (cf. [34]) for a planetary orbit in the Brans-Dicke theory is:

$$\dot{\omega}_{\text{BD}} = \left(\frac{4+3\omega}{6+3\omega} \right) \dot{\omega}_{\text{GR}}$$

where $\dot{\omega}_{\text{GR}}$ is the precession predicted by General Relativity. This result provides a direct way to test the theory against observation.

3.1.5 Comparison with Solar System Observations

The predictions of Brans-Dicke theory have been subjected to rigorous tests using high-precision Solar System observations. These tests provide some of the strongest constraints on any alternative theory of gravity.

Perihelion Advance of Mercury The observed perihelion advance of Mercury, after accounting for all Newtonian perturbations, is in excellent agreement with the prediction of General Relativity. The Brans-Dicke prediction, however, depends on the value of ω . The historical debate on this topic centered on the Sun's oblateness (its equatorial bulge), which would add a classical contribution to the precession.

In 1967, Dicke and Goldenberg published measurements suggesting a significant solar oblateness, which, if correct, would have reduced the anomalous precession that relativity needed to explain [35]. This smaller value would have been consistent with a Brans-Dicke theory with $\omega \approx 6$. However, subsequent and more precise measurements, such as those by Hill and Stebbins in 1975, found a much smaller oblateness, which supported the GR prediction [36].

The issue is now considered settled. Modern data from the MESSENGER mission has provided unprecedentedly accurate measurements of the Sun's gravitational field, confirming that its oblateness is small and its contribution to Mercury's precession is minimal [37]. This confirms that the observed precession is almost entirely a relativistic effect, in strong agreement with General Relativity and ruling out low- ω Brans-Dicke theories.

Deflection of Light and Shapiro Time Delay The deflection of light by the Sun and the Shapiro time delay (the delay of radar signals passing near the Sun) are both sensitive to the post-Newtonian parameter γ . As shown, in Brans-Dicke theory, $\gamma = (1 + \omega)/(2 + \omega)$, while General Relativity predicts $\gamma = 1$.

The most precise measurement of γ to date comes from the Cassini mission, which tracked radio signals sent to and from the spacecraft as it passed behind the Sun. The results, published by Bertotti, Iess, and Tortora in 2003, yielded a value of:

$$\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$$

This measurement is highly consistent with General Relativity [38]. As detailed in comprehensive reviews on experimental gravity, this places a very strong lower-bound constraint on the Brans-Dicke parameter [39]:

$$\omega > 40,000$$

This implies that if a scalar field of the Brans-Dicke type exists, its coupling to gravity must be extremely weak, rendering the theory practically indistinguishable from General Relativity in the Solar System.

3.2 $f(R)$ Theories of Gravity

A significant generalization of General Relativity can be achieved by modifying the geometric part of the Einstein-Hilbert action. Instead of a Lagrangian that is linear in the Ricci scalar R , one can consider a more general function of R . This class of theories is known as $f(R)$ gravity. The motivation for such theories is twofold. From a high-energy physics perspective, quantum corrections to General Relativity, as required by renormalization or predicted by string theory, naturally introduce higher-order curvature invariants into the effective action. From a cosmological standpoint, these modifications offer a potential alternative to dark energy, aiming to explain the observed accelerated expansion of the universe by altering the laws of gravity on large scales [40].

The fundamental action for $f(R)$ gravity is a direct generalization of the Einstein-Hilbert action:

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} f(R) + S_M(g_{\mu\nu}, \psi)$$

where $\kappa = 8\pi G$, S_M is the matter action, and $f(R)$ is an arbitrary function of the Ricci scalar. The specific dynamics and physical interpretation of the theory depend critically on the variational principle, or "formalism," used to derive the field equations. The two most prominent formalisms are the metric and the Palatini approaches [40].

3.2.1 The Metric and Palatini Formalisms

The distinction between the two main versions of $f(R)$ gravity lies in the set of independent variables with respect to which the action is varied.

Metric Formalism In the metric formalism, the connection is assumed to be the Levi-Civita connection of the metric $g_{\mu\nu}$ from the outset. The action is varied only with respect to the metric. This procedure yields the field equations (Derivation of the Field Equations for Metric $f(R)$ Gravity in the Appendix) :

$$f'(R)R_{\mu\nu} - \frac{1}{2}f(R)g_{\mu\nu} - [\nabla_\mu \nabla_\nu - g_{\mu\nu} \nabla^\lambda \nabla_\lambda]f'(R) = \kappa T_{\mu\nu}$$

where a prime denotes differentiation with respect to R , and $T_{\mu\nu}$ is the standard energy-momentum tensor. These are fourth-order partial differential equations for the metric, a significant departure from the second-order equations of General Relativity. The fourth-order terms vanish only when $f(R)$ is linear in R , in which case the theory reduces to GR. The trace of this equation,

$$f'(R)R - 2f(R) + 3\nabla^\mu \nabla_\mu f'(R) = \kappa T$$

shows that the Ricci scalar R is a dynamical degree of freedom, unlike in GR where it is algebraically determined by the trace of the energy-momentum tensor [40].

Palatini Formalism In the Palatini formalism, the metric $g_{\mu\nu}$ and the connection $\Gamma_{\mu\nu}^\lambda$ are treated as independent fields. The Ricci tensor, denoted here as $\mathcal{R}_{\mu\nu}$, is constructed solely from the independent connection. A crucial assumption is that the matter action S_M does not depend on this connection. Varying the action with respect to the metric and the connection yields two sets of equations. After simplification, they can be written as:

$$f'(\mathcal{R})\mathcal{R}_{(\mu\nu)} - \frac{1}{2}f(\mathcal{R})g_{\mu\nu} = \kappa T_{\mu\nu}$$

$$\bar{\nabla}_\lambda (\sqrt{-g} f'(\mathcal{R}) g^{\mu\nu}) = 0$$

where $\mathcal{R} = g^{\mu\nu} \mathcal{R}_{\mu\nu}$ and $\bar{\nabla}$ is the covariant derivative associated with Γ . The second equation implies that the independent connection is the Levi-Civita connection of a metric $h_{\mu\nu} = f'(\mathcal{R}) g_{\mu\nu}$, which is conformally related to the physical metric. The trace of the first field equation gives an algebraic, rather than differential, relation between $\mathcal{R}$ and the trace of the energy-momentum tensor, T :

$$f'(\mathcal{R}) \mathcal{R} - 2f(\mathcal{R}) = \kappa T$$

This implies that $\mathcal{R}$ is not an independent dynamical degree of freedom but is instead a function of the matter content. This makes Palatini $f(R)$ gravity fundamentally different from its metric counterpart [40].

3.2.2 Equivalence with Brans-Dicke Theory

A powerful tool for understanding $f(R)$ theories is their dynamical equivalence to specific classes of Brans-Dicke theory. This equivalence is established by introducing an auxiliary scalar field and performing field redefinitions.

Metric Formalism Equivalence Metric $f(R)$ gravity can be shown to be dynamically equivalent to a Brans-Dicke theory with a potential and a Brans-Dicke parameter $\omega_0 = 0$. This is achieved by defining a scalar field $\phi = f'(R)$ and a potential $V(\phi) = R\phi - f(R)$. The action can then be rewritten in the Jordan frame as:

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} [\phi R - V(\phi)] + S_M(g_{\mu\nu}, \psi)$$

The field equation for ϕ is algebraic, $R = V'(\phi)$, but the trace of the gravitational field equations reveals its dynamics: $3\nabla^\mu \nabla_\mu \phi + 2V(\phi) - \phi V'(\phi) = \kappa T$. This confirms that the scalar field is dynamical [40].

Palatini Formalism Equivalence Following a similar procedure, Palatini $f(R)$ gravity can be shown to be equivalent to a Brans-Dicke theory with a potential and a Brans-Dicke parameter $\omega_0 = -3/2$. The action in the Jordan frame is:

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left[\phi R + \frac{3}{2\phi} \partial_\mu \phi \partial^\mu \phi - V(\phi) \right] + S_M(g_{\mu\nu}, \psi)$$

Despite the presence of a kinetic term for ϕ in the action, the field equations reveal that the scalar is non-dynamical and is determined algebraically by the matter trace: $2V(\phi) - \phi V'(\phi) = \kappa T$. This lack of dynamics is a key feature of the Palatini formalism and the root of many of its phenomenological challenges [40].

3.2.3 The Chameleon Mechanism: Hiding the Fifth Force

A generic scalar field that couples to matter, such as the one that arises in metric $f(R)$ gravity, inevitably mediates a fifth force. This force would lead to violations of the equivalence principle and deviations from General Relativity that are strongly constrained by Solar System experiments [38]. The **chameleon mechanism** provides an elegant solution to this problem by allowing the scalar field to acquire an environment-dependent effective mass, making it heavy and short-ranged in high-density regions (like the Solar System) but light and long-ranged on cosmological scales [41].

The Effective Potential The core of the mechanism lies in the direct coupling between the scalar field ϕ and the matter density. To see this mathematically, it is clearest to work in the *Einstein frame*, where the gravitational action takes the standard Einstein-Hilbert form, and the scalar field's coupling to matter is made explicit. The action is:

$$S = \int d^4x \sqrt{-\tilde{g}} \left[\frac{\tilde{R}}{2\kappa} - \frac{1}{2} (\tilde{\nabla} \phi)^2 - V(\phi) \right] + S_m(g_{\mu\nu}, \psi)$$

The metric in the Einstein frame, $\tilde{g}_{\mu\nu}$, is related to the physical metric in the Jordan frame, $g_{\mu\nu}$, by a conformal transformation involving the scalar field: $g_{\mu\nu} = A^2(\phi) \tilde{g}_{\mu\nu}$. The matter action S_m is defined with the physical metric.

The equation of motion for the scalar field ϕ is derived from this action. Crucially, the coupling to matter introduces a new term into the field's potential. The full dynamics of the scalar field are governed by an **effective potential**, $V_{\text{eff}}(\phi)$, which is the sum of its bare potential $V(\phi)$ and a term dependent on the local matter density, ρ_m :

$$V_{\text{eff}}(\phi) = V(\phi) + \rho_m A(\phi) \tag{43}$$

The presence of the $\rho_m A(\phi)$ term means that the shape of the potential, and specifically the location of its minimum, now depends on the local density of matter.

Environment-Dependent Mass and Screening For a given environment with density ρ_m , the scalar field will settle to the minimum of this effective potential, $\phi_{\min}(\rho_m)$. The effective mass of perturbations of the field around this minimum is given by the second derivative of the potential:

$$m_{\text{eff}}^2 = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \phi^2} \right|_{\phi=\phi_{\min}}$$

This leads to two distinct behaviors:

- **In High-Density Environments** (e.g., inside the Earth or the Sun), ρ_m is large. This makes the effective mass m_{eff} large. A massive scalar field mediates a Yukawa-type force, $F \propto \frac{e^{-m_{\text{eff}} r}}{r^2}$, which has an extremely short range $\lambda = 1/m_{\text{eff}}$. The fifth force is therefore suppressed, or "screened," and the theory becomes indistinguishable from General Relativity in local experiments.
- **In Low-Density Environments** (e.g., intergalactic space), $\rho_m \approx 0$. The effective potential reduces to the bare potential, $V_{\text{eff}}(\phi) \approx V(\phi)$. If the bare potential is shallow, the field is very light (small m_{eff}). It can then mediate a long-range force and have a significant impact on the expansion of the universe, potentially acting as dark energy.

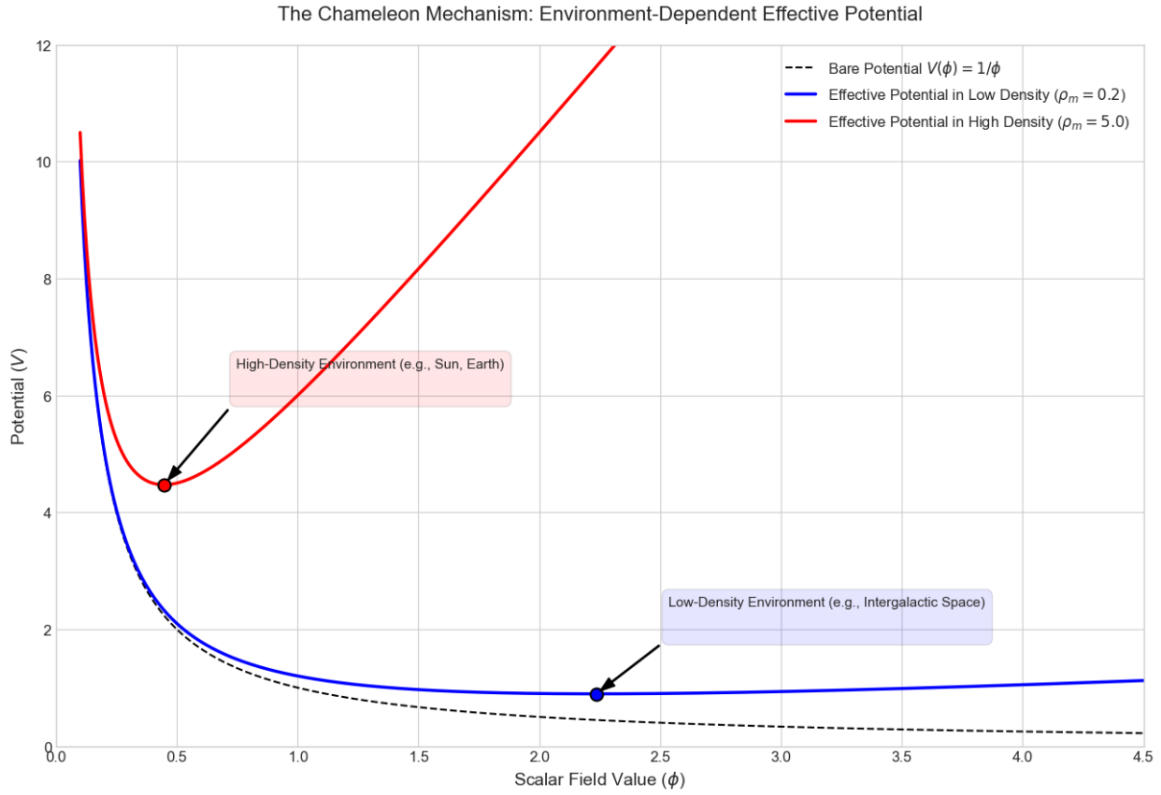


Figure 14: This figure illustrates the core principle of the chameleon mechanism, where a scalar field (ϕ) acquires an effective mass that depends on the local matter density (ρ_m). The plot shows the bare potential $V(\phi)$ (dashed line) and the effective potential $V_{\text{eff}} = V(\phi) + \rho_m \phi$ for two distinct environments. In **high-density environments** (red line), such as the Sun, the potential well becomes deep and steep. The large curvature at the minimum gives the field a large effective mass (m_{eff}), resulting in a short-range force that is "screened" and hidden from local experiments. Conversely, in **low-density environments** (blue line), like intergalactic space, the potential is much shallower. This implies a small effective mass, allowing the field to mediate a long-range fifth force that can influence the expansion of the universe on cosmological scales.

The Thin-Shell Condition For an object (like the Earth) to be screened from the influence of the ambient scalar field of its environment (the Solar System), it is not sufficient for the field to be massive inside the object. The field value within the object must actually be close to the minimum of its high-density potential. Whether this occurs is determined by the **thin-shell condition**.

An object with radius R_c and surface Newtonian potential Φ_c is considered to have a "thin shell" if its self-gravity is strong enough to drive the scalar field to its local minimum. The condition for this is given by

the thin-shell parameter:

$$\frac{\Delta R_c}{R_c} \approx \frac{\phi_{\text{bg}} - \phi_c}{6M_{Pl}\Phi_c} \ll 1$$

where ϕ_{bg} and ϕ_c are the values of the scalar field in the background and at the center of the object, respectively, and M_{Pl} is the Planck mass. If this condition is met, only a thin outer layer of the object contributes to its scalar "charge," and its influence on external bodies is suppressed. Objects with deep gravitational potentials (large Φ_c) are more easily screened. The Earth, the Sun, and laboratory test masses are all deeply in the thin-shell regime for viable chameleon models, which explains why no fifth force is detected [41, 40].

Experimental Verdict from the MICROSCOPE Mission The core prediction of a detectable Equivalence Principle (EP) violation in space was put to the test by the French CNES mission, MICROSCOPE. Operating from 2016 to 2018, this satellite experiment compared the free-fall of two test masses of different compositions (titanium and platinum-rhodium) to unprecedented precision. The final results, published in 2022, found no violation of the Equivalence Principle. The mission constrained the Eötvös parameter to $\eta = (-1.5 \pm 2.8) \times 10^{-15}$ [42]. This null result stands in stark contrast to the optimistic "smoking gun" signal predicted by the chameleon model [41]. Rather than discovering a large EP violation that was screened on Earth, MICROSCOPE's findings place severe constraints on the chameleon parameter space, effectively ruling out models with order-unity couplings that would have produced the dramatic effects anticipated by Khoury and Weltman [41].

3.2.4 Viability and Observational Constraints

While $f(R)$ theories are compelling as alternatives to dark energy, they must satisfy stringent observational and theoretical viability criteria before they can be considered serious contenders to the Λ CDM model.

Constraints on Metric $f(R)$ Gravity The weak-field limit of metric $f(R)$ gravity can be problematic. For a broad class of models, the theory introduces a scalar degree of freedom that mediates a fifth force, predicting a value for the post-Newtonian parameter of $\gamma = 1/2$. This is in strong contradiction with Solar System observations, most notably from the Cassini mission, which tracked radio signals passing near the Sun and constrained $\gamma - 1 = (2.1 \pm 2.3) \times 10^{-5}$ [38]. This makes simple metric $f(R)$ models non-viable.

However, this conclusion can be evaded if the scalar degree of freedom acquires a large, environment-dependent mass in high-density regions like the Solar System. This is achieved through the **chameleon mechanism** 3.2.3, which screens the fifth force locally. This would make the scalar short-ranged and effectively invisible to local tests, while allowing it to be light and have cosmological effects at the low densities of the universe. This mechanism, while elegant, severely restricts the viable functional forms of $f(R)$ [40].

A second critical constraint is stability. It has been shown that for a model to be free of a catastrophic matter instability (the Dolgov-Kawasaki instability) on very short timescales, the condition $f''(R) \geq 0$ must be satisfied. This ensures that the scalar degree of freedom is not a tachyon (i.e., it has a non-negative mass squared) [43]. Furthermore, cosmological data from the Planck satellite and large-scale structure surveys place tight constraints on deviations from General Relativity, further limiting the parameter space of viable $f(R)$ models [44].

Constraints on Palatini $f(R)$ Gravity The Palatini formalism, despite its appealing second-order field equations, faces severe challenges that likely render it non-viable in its standard form. The algebraic relationship between the curvature $\mathcal{R}$ and the matter trace T leads to several fundamental problems:

- **Conflict with the Standard Model:** The dependence on T introduces non-perturbative corrections to the matter action and strong, unacceptable couplings between gravity and matter fields at microscopic scales. This leads to violations of the equivalence principle that are inconsistent with experimental physics [45, 40].
- **Surface Singularities:** When modeling stellar objects with a sharp surface (i.e., a rapid change in matter density), the algebraic dependence of $\mathcal{R}$ on T can cause the curvature to diverge at the surface. This leads to infinite tidal forces, making it impossible to construct realistic models of stars with a standard polytropic equation of state [46].
- **Cauchy Problem:** The presence of higher derivatives of the matter fields in the effective field equations that emerge from the theory makes it highly unlikely that the theory admits a well-posed initial value problem in the presence of matter. This means the theory may lack predictive power [40].

These issues are fundamental to the structure of the theory and are largely independent of the specific choice for the function $f(R)$, making the Palatini formalism a less promising avenue for explaining cosmic acceleration.

3.2.5 Gravity with Perturbative Constraints: An Alternative Formalism

An alternative approach to handling the higher-order dynamics of $f(R)$ gravity, which aims to produce modified cosmological behavior without introducing new dynamical degrees of freedom, is the formalism of **gravity with perturbative constraints** [47, 48]. This method provides a consistent way to obtain second-order field equations from a higher-order Lagrangian by treating the modifications to General Relativity as perturbative and preventing the associated new degrees of freedom from evolving independently.

The method is applied to $f(R)$ models with a Lagrangian that deviates slightly from the Einstein-Hilbert action, such as:

$$f(R) = R - 2\Lambda + \mu^4 R^{-n}$$

where μ^4 is a small parameter controlling the deviation. For a flat Friedmann-Robertson-Walker (FRW) (9) metric, the general field equations for $f(R)$ gravity lead to a modified Friedmann equation that contains higher-order time derivatives of the scale factor $a(t)$ [47]:

$$-3f'\frac{\ddot{a}}{a} + \frac{1}{2}f + 3f''\dot{R}\frac{\dot{a}}{a} = \kappa\rho \quad (44)$$

$$Rf' - 2f - 3\left(f''\ddot{R} + \dot{R}^2 f''' + 3\frac{\dot{a}}{a}\dot{R}f''\right) = -\kappa(\rho + 3P) \quad (45)$$

The core of the perturbative constraint method is to reduce these higher-order equations to a tractable form through a two-step process:

1. **Zeroth-Order Solution:** First, one solves the equations with $\mu^4 = 0$, which corresponds to standard General Relativity. This provides the background solution, $a^{(0)}(t)$, and the corresponding Ricci scalar, $R^{(0)}(t)$. For a Λ CDM universe, the background scale factor is given by [47]:

$$a^{(0)}(t) = \left(\frac{\rho_{m,0}}{\Lambda}\right)^{1/3} \sinh^{2/3}\left(\frac{\sqrt{3\Lambda}}{2}t\right)$$

where the equation is expressed in units where $8\pi G = 1$.

2. **Perturbative Correction:** This background solution, $a^{(0)}(t)$, is then substituted into all the higher-derivative terms of the full field equations (i.e., the terms multiplied by μ^4). This procedure effectively "freezes" the new degrees of freedom, reducing the full system to a first-order ordinary differential equation (in time) for the corrected scale factor, $a(t)$.

For the specific and cosmologically relevant case where $n = 1$ (an inverse-Ricci term) and $\Lambda \neq 0$, this procedure yields a modified Friedmann equation, valid to first order in μ^4 [47]:

$$3H^2 = \rho_m + \Lambda - \mu^4 \frac{3\Lambda - 6(\rho_m + \Lambda)\frac{\rho_m}{\rho_m + 4\Lambda}}{(\rho_m + 4\Lambda)^2}$$

The corresponding acceleration equation is:

$$6\frac{\ddot{a}}{a} = -\rho_m + 2\Lambda - \mu^4 \frac{6\Lambda + 24(\rho_m + \Lambda)\frac{\rho_m}{\rho_m + 4\Lambda} - 54(\rho_m + \Lambda)\left(\frac{\rho_m}{\rho_m + 4\Lambda}\right)^2}{(\rho_m + 4\Lambda)^2}$$

These equations can be mapped to an effective dark energy component with a time-varying equation of state parameter, w_X (cf. figure 15). To first order in μ^4 , this parameter is given by [47]:

$$w_X \approx -1 + \frac{\mu^4}{(\rho_m + 4\Lambda)^2} \frac{\rho_m + \Lambda}{\rho_X} \left(12\frac{\rho_m}{\rho_m + 4\Lambda} - 18\left(\frac{\rho_m}{\rho_m + 4\Lambda}\right)^2\right)$$

where $\rho_X = 3H^2 - \rho_m$ is the total effective energy density of the dark energy component in the newly introduced XCDM model.

This formalism is distinct from the Palatini approach. By design, it is only valid in the perturbative regime and thus decouples cosmological predictions from physics at high densities (e.g., atomic scales), where the perturbative expansion would break down. This avoids some of the fundamental issues that challenge Palatini gravity, such as its problematic behavior with discrete matter sources [45, 47]. The framework demonstrates that it is possible to generate a non-trivial dark energy equation of state, consistent with observational constraints, without invoking new dynamical fields.

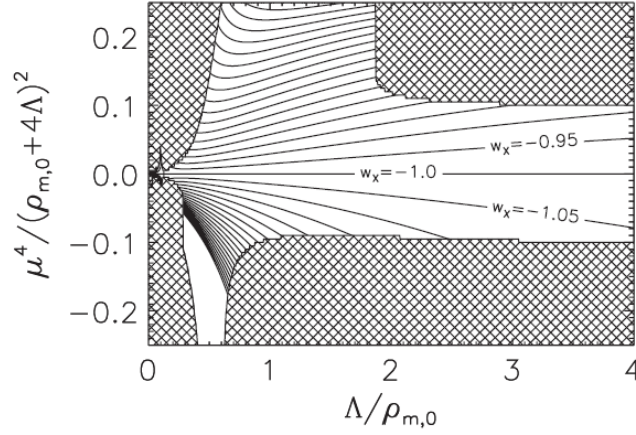


Figure 15: Contours of constant values of the parameter w_X , adapted from [47]. The current observed limits are approximately $-1.14 < w_X < -0.88$. The normalization constant $\rho_{m,0}$ is the density of matter at the present epoch. The hatched area shows the region of the parameter space where the calculation of w_X is not perturbatively valid.

3.3 Einstein-Gauss-Bonnet Gravity and its 4D Formulation

Another fascinating avenue in modifying gravity stems from **Lovelock's theorem**, which identifies the most general tensor that can be constructed from the metric and its derivatives while yielding second-order equations of motion [49]. In dimensions $D > 4$, the Einstein-Hilbert action is not the only possibility. The next-to-leading order term is the **Gauss-Bonnet** term, $\mathcal{G}$, defined as:

$$\mathcal{G} = R^2 - 4R_{\mu\nu}R^{\mu\nu} + R_{\mu\nu\rho\sigma}R^{\mu\nu\rho\sigma} \quad (46)$$

The Einstein-Gauss-Bonnet (EGB) action is given by:

$$S = \int d^D x \sqrt{-g} (R + \alpha_{GB} \mathcal{G})$$

where α_{GB} is the coupling constant. In exactly four dimensions, the Gauss-Bonnet term is a topological invariant (its integral gives the Euler characteristic of the manifold) and therefore does not contribute to the equations of motion. For this reason, it was long thought to be irrelevant for 4D gravitational dynamics.

3.3.1 The 4D Einstein-Gauss-Bonnet Proposal

This established view was challenged in 2020 by Glavan Lin, who proposed a novel method to elicit a non-trivial contribution from the Gauss-Bonnet term in four dimensions [50]. Their idea was to rescale the coupling constant as

$$\alpha_{GB} \rightarrow \frac{\alpha}{D-4}$$

and then take the limit $D \rightarrow 4$. The motivation was that the contribution of the Gauss-Bonnet term to the field equations is proportional to $(D-4)$, and this factor might be precisely cancelled by the singular rescaling.

While this procedure is not without mathematical subtleties [51, 52], it leads to a remarkably simple and interesting set of modified field equations:

$$G_{\mu\nu} + \alpha H_{\mu\nu} = 8\pi G T_{\mu\nu}$$

where the contribution from the Gauss-Bonnet term $\mathcal{G}$, $H_{\mu\nu}$, can be derived and is non-vanishing. This theory, dubbed 4DEGB gravity, produced intriguing new solutions for black holes and cosmology. For instance, the static, spherically symmetric vacuum solution is no longer the Schwarzschild metric (5), but is given by [50]:

$$ds^2 = -f(r)dt^2 + \frac{dr^2}{f(r)} + r^2 d\Omega^2 \quad \text{with} \quad f(r) = 1 + \frac{r^2}{2\alpha} \left(1 \pm \sqrt{1 + \frac{8GM\alpha}{r^3}} \right) \quad (47)$$

Similarly, the Friedmann equation for a flat FLRW (9) universe is modified to:

$$H^2 + \alpha H^4 = \frac{8\pi G}{3} \rho \quad (48)$$

These results sparked enormous interest, as they suggested a simple yet powerful modification of GR.

3.3.2 Regularization and the Scalar-Tensor Equivalence

Subsequent investigations revealed that the Glavan–Lin procedure is ill-defined at the level of the action and cannot be consistently formulated without introducing new degrees of freedom, thus respecting Lovelock’s theorem [51, 53]. However, the core idea can be made rigorous through several “regularization” schemes. These methods invariably show that a well-defined 4DEGB theory is equivalent to a specific scalar-tensor theory belonging to the Horndeski class.

One of the clearest methods is the **counter-term regularization** [54, 55]. One starts with the D-dimensional EGB action and adds a specific counter-term designed to cancel the divergences as $D \rightarrow 4$. The regularized action, in the limit, becomes:

$$S = \int d^4x \sqrt{-g} [R + \alpha (4G^{\mu\nu} \nabla_\mu \phi \nabla_\nu \phi - \phi \mathcal{G} + 4(g^{\rho\sigma} \nabla_\rho \nabla_\sigma \phi)(\nabla \phi)^2 + 2((\nabla \phi)^2)^2)] + S_m \quad (49)$$

where ϕ is a new scalar degree of freedom. This action is a member of the Horndeski class (52) with a specific set of functions [56]:

$$\begin{aligned} G_2(\phi, X) &= 8\alpha X^2 \\ G_3(\phi, X) &= 8\alpha X \\ G_4(\phi, X) &= 1 + 4\alpha X \\ G_5(\phi, X) &= 4\alpha \ln(X) \end{aligned}$$

where $X = -\frac{1}{2}(\nabla \phi)^2$. This demonstrates that the well-defined version of 4DEGB gravity is not a pure geometric modification but a particular scalar-tensor theory. The simple solutions like (47) and (48) are recovered in this theory for specific configurations of the scalar field (e.g., $\dot{\phi} = -H(t)$ in cosmology).

3.3.3 Phenomenology and Constraints

The regularized 4DEGB theory has a rich phenomenology that has been constrained by a variety of observations.

Black Holes The black hole solution (47) has two branches. The “GR branch” (minus sign) approaches the Schwarzschild solution (6) for large r and is asymptotically flat, while the “Gauss-Bonnet branch” (plus sign) is not and is typically disregarded. A key feature of the GR branch is the existence of a minimum mass for a given α . For a black hole to have a real-valued horizon radius, its mass must satisfy $M^2 \geq \alpha$. This “reality condition” provides a powerful constraint. For the black holes observed by LIGO/Virgo, their masses imply an upper bound on the coupling [57, 58]:

$$\sqrt{\alpha} 10^4 \text{ m} \implies \alpha 10^8 \text{ m}^2$$

Cosmology The modified Friedmann equation (48) implies that for $\alpha < 0$, there is a maximum energy density $\rho_{max} = 3/(32\pi G|\alpha|)$ that the universe can reach. This places a very strong bound from Big Bang Nucleosynthesis and the requirement that inflation could have occurred, effectively ruling out negative α for most practical purposes [57]. For $\alpha > 0$, the theory modifies the very early universe but quickly approaches standard GR as the energy density drops.

Weak-Field Tests In the post-Newtonian limit, the theory is nearly indistinguishable from GR. The PPN parameters γ and β are identical to their GR values ($\gamma = \beta = 1$). This means that classic tests like the deflection of light and the Shapiro time delay do not constrain α . However, the precession of periaapsis is modified [57]:

$$\delta\phi = \frac{6\pi GM}{a(1-e^2)} \left[1 \mp \frac{\alpha(4+e^2)}{a^2(1-e^2)^2} \right]$$

The tightest constraint from orbital dynamics comes from the LAGEOS satellites, which gives an upper bound of $\alpha 10^{10} \text{ m}^2$ [59].

Overall, the 4DEGB theory, when properly formulated as a scalar-tensor theory, is a viable and interesting modification of gravity. Current observational constraints, primarily from black hole mergers and satellite experiments, restrict its effects to be significant only in the very early universe or in the extreme gravity regime near black hole horizons.

3.4 Horndeski Theory: The Most General Second-Order Scalar-Tensor Gravity

While theories like Brans-Dicke and $f(R)$ gravity provide compelling modifications to General Relativity, a more systematic approach is to ask: what is the most general scalar-tensor theory of gravity that can be constructed without introducing dangerous instabilities? The primary guiding principle is the avoidance of the **Ostrogradsky instability**, a ghost-like instability that plagues theories with higher-order time derivatives in their equations of motion [60, 61]. To generalize GR while maintaining second-order dynamics (forced by the Lovelock's theorem [49]), one must therefore introduce new degrees of freedom, the simplest of which is a scalar field.

The search for the most general, healthy scalar-tensor theory leads to the framework first discovered by Gregory Horndeski in 1974 [62] and rediscovered much later in the context of Galileon theories.

3.4.1 The Galileon and its Covariantization

The modern path to the Horndeski theory begins with the concept of the **Galileon**, a scalar field ϕ defined on a flat Minkowski background that respects a "Galilean" shift symmetry [63]:

$$\phi \rightarrow \phi + c + b_\mu x^\mu$$

where c is a constant and b_μ is a constant vector. The remarkable property of the Galileon is that it is possible to construct a Lagrangian with higher derivatives of ϕ that still yields second-order equations of motion, thereby avoiding the Ostrogradsky ghost⁵. In four dimensions, the most general such Lagrangian is a sum of five terms:

$$\mathcal{L}_{\text{Gal}} = \sum_{i=1}^5 c_i \mathcal{L}_i \quad (50)$$

where, for instance, the cubic and quartic terms are given by:

$$\begin{aligned} \mathcal{L}_3 &= -X(\eta^{\mu\nu} \partial_\mu \partial_\nu \phi) \\ \mathcal{L}_4 &= X[(\eta^{\mu\nu} \partial_\mu \partial_\nu \phi)^2 - (\partial_\mu \partial_\nu \phi)(\partial^\mu \partial^\nu \phi)] \end{aligned}$$

with $X = -\frac{1}{2}\eta^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$.

The next step is to promote this theory to a curved spacetime by replacing the Minkowski metric $\eta_{\mu\nu}$ with a dynamical metric $g_{\mu\nu}$ and partial derivatives with covariant derivatives. This "covariantization" is non-trivial. A naive replacement leads to higher-order derivatives in the equations of motion because covariant derivatives do not commute (e.g., $[\nabla_\mu, \nabla_\nu]\phi_\alpha = R^\beta_{\alpha\mu\nu}\phi_\beta$). To cancel these unwanted terms and preserve the second-order nature of the dynamics, unique non-minimal coupling terms involving the curvature must be added. This procedure leads to the **covariant Galileon** [64]:

$$\mathcal{L}_{\text{covGal}} = c_2 X - c_3 X(g^{\mu\nu} \nabla_\mu \nabla_\nu \phi) + \frac{c_4}{2} X^2 R + c_4 X[(g^{\mu\nu} \nabla_\mu \nabla_\nu \phi)^2 - \phi_{\mu\nu} \phi^{\mu\nu}] + \dots \quad (51)$$

where $\phi_{\mu\nu} = \nabla_\mu \nabla_\nu \phi$. The term $\frac{c_4}{2} X^2 R$ is the unique "counter-term" that ensures the equations remain second-order.

3.4.2 The Generalized Galileon and the Horndeski Action

The covariant Galileon can be further extended by allowing the constant coefficients c_i to be arbitrary functions of ϕ and its canonical kinetic term $X = -\frac{1}{2}g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi$. This leads to the **generalized Galileon**, whose Lagrangian in four dimensions is given by [65]:

$$\begin{aligned} \mathcal{L} &= G_2(\phi, X) - G_3(\phi, X)(g^{\mu\nu} \nabla_\mu \nabla_\nu \phi) + G_4(\phi, X)R \\ &\quad + G_{4X}[(g^{\mu\nu} \nabla_\mu \nabla_\nu \phi)^2 - \phi_{\mu\nu} \phi^{\mu\nu}] \\ &\quad + G_5(\phi, X)G_{\mu\nu} \phi^{\mu\nu} \\ &\quad - \frac{G_{5X}}{6}[(g^{\mu\nu} \nabla_\mu \nabla_\nu \phi)^3 - 3(g^{\mu\nu} \nabla_\mu \nabla_\nu \phi)\phi_{\alpha\beta} \phi^{\alpha\beta} + 2\phi_{\mu\nu} \phi^{\nu\lambda} \phi^\mu_\lambda] \end{aligned} \quad (52)$$

where $G_{iX} = \partial G_i / \partial X$. This action is precisely the theory derived by Horndeski in 1974, and is recognized as the most general scalar-tensor theory in four dimensions that yields second-order equations of motion for both the metric and the scalar field [66].

⁵An Ostrogradsky ghost is a pathological, unphysical degree of freedom that arises in theories with higher-order time derivatives (e.g., terms depending on acceleration, $\ddot{q}$, or higher). Its presence implies that the energy of the system is not bounded from below. This would allow the vacuum to spontaneously decay into an infinite number of positive-energy particles and negative-energy ghosts while conserving total energy, leading to a catastrophic instability. Therefore, any viable physical theory must be free of such ghosts, which is the primary motivation for seeking theories with at most second-order equations of motion.

The Horndeski framework is remarkably comprehensive. It contains nearly all well-known scalar-tensor theories as special cases:

- **General Relativity with a scalar field:** $G_2 = X - V(\phi)$, $G_4 = M_{Pl}^2/2$, $G_3 = G_5 = 0$.
- **Brans-Dicke Theory:** $G_2 = -\frac{\omega}{\phi}(\partial\phi)^2$, $G_4 = \phi$.
- **f(R) Gravity:** Can be mapped to a scalar-tensor theory with $G_2 = \frac{1}{2}(Rf'(R) - f(R))$ and $G_4 = \frac{1}{2}f'(R)$, where $\phi = R$.
- **Gauss-Bonnet Gravity:** A non-minimal coupling to the Gauss-Bonnet term $\mathcal{G}$ (46) which is known to yield second-order equations, can also be expressed in the Horndeski form as we showed in 3.3.2

3.4.3 Cosmological Perturbations and the Speed of Gravitational Waves

One of the most powerful applications of the Horndeski framework is in cosmology, where it provides a unified way to study cosmological perturbations. The stability and propagation speed of these perturbations place strong constraints on the theory. Of particular importance are tensor perturbations, h_{ij} , which represent gravitational waves (cf. chapter 1.16).

By expanding the Horndeski action (52) to second order around a FLRW background (9), one can derive the quadratic action for tensor perturbations:

$$S_{\text{tensor}}^{(2)} = \frac{1}{8} \int dt d^3x a^3 \left[G_T \dot{h}_{ij}^2 - \frac{F_T}{a^2} (\partial_k h_{ij})^2 \right] \quad (53)$$

The coefficients G_T and F_T determine the kinetic and gradient energies of the gravitational waves, respectively. From this action, the propagation speed squared of gravitational waves, c_T^2 , is immediately identified as:

$$c_T^2 = \frac{F_T}{G_T}$$

For the theory to be free of ghost and gradient instabilities, one must require $G_T > 0$ and $F_T > 0$. In the full Horndeski theory, these coefficients are given by [66]:

$$G_T = 2 \left[G_4 - 2XG_{4X} - X(\dot{\phi}HG_{5X} - G_{5\phi}) \right] \quad (54)$$

$$F_T = 2 \left[G_4 - X(\ddot{\phi}G_{5X} + G_{5\phi}) \right] \quad (55)$$

In general, c_T can deviate from the speed of light, c . However, the landmark observation of the binary neutron star merger GW170817 and its electromagnetic counterpart GRB 170817A showed that the speed of gravitational waves is equal to the speed of light to extraordinary precision, $|c_T/c - 1|10^{-15}$ [67, 68].

This single measurement has profound consequences. For $c_T^2 = 1$ to hold robustly on cosmological scales (where $H \neq 0$ and $\ddot{\phi} \neq 0$), the coefficients must satisfy $G_T = F_T$. This forces the following conditions on the Horndeski functions [69, 70]:

$$G_5 = \text{constant} \quad \text{and} \quad G_{4X} = 0 \implies G_4 = G_4(\phi)$$

Setting the constant for G_5 to zero to avoid other issues, the surviving part of the Horndeski Lagrangian, which can act as a model for dark energy today, is significantly simplified:

$$\mathcal{L}_{\text{viable}} = G_2(\phi, X) - G_3(\phi, X)(g^{\mu\nu}\nabla_\mu\nabla_\nu\phi) + G_4(\phi)R \quad (56)$$

This result has ruled out a large class of previously considered modified gravity models, while leaving a substantial and interesting parameter space still open for exploration.

3.4.4 The Vainshtein Screening Mechanism

For any scalar-tensor theory to be viable, it must suppress the scalar-mediated fifth force in high-density regions like the Solar System, where General Relativity has been tested to high precision. While f(R) gravity often relies on the chameleon mechanism, Galileon and Horndeski theories typically employ the **Vainshtein mechanism** [71].

The core idea of Vainshtein screening is that non-linearities in the scalar field's derivative interactions, rather than its potential, become dominant near a massive source. To illustrate, consider a simplified model with a cubic Galileon interaction (in a flat background):

$$\mathcal{L} = -\frac{1}{2}(\partial\phi)^2 - \frac{1}{\Lambda^3}(\partial\phi)^2(\eta^{\mu\nu}\partial_\mu\partial_\nu\phi) - \frac{\alpha}{M_{Pl}}\phi T$$

Here, Λ is an energy scale and the last term represents the coupling of the scalar to the trace of the matter energy-momentum tensor T . The equation of motion for ϕ is:

$$\eta^{\mu\nu}\partial_\mu\partial_\nu\phi + \frac{2}{\Lambda^3} [(\eta^{\alpha\beta}\partial_\alpha\partial_\beta\phi)^2 - (\partial_\mu\partial_\nu\phi)^2] = \frac{\alpha}{M_{Pl}}T$$

For a static, spherically symmetric source of mass M , this equation becomes an algebraic equation for $\phi'(r)$. Far from the source, the linear term dominates and the scalar force is unscreened. However, close to the source, the non-linear term $\sim (\phi')^2/r$ dominates. This leads to a suppression of the scalar field gradient compared to the Newtonian potential. The transition occurs at the **Vainshtein radius**, r_V , defined as:

$$r_V = \left(\frac{\alpha M}{4\pi M_{Pl}\Lambda^3} \right)^{1/3}$$

For $r \ll r_V$, the scalar fifth force is suppressed by a factor of $(r/r_V)^3$, effectively hiding the modification and recovering the predictions of General Relativity. For typical dark energy models, this radius is large enough to encompass the entire Solar System, thus satisfying experimental constraints [72].

4 Observing the Universe

4.1 Supernova Measurements

4.1.1 Supernovae Apparent Magnitude and Cosmological Observations (SNIa)

Recent astronomical surveys of distant supernovae have provided compelling evidence that the expansion rate of the universe is increasing. These observations are largely based on measurements of the distance modulus for a particular class of supernovae—Type Ia. These events, often referred to as SNIa, act as remarkably consistent “standard candles” due to their uniform peak luminosity, making them tools for measuring cosmological distances that will allow us to study the influence of dark energy.



Figure 16: This image shows Cassiopeia A (Cas A), the remnant of a supernova located approximately 11,000 light-years from Earth. Captured by NASA’s *Chandra X-ray Observatory*, the image reveals an expanding shell of stellar debris. X-ray emission, shown in blue, maps the hot plasma from the explosion, while infrared data from the *James Webb Space Telescope* (in orange and white) highlight cooler dust structures. Adapted from: NASA.gov

4.1.2 Nature of Supernovae

A supernova is an immensely powerful stellar explosion that emits intense electromagnetic radiation across a broad spectrum over a short timescale. This phenomenon is capable of outshining an entire galaxy for weeks due to the sheer amount of energy released, which can reach levels on the order of 10^{44} J—comparable to the total energy our Sun will emit over its entire lifetime.

There are two principal scenarios under which a supernova can occur:

- **Gravitational Core Collapse:** This mechanism is triggered when the core of a massive star—exceeding the Chandrasekhar mass limit of approximately $1.4M_{\odot}$ —collapses under its own gravity. The implosion results in a shock wave that expels the outer layers of the star, producing a Type II, Ib, or Ic supernova.
- **Thermonuclear Runaway in Compact Stars:** In this case, the progenitor is a compact object such as a white dwarf. When sufficient matter is accreted from a companion or merged from another star, the core temperature rises. Upon reaching critical conditions, runaway nuclear fusion ignites—particularly of carbon—leading to a cataclysmic explosion. This pathway typically results in a Type Ia supernova.

4.1.3 Classification of Supernovae

Supernovae can be divided based on their observed spectral features and the shape of their light curves. A key distinction is made depending on the presence of hydrogen lines in the spectrum. If hydrogen is present, the event is classified as a Type II supernova. In the absence of hydrogen lines, the event falls into the Type I category.

Further subdivisions exist within Type I supernovae:

- **Type Ia:** Identified by the presence of singly ionized silicon (Si II) near 615 nm.
- **Type Ib:** Lacks silicon but shows prominent helium (He I) lines around 587.6 nm.
- **Type Ic:** Lacks both silicon and helium lines.

Type II supernovae may also be further subclassified based on the morphology of their light curves, though such refinements lie beyond the scope of this discussion. Notably, only Type Ia supernovae originate from a fundamentally different progenitor system than the rest, which typically result from the gravitational collapse of massive stars.

4.1.4 Type Ia Supernovae: Origin and Mechanism

Progenitor System Type Ia supernovae result from the thermonuclear disruption of a white dwarf in a binary star system. The white dwarf, initially stable and below the Chandrasekhar mass limit (~ 1.4 solar masses), begins to accrete matter from its companion, which is often a red giant.

Instability and Explosion As mass transfer continues, the white dwarf grows more massive. Once it surpasses the Chandrasekhar threshold, electron degeneracy pressure is no longer sufficient to counteract gravitational collapse. The core becomes denser and hotter, ultimately reaching the conditions necessary for runaway carbon fusion. This ignition results in a catastrophic thermonuclear explosion that disrupts the star entirely, leaving no remnant.

The resulting event emits a characteristic light curve and spectrum, which serve as key tools in cosmology, particularly in measuring extragalactic distances. An example of such a light curve is depicted in Figure(17), showcasing the typical luminosity evolution over time.

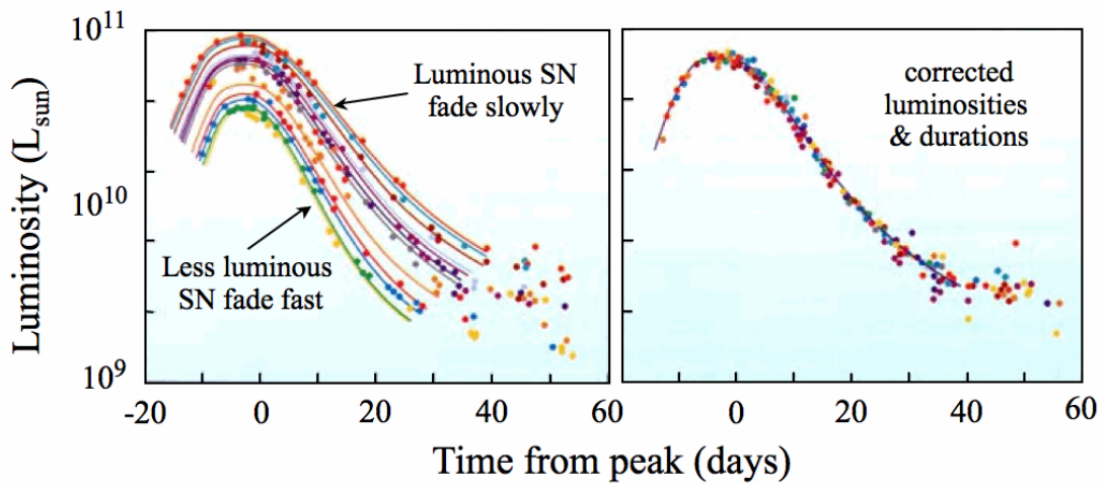


Figure 17: Left: Observed light curves of Type Ia supernovae show a range of luminosities and decline rates. More luminous SNe Ia fade slowly, while less luminous ones fade faster. Right: After applying a correction known as the “stretch factor,” which adjusts both the duration and brightness, the light curves align and reveal a standard profile. Adapted from: Durham University – SN2015M Bridge Project.

4.1.5 Light Curve Properties under Standard Gravity

Under the assumption of standard General Relativity, where the gravitational constant G is a true constant, the physics governing Type Ia supernova light curves provides a remarkably consistent picture that allows them to be used as standardizable candles. The shape and peak luminosity of the light curve are primarily determined by three key physical parameters: the mass of synthesized radioactive nickel-56 ($M_{56\text{Ni}}$), the total mass of the supernova ejecta (M_{ej}), and the opacity of the ejecta (κ) [73, 74].

The Role of Radioactive Decay The dominant power source for the light curve is the radioactive decay chain of ^{56}Ni to ^{56}Co and then to stable ^{56}Fe [75, 76, 77]. The energy released by these decays is deposited into the expanding ejecta, thermalized, and then radiated away. The instantaneous luminosity is therefore directly tied to the amount of ^{56}Ni produced in the thermonuclear explosion. A seminal result in this area is **Arnett’s Rule** [78, 79], which states that, to a good approximation, the peak bolometric luminosity of the supernova is equal to the instantaneous rate of energy generation from radioactive decay at the time of the peak:

$$L_{\text{peak}} \approx \dot{E}_{\text{rad}}(t_{\text{peak}})$$

where $\dot{E}_{\text{rad}}$ is the energy deposition rate from the decay chain. This rule provides a direct physical link between the observable peak brightness and the synthesized nickel mass, $M_{56\text{Ni}}$ [80, 81]. More detailed radiative transfer models have shown that this rule is a reasonable approximation, though it can deviate depending on the distribution of nickel within the ejecta [79].

Light Curve Shape and the Diffusion Timescale The width of the light curve—how quickly it rises and falls—is governed by the timescale over which the trapped radiation can diffuse out of the expanding ejecta. This diffusion timescale, τ_d , depends on the ejecta mass and opacity. A simple analytical model yields the following dependencies [73, 74]:

$$\tau_d \propto \sqrt{\frac{\kappa M_{\text{ej}}}{v_{\text{exp}}}}$$

where v_{exp} is the expansion velocity of the ejecta.

- **Higher Ejecta Mass (M_{ej}):** A more massive ejecta is more opaque and traps photons for longer. This leads to a broader light curve (slower decline) and, for a fixed nickel mass, a lower peak luminosity because the energy is released over a longer period [73].
- **Higher Opacity (κ):** Similarly, higher opacity increases the diffusion time, resulting in a wider and dimmer light curve.

This interplay between nickel mass, ejecta mass, and opacity gives rise to the observed empirical correlation known as the **Phillips relation**: brighter supernovae have broader light curves (they decline more slowly) [82]. This relation is what allows SNIa to be “standardized.” By measuring the shape of the light curve (the “stretch factor”), one can correct for the intrinsic variations in luminosity and derive a standardized distance.

The Four Phases of Radiation Transport The evolution of the light curve can be understood through four distinct phases of radiation transport, as detailed in models like those described by Kasen and Woosley [83] and simulated with Monte Carlo codes like ARTIS [84, 85, 86]:

- **Phase I (Initial Expansion):** Immediately after the explosion, the ejecta is extremely dense and optically thick. The energy from the explosion is trapped, and the supernova is faint.
- **Phase II (Rise to Peak):** As the ejecta expands and cools, it becomes more transparent. Photons from the thermalized decay energy begin to diffuse out. The observed luminosity rises sharply as this diffusion wave reaches the outer layers. The peak of the light curve occurs when the diffusion timescale becomes comparable to the expansion timescale.
- **Phase III (Post-Peak Decline):** After the peak, the ejecta continues to expand and thin out. The rate of energy escape starts to exceed the rate of energy deposition from the ongoing decay of ^{56}Co . The luminosity begins its decline, with the slope determined by the diffusion properties.
- **Phase IV (Nebular Phase):** Months after the explosion, the ejecta becomes largely transparent to gamma-rays. The light curve enters a late-time “tail” where its decline rate is primarily governed by the 77-day half-life of ^{56}Co decay, as the energy is deposited and radiated away almost instantaneously.

Within the standard Λ CDM model 1.12.4, these physical processes are assumed to be universal. The underlying physics, rooted in constant G and standard nuclear physics, is expected to be the same for a supernova at redshift $z = 0.1$ and one at $z = 1$. Any observed differences in apparent brightness are therefore attributed solely to cosmological effects like distance and redshift, allowing for the measurement of cosmological parameters [87].

4.1.6 Light Curve Properties under Modified Gravity

In theories of modified gravity, such as those discussed in Chapter 2, the fundamental assumption that the laws of physics governing a supernova explosion are constant throughout cosmic history is challenged. These frameworks can introduce a time- or environment-dependent effective gravitational constant, G_{eff} , or new scalar fields that couple to matter. Such modifications can systematically alter the intrinsic properties of Type Ia supernovae, potentially biasing cosmological measurements or, conversely, providing a new observational window to test gravity itself [73, 74].

The Chandrasekhar Mass and Varying G_{eff} The cornerstone of the Type Ia supernova mechanism is the explosion of a white dwarf as it approaches the Chandrasekhar mass, M_{Ch} . As established by foundational work in stellar structure, this critical mass is not a universal constant but depends on the strength of gravity. In a theory with a varying gravitational coupling G_{eff} , the Chandrasekhar mass scales as [88, 89]:

$$M_{\text{Ch}} = C \left(\frac{\hbar c}{G_{\text{eff}}} \right)^{3/2} \frac{1}{m_p^2} \propto G_{\text{eff}}^{-3/2}$$

where C is a dimensionless constant and m_p is the proton mass. Since the peak luminosity (L_{peak}) is directly proportional to the amount of synthesized ^{56}Ni , which in turn is proportional to the exploding mass M_{Ch} , any evolution in G_{eff} induces a redshift-dependent intrinsic luminosity [73, 74]:

$$L_{\text{peak}}(z) \propto M_{^{56}\text{Ni}}(z) \propto M_{\text{Ch}}(z) \propto G_{\text{eff}}(z)^{-3/2} \quad (57)$$

This implies that if G_{eff} was different in the past, the intrinsic brightness of high-redshift supernovae would be systematically different from their low-redshift counterparts.

Constraints on Brans-Dicke Theory The **Brans-Dicke theory** 3.1 provides a clear example of this effect. In this theory, $G_{\text{eff}} \propto 1/\phi$, where ϕ is the dynamical scalar field. The evolution of ϕ is sourced by the trace of the universe's energy-momentum tensor, causing G_{eff} to vary over cosmic time. While the exponent in Eq. (57) is fixed at $-3/2$ by stellar physics, the theory itself is constrained by the PPN parameter $\gamma = (1 + \omega)/(2 + \omega)$. A value of γ close to 1, as required by Solar System tests, implies a very large Brans-Dicke parameter ω . This, in turn, forces the scalar field ϕ to be very stable, leading to a nearly constant G_{eff} . Therefore, precise measurements of supernova luminosities over a range of redshifts can provide constraints on the evolution of G_{eff} and thus on the parameter ω , complementing the static, weak-field tests from the Solar System [90, 91].

f(R) Gravity and the Chameleon Screening For **f(R) theories of gravity** 3.2 the situation is more subtle. As these theories are equivalent to a scalar-tensor theory, one would expect a similar evolution of G_{eff} . However, viable $f(R)$ models must employ a screening mechanism like the **chameleon mechanism** 3.2.3 to satisfy Solar System constraints. This same mechanism can operate in the progenitor systems of supernovae. If the white dwarf resides in a sufficiently dense environment (e.g., within a host galaxy), the scalar degree of freedom (the "scalaron") may acquire a large effective mass, causing it to decouple from matter. This "screening" would force G_{eff} to revert to the standard Newtonian value G_N within the system, regardless of the cosmological value of the scalar field [92]. Consequently, the Chandrasekhar mass would remain constant, and SNIa would not show any intrinsic evolution. The absence of such an evolutionary effect can thus be used to constrain the parameters of the chameleon mechanism itself [93].

General Frameworks: Horndeski and 4DEGB Gravity In more general frameworks like **Horndeski theory** 3.4, modifications to gravity affect supernova observations in two ways. First, through the evolution of G_{eff} as described above. Second, by directly altering the cosmological expansion history, which changes the luminosity distance-redshift relation, $d_L(z)$ 21. The landmark detection of GW170817 severely constrained the Horndeski parameter space, but the surviving theories still permit modifications to the cosmic background. Similarly, the **4D Einstein-Gauss-Bonnet (4DEGB) gravity** 3.3, when regularized as a scalar-tensor theory, modifies the Friedmann equation (48). Cosmological analyses primarily use SNIa data to constrain the theory's coupling constant α by fitting the predicted $d_L(z)$ to the observed Hubble diagram [94, 95].

The Degeneracy Problem A critical challenge in this field is the degeneracy between an evolving intrinsic luminosity $L_{\text{peak}}(z)$ and the effects of a dark energy component. A systematic change in supernova brightness due to a varying G_{eff} can precisely mimic the Hubble diagram of a standard GR universe with a different cosmic composition [89, 96].

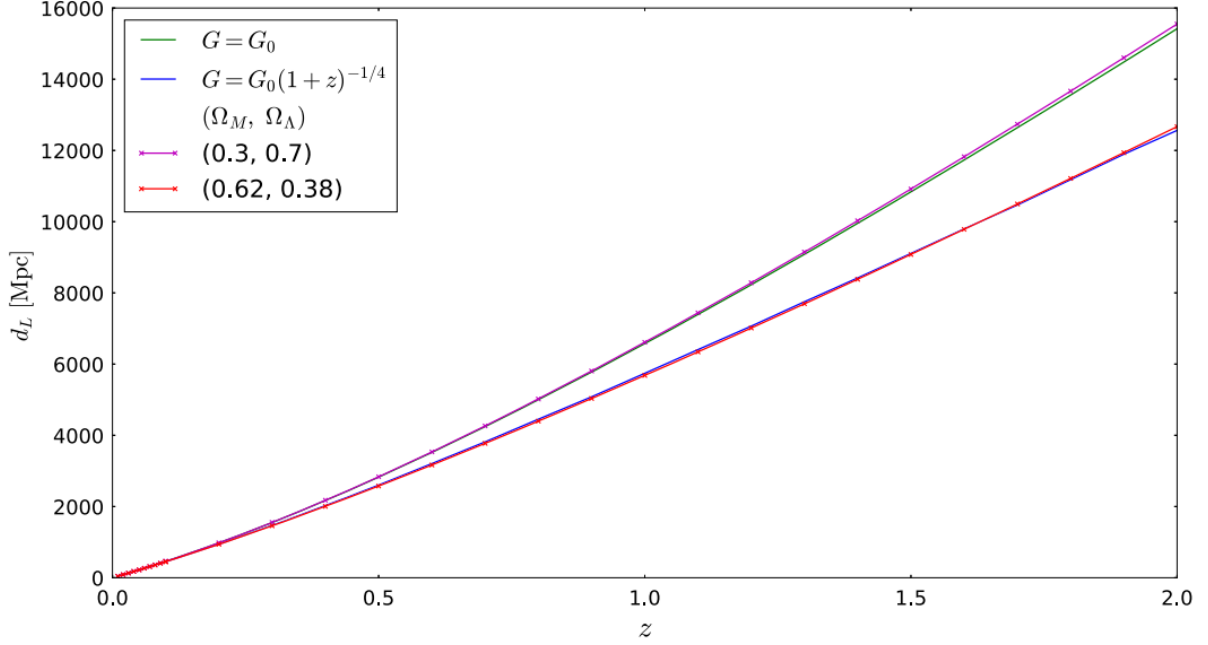


Figure 18: **A direct visualization of the degeneracy between a standard dark energy model and a modified gravity model.** The plot shows the luminosity distance (d_L) versus redshift (z). The **purple line** represents the standard Λ CDM model, which assumes a constant gravitational constant ($G = G_0$) and fits the supernova data with cosmological parameters $(\Omega_M, \Omega_\Lambda) = (0.3, 0.7)$. The **red line** represents a model where gravity evolves as $G(z) = G_0(1+z)^{-1/4}$. When the supernova data is re-interpreted with this assumption, it yields a different best-fit cosmology of $(\Omega_M, \Omega_\Lambda) = (0.62, 0.38)$. The critical feature is the **near-perfect overlap** of these two lines. This demonstrates that both physically distinct models produce identical observational predictions for the supernova Hubble diagram, making them indistinguishable with this probe alone. *Figure adapted from [96].*

A powerful numerical example from [96] illustrates this ambiguity perfectly (cf. Figure 18). The process begins by postulating a toy model for a time-varying gravitational, note that this is a toy model intended to describe the late-time variation of G and we do not assume that it works for $z > 2 - 3$, such as:

$$G(z) = G_0(1+z)^{-1/4} \quad (58)$$

This evolution of G directly impacts the Chandrasekhar mass and, consequently, the peak supernova luminosity, as described by Eq. (57). By using the Eq. (20) for the flux and the Eq. (22) for the apparent magnitude, we find the relationship between the intrinsic luminosity L and the absolute magnitude M to be given by:

$$M = M_\odot - 2.5 \log_{10} \left(\frac{L}{L_\odot} \right) \quad (59)$$

where $M_\odot$ and $L_\odot$ are reference values. A varying $G(z)$ thus implies that both the intrinsic luminosity L and the absolute magnitude M become redshift-dependent.

The numerical experiment described in the paper proceeds as follows:

1. A mock supernova dataset is first generated within a standard Λ CDM framework, assuming a constant $G = G_0$ and cosmological parameters $(\Omega_{M,0}, \Omega_{\Lambda,0}) = (0.3, 0.7)$. The theoretical distance for this model is calculated using the standard formula for luminosity distance (cf. Derivation of the Luminosity Distance in a Flat Λ CDM Universe in the Appendix) in a flat, is given by:

$$d_L(z; \Omega_M, \Omega_\Lambda, H_0) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_{M,0}(1+z')^3 + \Omega_{\Lambda,0}}} \quad (60)$$

2. This mock dataset of apparent magnitudes is then reinterpreted under the new assumption that gravity evolves according to Eq. (58). Because supernovae at higher redshift would be intrinsically fainter in this model (since G was weaker), the same observed apparent magnitudes now imply smaller luminosity distances when inferred using the distance modulus (23).

3. Finally, this reinterpreted distance-redshift data is fit again using the standard cosmological model (Eq. (60)) to find the cosmological parameters that best describe it. The analysis yields a new best-fitting cosmology of $(\Omega_{M,0}, \Omega_{\Lambda,0}) = (0.62, 0.38)$.

The remarkable result is that this new, matter-dominated model produces a Hubble diagram that is almost identical to the original dark-energy-dominated model. This demonstrates that supernova data alone cannot distinguish between cosmic acceleration caused by a dark energy component and an apparent acceleration caused by an evolution in fundamental constants. Breaking this degeneracy requires combining supernova data with independent cosmological probes, especially those sensitive to the growth of large-scale structure (e.g., galaxy clustering and weak lensing), which is affected differently in modified gravity than in GR with a dark energy fluid [97, 98].

4.2 CMB Measurement as a Cosmological Probe

The Cosmic Microwave Background (CMB) provides a snapshot of the universe at the epoch of photon decoupling, $z_* \approx 1090$. The anisotropies observed in the CMB temperature map are the direct descendants of primordial quantum fluctuations, processed by the physics of the early universe. Analyzing the statistical properties of these anisotropies allows for the determination of fundamental cosmological parameters with unprecedented precision.

4.2.1 The Acoustic Scale: A Standard Ruler in the Early Universe

The physics of the primordial photon-baryon fluid imprints a characteristic physical scale onto the CMB: the **sound horizon at recombination**, $r_s(z_*)$. This is the maximum comoving distance that a sound wave could have propagated in the plasma from the Big Bang until the moment photons decoupled. This scale acts as a standard ruler. We observe this ruler on the sky under a characteristic angle, θ_A , which corresponds to the location of the first peak in the CMB power spectrum. This angle is given by the simple geometric relation:

$$\theta_A = \frac{r_s(z_*)}{D_M(z_*)}$$

where $D_M(z_*)$ is the comoving angular diameter distance to the surface of last scattering. The corresponding multipole moment, l_A , which is a more direct observable in the power spectrum, is $l_A = \pi/\theta_A$.

To use this ruler, we must be able to theoretically predict its physical size and the distance to it.

The Sound Horizon $r_s(z_*)$ The comoving sound horizon is calculated by integrating the sound speed, $c_s(z)$, over the expansion history of the universe up to the redshift of decoupling, z_* :

$$r_s(z_*) = \int_0^{\tau_*} c_s(\tau) d\tau = \int_{z_*}^{\infty} \frac{c_s(z)}{H(z)} dz$$

The sound speed depends on the ratio of the baryon density to the photon density. Therefore, the size of the sound horizon is primarily sensitive to the physical baryon density, $\omega_b \equiv \Omega_b h^2$, and the physical matter density, $\omega_m \equiv \Omega_m h^2$.

The Angular Diameter Distance $D_M(z_*)$ This is the comoving distance a photon travels from the last scattering surface to us today. In a flat universe, it is given by:

$$D_M(z_*) = \int_0^{z_*} \frac{c}{H(z')} dz'$$

This distance depends on the expansion history of the universe *after* recombination, and is therefore sensitive to the total matter density Ω_m , the dark energy density Ω_Λ , and the spatial curvature Ω_k .

The Redshift of Decoupling z_* The precise redshift of decoupling is a complex function of recombination physics. A highly accurate fitting formula, derived by Hu and Sugiyama, is often used [99]:

$$z_* = 1048 \left[1 + 0.00124(\Omega_b h^2)^{-0.738} \right] \left[1 + g_1(\Omega_m h^2)^{g_2} \right]$$

where g_1 and g_2 are functions of the baryon and matter densities themselves. This shows that all fundamental quantities depend on the underlying cosmological parameters.

4.2.2 The Angular Power Spectrum from the Boltzmann Hierarchy

To move beyond the simple standard ruler picture and utilize the full statistical information in the CMB map, we must model the evolution of the temperature fluctuations, $\Theta(\vec{x}, \hat{n}, \tau) \equiv \delta T/T$. This is achieved with the **linearized Boltzmann equation** (33) which is intractable due to its angular dependence. The standard technique is to expand the temperature fluctuation Θ in a basis of Legendre polynomials, $P_l(\mu)$, which are appropriate for the azimuthal symmetry of the problem:

$$\Theta(\vec{k}, \mu, \tau) = \sum_{l=0}^{\infty} (-i)^l (2l+1) \Theta_l(\vec{k}, \tau) P_l(\mu) \quad (61)$$

Substituting this expansion (61) into Equation (33) transforms the single partial differential equation into an infinite hierarchy of coupled ordinary differential equations for the multipole moments Θ_l . The first two equations in this hierarchy are:

$$\dot{\Theta}_0 + k\Theta_1 = -\dot{\Phi} \quad (62)$$

$$\dot{\Theta}_1 - \frac{k}{3}\Theta_0 + \frac{2k}{3}\Theta_2 = -\frac{k}{3}\Psi - \dot{\tau}_c\Theta_1 \quad (63)$$

where $\dot{\tau}_c$ is the differential optical depth from the collision term. The solution to this full hierarchy, coupled with the perturbed Einstein equations, yields the theoretical values for $\Theta_l(\tau_0)$ today.

The primary observable is the **angular power spectrum**, C_l . It is defined as the variance of the coefficients, a_{lm} , of a spherical harmonic expansion of the temperature map, $T(\hat{n}) = \sum_{lm} a_{lm} Y_{lm}(\hat{n})$. The C_l is related to the theoretical multipole moments by the **Sachs-Wolfe formula** [100]:

$$C_l = 4\pi \int \frac{d^3k}{(2\pi)^3} P(k) |\Theta_l(k, \tau_0) + \dots|^2$$

where $P(k)$ is the primordial power spectrum of fluctuations from inflation. Computer codes like CAMB and CLASS solve this system of equations to compute the theoretical C_l spectrum for any given set of cosmological parameters.

4.2.3 Parameter Estimation and Comparison with Data

The process of measuring cosmological parameters involves finding the theoretical model that best fits the observed data. The Planck satellite has measured the CMB power spectrum with unprecedented accuracy, as shown by the red data points in Figure 19.

The comparison is done via a statistical **likelihood analysis**. Assuming the primordial fluctuations are Gaussian, the likelihood of observing a particular set of data given a theoretical model with parameters $\{p_\alpha\}$ is:

$$\mathcal{L}(\text{data}|\{p_\alpha\}) \propto \exp \left(-\frac{1}{2} \sum_{l,m} \frac{|a_{lm}^{\text{obs}}|^2}{C_l^{\text{th}}(\{p_\alpha\})} \right)$$

By maximizing this likelihood (or, more commonly, minimizing $-\ln \mathcal{L}$) over the parameter space, one finds the best-fit cosmological model.

A compressed parameter often used to capture the geometric information in the CMB is the **shift parameter**, R , which is constructed to be independent of H_0 :

$$R \equiv \sqrt{\Omega_m H_0^2} D_M(z_*) = \sqrt{\omega_m} \int_0^{z_*} \frac{dz}{H(z)}$$

This parameter effectively measures the angular position of the acoustic peaks and is a powerful tool for comparing the CMB with other geometric probes like BAO from galaxy surveys.

4.2.4 Results from CMB Measurements

The analysis of CMB power spectra, particularly from the Planck mission, has led to the establishment of the standard model of cosmology, the **six-parameter flat Λ CDM model**.

- **Flat Geometry:** The position of the first acoustic peak at $l \approx 220$ provides the strongest evidence that the universe is spatially flat to a high degree of precision, with $\Omega_k = 0.001 \pm 0.002$.
- **Cosmic Composition:** The morphology of the acoustic peaks provides robust measurements of the physical densities of baryons and cold dark matter:

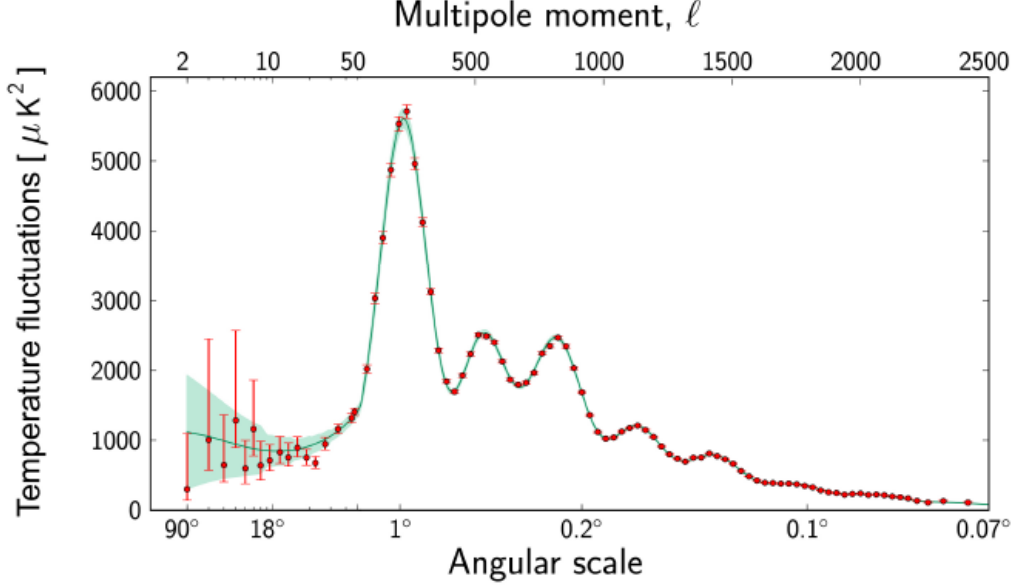


Figure 19: The CMB angular power spectrum as measured by the Planck satellite (red points with error bars), compared with the best-fit six-parameter Λ CDM theoretical model (green curve). The remarkable agreement is a cornerstone of modern cosmology.

- Baryon Density: $\Omega_b h^2 = 0.0224 \pm 0.0001$
- Cold Dark Matter Density: $\Omega_c h^2 = 0.120 \pm 0.001$
- **Primordial Spectrum:** The shape of the spectrum, particularly the Sachs-Wolfe plateau and the damping tail, constrains the properties of the primordial fluctuations, confirming a nearly scale-invariant spectrum ($n_s \approx 0.965$) as predicted by simple models of inflation.
- **Confirmation of Dark Energy:** Combining the CMB measurement of a flat universe ($\Omega_m + \Omega_\Lambda = 1$) with the measured matter density of $\Omega_m \approx 0.315$ provides powerful, independent confirmation for a universe dominated by dark energy, with $\Omega_\Lambda \approx 0.685$.

4.3 Baryon Acoustic Oscillations (BAO)

Baryon Acoustic Oscillations (BAO) provide a "standard ruler" to map the expansion history of the Universe with high precision, representing one of the most robust probes in modern cosmology. This feature, imprinted on the large-scale distribution of matter, is a direct relic of sound waves that propagated through the hot, dense plasma of the early Universe before recombination.

4.3.1 Physical Origin of the BAO Standard Ruler

In the early Universe (prior to $z \approx 1100$), photons and baryons were tightly coupled via Compton scattering, forming a single photon-baryon fluid. Primordial density fluctuations, seeded during inflation, acted as gravitational potential wells. While the collisionless dark matter began to collapse into these wells, the photon-baryon fluid experienced both the inward pull of gravity and a strong, counteracting outward radiation pressure. This interplay launched spherical acoustic waves that propagated outward from every initial overdensity.

These waves traveled until the Universe cooled enough for protons and electrons to combine into neutral hydrogen, an event known as recombination. At this point, photons decoupled from baryons and began to stream freely, forming the Cosmic Microwave Background (CMB) we observe today. The abrupt drop in pressure stalled the acoustic waves, effectively "freezing" the baryons in a spherical shell at a characteristic comoving radius from the initial perturbation, as simulated in Fig.20.

This radius is the **sound horizon** at the **drag epoch**, r_d , which is the maximum comoving distance a sound wave could travel before the baryons were released from the photon drag. Its size is determined by integrating the sound speed, $c_s(z)$, over the expansion history up to the drag epoch, z_d :

$$r_d = \int_{z_d}^{\infty} \frac{c_s(z)}{H(z)} dz \quad (64)$$

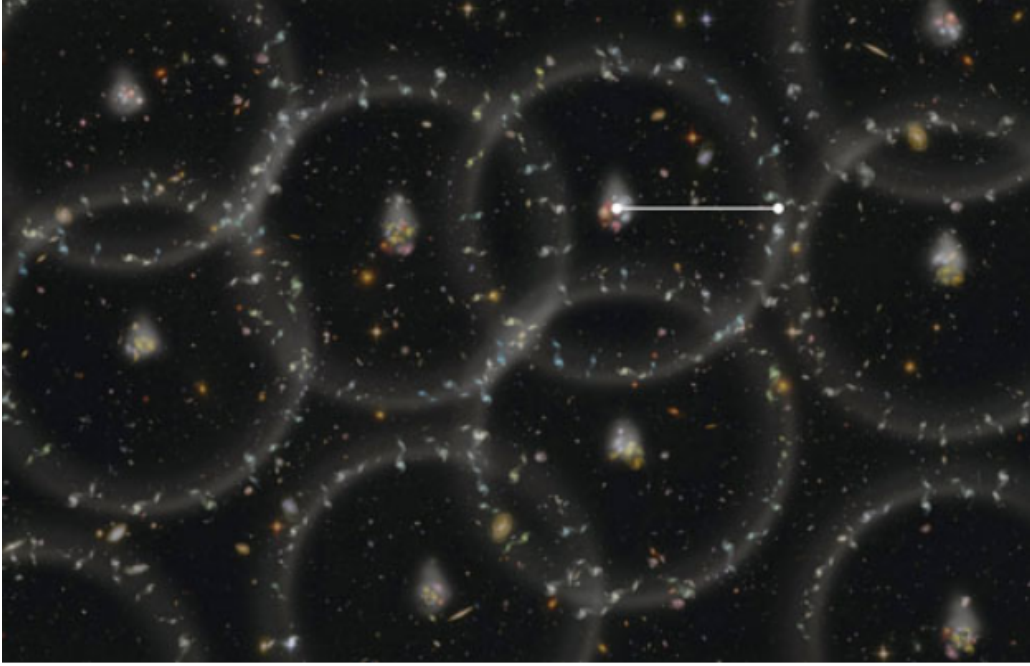


Figure 20: **Visualization of the BAO Imprint on Large-Scale Structure.** This illustration, produced by the Baryon Oscillation Spectroscopic Survey (BOSS) team, provides a visual metaphor for the BAO phenomenon. The central bright spots represent the initial dark matter potential wells where galaxies begin to form. The surrounding faint rings, at a fixed radius, represent the spherical shells where the baryon density was enhanced by the stalled acoustic waves at the drag epoch. Because galaxies preferentially form in both the central clumps and these overdense shells, this creates a characteristic preferred separation scale in the Universe, which is detected as the BAO peak in the galaxy correlation function. Adapted from [101].

where the sound speed depends on the baryon-to-photon ratio, $c_s = c/\sqrt{3(1 + 3\rho_b/4\rho_\gamma)}$. Because the physical densities that determine r_d are precisely measured by the CMB, the sound horizon provides a robustly calibrated standard ruler of size $r_d \approx 147$ Mpc.

After decoupling, both the central dark matter overdensity and the baryonic shell at radius r_d acted as seeds for gravitational collapse, leading to enhanced galaxy formation at these locations. Consequently, there is a small but statistically significant excess probability of finding pairs of galaxies separated by the comoving distance r_d . This signature appears as a peak in the two-point galaxy correlation function and as oscillations in the matter power spectrum, as illustrated in Fig. 21.

4.3.2 BAO as a Cosmological Probe

The measurement of the BAO standard ruler manifests differently in the directions parallel (radial) and perpendicular (transverse) to the observer's line of sight. The observed angular size, $\Delta\theta$, and redshift separation, Δz , of the sound horizon scale are given by :

$$\Delta\theta = \frac{r_d}{D_A(z)} = \frac{r_d(1+z)}{D_M(z)} \quad (65)$$

$$\Delta z = \frac{H(z)r_d}{c} \quad (66)$$

where $D_A(z)$ is the proper angular diameter distance and $D_M(z)$ is the comoving angular diameter distance. These relations allow BAO measurements to constrain two key cosmological distances:

- The **Hubble distance**, $D_H(z) \equiv c/H(z)$, constrained by the radial measurement.
- The **comoving angular diameter distance**, $D_M(z)$, constrained by the transverse measurement.

When analyzing the spherically-averaged clustering signal, these are often combined into the volume-averaged distance, $D_V(z) \equiv [D_M^2(z)D_H(z)]^{1/3}$ [102].

Since the physical size of r_d is calibrated by the CMB, measuring its apparent size at different redshifts allows for a precise mapping of the cosmic expansion history. This geometric test is a powerful tool for constraining the properties of dark energy, as it is sensitive to the evolution of $H(z)$ over cosmic time [103].

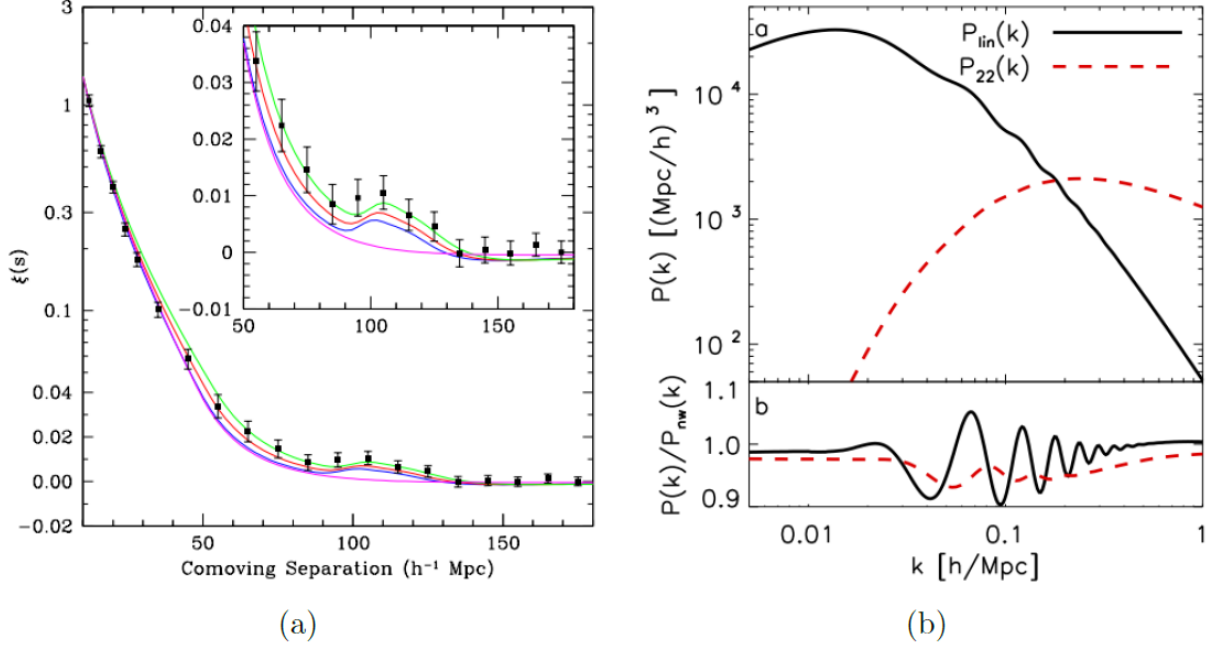


Figure 21: **The Baryon Acoustic Oscillation Feature in Real and Fourier Space.** **Left Panel (a):** The BAO peak detected in the two-point correlation function, $\xi(s)$, of the SDSS Luminous Red Galaxy sample. The data points clearly show an excess over the smooth model at a comoving separation of $\approx 100 h^{-1} \text{Mpc}$ ($\approx 150 \text{Mpc}$), confirming the existence of the standard ruler. **Right Panel (b):** The corresponding feature in the matter power spectrum, $P(k)$. The bottom sub-panel shows the ratio of the full power spectrum to a smooth, “no-wiggles” reference spectrum, isolating the characteristic baryonic oscillations. The peak in configuration space is the Fourier transform of these oscillations in k -space. Adapted from:[102]

4.3.3 The DESI 2024 Results and Dynamical Dark Energy

The Dark Energy Spectroscopic Instrument (DESI) has provided the most precise BAO measurements to date. The first year of data (DR1) has been used in a “Full-Shape” analysis, which utilizes the entire broadband shape of the power spectrum beyond just the BAO peaks, including information from Redshift-Space Distortions (RSD) [104]. This analysis, when combined with CMB and supernova data, yielded a hint for a dynamical dark energy component, showing a $\approx 4\sigma$ preference for a dark energy equation of state that evolves with time over the simple cosmological constant (ΛCDM) model [105].

Subsequent analyses have sought to understand the origin of this signal. It has been shown that the preference for a “phantom” dark energy equation of state ($w < -1$) is likely an artifact of the simple CPL parametrization, $w(a) = w_0 + (1 - a)w_a$, used in the main analysis [106, 107]. More physically motivated quintessence models can fit the data equally well without requiring phantom behavior.

Crucially, model-independent reconstructions of the dark energy density reveal that the deviation from ΛCDM is driven almost entirely by the combination of DESI BAO with low-redshift ($z < 0.1$) supernovae from the DES-Y5 catalog [106, 107]. Removing the low-redshift SNe data completely removes the tension with ΛCDM . This raises a critical question: is the hint of dynamical dark energy a sign of new fundamental physics, or is it the result of unaccounted-for systematic effects in our understanding of the local Universe? The remarkable precision of DESI has thus shifted the focus of the dark energy debate to the intricate details of our observational datasets.

5 Dark Energy

The confirmation of the universe’s accelerated expansion [108, 109] has established a new standard model of cosmology, but has also posed one of the most profound questions in modern physics: what is the nature of the “dark energy” that drives this acceleration? While the cosmological constant, Λ , provides the simplest explanation, it is beset by significant theoretical challenges. This has motivated a vast field of research into dynamical dark energy, where the component driving acceleration is not a constant but evolves over cosmic time. This chapter explores the standard ΛCDM model, its challenges, and the rich theoretical landscape of dynamical alternatives, with a focus on scalar fields and their evolution.

5.1 The Cosmological Constant: Successes and Tensions

The simplest candidate for dark energy is the cosmological constant Λ , first introduced by Einstein. Within the framework of quantum field theory, Λ can be interpreted as the energy of the vacuum. This leads to the Standard Model of Cosmology, known as the Λ CDM model, which posits a universe composed of a cosmological constant, cold dark matter, and baryonic matter.

5.1.1 The Λ CDM Model as the Standard Paradigm

The Λ CDM model is described by the Friedmann equation for a flat universe, as given in 1.12.4, with the dark energy component having a constant equation of state parameter $w = -1$. This simple, six-parameter model has been remarkably successful, providing an excellent fit to a wide range of high-precision cosmological data, including the Cosmic Microwave Background (CMB) anisotropies [5], Baryon Acoustic Oscillations (BAO), and the Hubble diagram from Type Ia supernovae [110]. Its success in describing the universe from the epoch of recombination to the present day makes it the undisputed benchmark for all cosmological theories.

5.1.2 Theoretical Challenges: Fine-Tuning and Coincidence

Despite its observational triumphs, the cosmological constant is plagued by deep theoretical problems, as reviewed extensively in [111, 112, 113].

The Fine-Tuning Problem. The most severe issue is the fine-tuning, or "cosmological constant," problem. Theoretical calculations of the vacuum energy density based on quantum field theory predict a value that is catastrophically larger than the observed value. As shown in the analysis of [110], depending on the energy scale used for the cutoff, the discrepancy can be between 55 to 121 orders of magnitude. For ρ_Λ to be as small as it is observed today, an exquisite and unexplained cancellation between the bare cosmological constant and the quantum vacuum contributions is required.

The Coincidence Problem. The second puzzle is the "coincidence" problem. The energy densities of matter and the cosmological constant evolve very differently with the expansion of the universe: $\rho_m \propto a^{-3}$ while ρ_Λ is constant. Observations indicate that today, their energy densities are of the same order of magnitude ($\Omega_{m,0} \approx 0.3, \Omega_{\Lambda,0} \approx 0.7$). This implies that we are living in a very special epoch in cosmic history. In the distant past, the cosmological constant was utterly negligible, and in the far future, matter will be completely diluted, leaving only Λ . Why the two are comparable precisely now, during the era of structure formation and human observers, is a coincidence for which the Λ CDM model offers no explanation [114].

5.1.3 Observational Tensions: The Hubble Constant and Growth of Structure

In recent years, the success of the Λ CDM model has been challenged by a growing number of tensions between different observational probes, as reviewed in [115].

The H_0 Tension. There is a significant discrepancy in the measured value of the Hubble constant, H_0 , between early-universe and late-universe probes. Measurements based on the CMB in the context of a Λ CDM model, such as those from the Planck satellite, infer a value of $H_0 = 67.4 \pm 0.5 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [5]. In contrast, direct local measurements using a distance ladder based on Cepheid variable stars and Type Ia supernovae consistently find a higher value, with the SH0ES team reporting $H_0 = 74.03 \pm 1.42 \text{ km s}^{-1} \text{ Mpc}^{-1}$ [116]. This discrepancy has now reached a statistical significance of over 4σ .

The Growth Tension. Another tension concerns the growth of large-scale structure, often parameterized by σ_8 , the amplitude of matter fluctuations on scales of $8h^{-1} \text{ Mpc}$. Probes that measure the amplitude of structure at late times, such as weak lensing surveys and redshift-space distortions, tend to find a lower value for σ_8 than that inferred from the CMB under the assumption of Λ CDM [117]. This suggests that the growth of structure in the late universe may be suppressed compared to the Λ CDM prediction.

5.2 Dynamical Dark Energy: Phenomenological Models

The theoretical and observational challenges facing Λ CDM motivate the exploration of dynamical dark energy models, where the equation of state parameter, w , is not fixed at -1.

5.2.1 The w CDM Model: A First Step Beyond Λ

The simplest extension to the standard model is the w CDM model, where dark energy is still treated as a perfect fluid but with a constant equation of state $w \neq -1$. The energy density of this fluid evolves according to (16) as $\rho_{DE} \propto a^{-3(1+w)}$. The universe still accelerates as long as $w < -1/3$. This model introduces one additional free parameter, w , which can be constrained by data. While simple, it serves as a crucial first test for deviations from the cosmological constant.

5.2.2 The Chevallier-Polarski-Linder (CPL) Parametrization

To allow for a time-varying equation of state, a simple and widely used phenomenological model is the Chevallier-Polarski-Linder (CPL) parametrization [118, 119]. It proposes a linear expansion of w in terms of the scale factor a :

$$w(a) = w_0 + w_a(1 - a) = w_0 + w_a \frac{z}{1 + z} \quad (67)$$

This two-parameter model captures a wide range of behaviors. w_0 represents the value of the equation of state today ($a = 1, z = 0$), while w_a represents its evolution. A positive w_a implies that w was higher (less negative) in the past, while a negative w_a implies it was lower. The Λ CDM model is recovered for $(w_0, w_a) = (-1, 0)$. The dark energy density in this model evolves according to (17), which integrates to:

$$\rho_{DE}(a) = \rho_{DE,0} a^{-3(1+w_0+w_a)} e^{3w_a(a-1)} \quad (68)$$

While extremely useful for fitting data, it is important to recognize that the CPL model is purely phenomenological and may not accurately represent the behavior of all physical dark energy models, an issue explored in detail in [120].

5.3 Quintessence: The Physics of a Scalar Field

The most physically motivated candidate for dynamical dark energy is a minimally coupled scalar field, dubbed **quintessence** [121, 122]. This framework proposes that the cosmic acceleration is driven by the potential energy of a slowly evolving scalar field, similar to the mechanism of inflation discussed in section 2.5.

5.3.1 The Canonical Scalar Field

For a homogeneous scalar field $\phi(t)$, the energy density ρ_ϕ and pressure p_ϕ are given by equations (35) and (36), respectively. The equation of state parameter is therefore:

$$w_\phi = \frac{p_\phi}{\rho_\phi} = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)}$$

This equation shows that w_ϕ can dynamically vary between $w_\phi = +1$ (when the kinetic energy dominates, $\dot{\phi}^2 \gg V(\phi)$) and $w_\phi = -1$ (when the potential energy dominates, $\dot{\phi}^2 \ll V(\phi)$). To drive acceleration ($w_\phi < -1/3$), the field's kinetic energy must be sub-dominant to its potential energy, a condition reminiscent of the slow-roll conditions for inflation, cf. subsection 2.5. The evolution of the field is governed by the Klein-Gordon equation (37) in an expanding background.

5.3.2 The Rolling Field as a Time-Varying Λ

The fundamental idea of quintessence, as originally proposed in foundational papers like [122], is to treat dark energy not as a static vacuum energy but as the energy of a dynamical scalar field slowly rolling down a potential $V(\phi)$. In this picture, the nearly constant energy density we observe today is not a fundamental constant of nature but rather a reflection of the fact that the field's evolution has become extremely slow in the current cosmological epoch.

The evolution of this field, governed by the Klein-Gordon equation (37), is critically affected by the expansion of the universe through the Hubble friction term, $3H\dot{\phi}$. At early times, when H was large, this friction term dominated, heavily damping the motion of ϕ . As the universe expanded and H decreased, the field began to roll more freely, allowing its energy density to evolve with time. The value of ρ_ϕ that we observe today is therefore not a finely-tuned initial condition but a result of this cosmic dynamical evolution.

5.3.3 Tracker Solutions and Inverse Power-Law Potentials

A key success of certain quintessence models is their ability to address the coincidence problem through the existence of **tracker solutions** [122, 123]. A tracker is a dynamical attractor in the phase space of solutions, meaning that for a very wide range of initial conditions, the scalar field's evolution will eventually converge onto a single, common trajectory.

On this tracker path, the energy density of the scalar field ρ_ϕ scales with the energy density of the background fluid (radiation or matter), maintaining a small but non-negligible fraction of the total energy. This "tracking" behavior ensures that the dark energy is not completely negligible in the past and naturally becomes comparable to the matter density in the present era, thus providing a dynamical explanation for the observed coincidence.

A classic potential that exhibits this behavior is the **inverse law potential**, proposed in [122]:

$$V(\phi) = M^{4+\alpha} \phi^{-\alpha} \quad (69)$$

where $\alpha > 0$ is a dimensionless constant and M is a mass scale. For a universe dominated by a background fluid with a constant equation of state w_m , this potential leads to a tracker solution where the effective equation of state of the scalar field becomes constant (cf. Derivation of the Tracker Solution for an Inverse Power-Law Potential in the Appendix):

$$w_\phi = \frac{\alpha w_m - 2}{\alpha + 2} \quad (70)$$

During the matter-dominated era ($w_m = 0$), this yields $w_\phi = -2/(\alpha+2)$. This is always greater than -1, meaning the model is quintessence, but it can be sufficiently negative to drive acceleration if α is small. For example, to achieve $w_\phi < -1/3$, one requires $\alpha < 4$. This model demonstrates how a simple, physically motivated potential can give rise to a tracking dark energy that dynamically evolves to an accelerating state.

5.3.4 Phantom Dark Energy and the Null Energy Condition

Observations do not exclude an equation of state with $w < -1$. A hypothetical substance with this property is known as **phantom energy** [124]. From the fluid equation (11), if $w < -1$, then $\rho + p < 0$, which implies that $\dot{\rho} > 0$. The energy density of a phantom fluid "increases" as the universe expands. This leads to a dramatic future scenario known as the "Big Rip," where the scale factor becomes infinite in a finite time.

Theoretically, phantom energy is problematic. A canonical scalar field, as described above, is restricted to $w_\phi \geq -1$. To achieve $w < -1$, one must introduce a field with a negative kinetic term in the Lagrangian, $\mathcal{L} = -\frac{1}{2}\partial_\mu\phi\partial^\mu\phi - V(\phi)$. Such a field is called a **ghost**. As reviewed in [125], theories with ghost fields are generically plagued by a catastrophic vacuum instability, as their existence implies that the energy of the system is not bounded from below. The boundary $w = -1$ is known as the **phantom divide line**. A model whose equation of state crosses this line presents a significant theoretical challenge.

5.4 The Dynamics of Scalar Fields: A Phase-Space Approach

The cosmological evolution of a scalar field coupled to a background fluid can be analyzed powerfully using the techniques of dynamical systems. This approach, detailed in Chapter 4 of [111] and in [126], allows one to understand the global behavior of the system without solving the equations of motion for specific initial conditions.

5.4.1 Autonomous Systems for Cosmology

The goal is to rewrite the cosmological equations as a set of first-order autonomous differential equations. For a flat FRW universe containing a scalar field ϕ and a barotropic fluid (e.g., matter) with equation of state w_m , we define the dimensionless variables:

$$x \equiv \frac{\kappa\dot{\phi}}{\sqrt{6}H}, \quad y \equiv \frac{\kappa\sqrt{V(\phi)}}{\sqrt{3}H}$$

The Friedmann constraint equation, $H^2 = \frac{\kappa^2}{3}(\rho_m + \rho_\phi)$, can be written in terms of these variables as:

$$\Omega_m = 1 - x^2 - y^2$$

The evolution equations for the scalar field can be transformed into a system of equations for x and y . We introduce the parameter $\lambda \equiv -V'/(KV)$. The system is (cf. Derivation of the Autonomous System Equations

in the Appendix):

$$\frac{dx}{dN} = -3x + \sqrt{\frac{3}{2}}\lambda y^2 + x \left(3x^2 + \frac{3}{2}(1+w_m)(1-x^2-y^2) \right) \quad (71)$$

$$\frac{dy}{dN} = -\sqrt{\frac{3}{2}}\lambda xy + y \left(3x^2 + \frac{3}{2}(1+w_m)(1-x^2-y^2) \right) \quad (72)$$

where $N = \ln a$ is the number of e-folds. For a given potential $V(\phi)$, λ is a function of x and y , making the system closed. For an exponential potential, $V(\phi) = V_0 e^{-\kappa\lambda\phi}$, λ is a constant.

5.4.2 Critical Points and Cosmological Solutions

The cosmological evolution corresponds to trajectories in the phase space defined by the dimensionless variables (x, y) . The long-term behavior of the universe is governed by the system's **critical (or fixed) points**, where the derivatives with respect to the number of e-folds vanish: $dx/dN = 0$ and $dy/dN = 0$. These special points represent exact cosmological solutions where the energy densities of the components maintain a constant ratio. The stability of these points determines which cosmological epochs are transient and which are the final attractors of the system. An analysis of the system for an exponential potential, $V(\phi) = V_0 e^{-\kappa\lambda\phi}$ where λ is constant, reveals several critical points corresponding to distinct cosmological solutions. The nature of the final attractor, and thus the ultimate fate of the universe, depends critically on the value of λ .

Case 1: Quintessence Attractor-Figure 22 ($\lambda^2 < 3(1+w_m)$). When the potential is sufficiently flat, a stable attractor solution exists where the scalar field comes to dominate the energy density of the universe and drives an accelerated expansion. This is the canonical quintessence scenario. The critical points for a matter-dominated background ($w_m = 0$) are:

- **Matter Domination:** Located at $(x, y) = (0, 1)$ (if $\Omega_m = 1$). This point is a saddle point, representing the unstable matter-dominated epoch of the past.
- **Scalar Field Domination:** Located at $(x, y) = (\lambda/\sqrt{6}, \sqrt{1-\lambda^2/6})$. This is the global attractor of the system. For this solution, $\Omega_\phi = 1$ and the equation of state is constant, $w_\phi = -1 + \lambda^2/3$. Acceleration occurs for $\lambda^2 < 2$.

Case 2: Scaling Solution Attractor-Figure 23 ($\lambda^2 > 3(1+w_m)$). If the potential is too steep, the scalar field cannot come to dominate the universe. Instead, the final state is a "scaling solution," where the scalar field's energy density decreases at the same rate as the background matter or radiation. The critical points for $w_m = 0$ are:

- **Scaling Solution:** Located at $(x, y) = (\sqrt{3/2}/\lambda, \sqrt{3/(2\lambda^2)})$. This is the stable attractor. In this solution, the scalar field's energy density is a constant fraction of the total, $\Omega_\phi = 3/\lambda^2$, and its equation of state mimics that of the background fluid, $w_\phi = w_m = 0$. This scenario does not produce late-time acceleration.

Case 3: Spiral Attractor-Figure 24. When λ is very small, the eigenvalues of the Jacobian matrix evaluated at the scalar-field dominated critical point become complex. This means that the point is a stable spiral. The physics remains the same—the universe is driven to a final state of dark-energy-driven acceleration—but the dynamical path to this state is different.

5.5 A Dynamical Classification: Thawing and Freezing Models

Scalar field models can be broadly classified into two dynamical categories based on their late-time behavior: freezing and thawing models [127]. This classification provides a physical basis for understanding the evolution of $w(z)$.

5.5.1 Defining the Models

- **Freezing Models:** In these models, the scalar field has been rolling down its potential for a long time. As the universe expands, the Hubble friction term $3H\dot{\phi}$ in the Klein-Gordon equation (37) decreases, allowing the field to accelerate. However, if the potential is shallow enough (e.g., exponential or inverse power-law), the field approaches a scaling or attractor solution where its equation of state "freezes" to a nearly constant value. These models, which include trackers, are characterized by $w'(z) = \frac{dw}{dz} < 0$ at late times; i.e., w approaches -1 from above. The potential for these models is typically convex ($V'' > 0$).

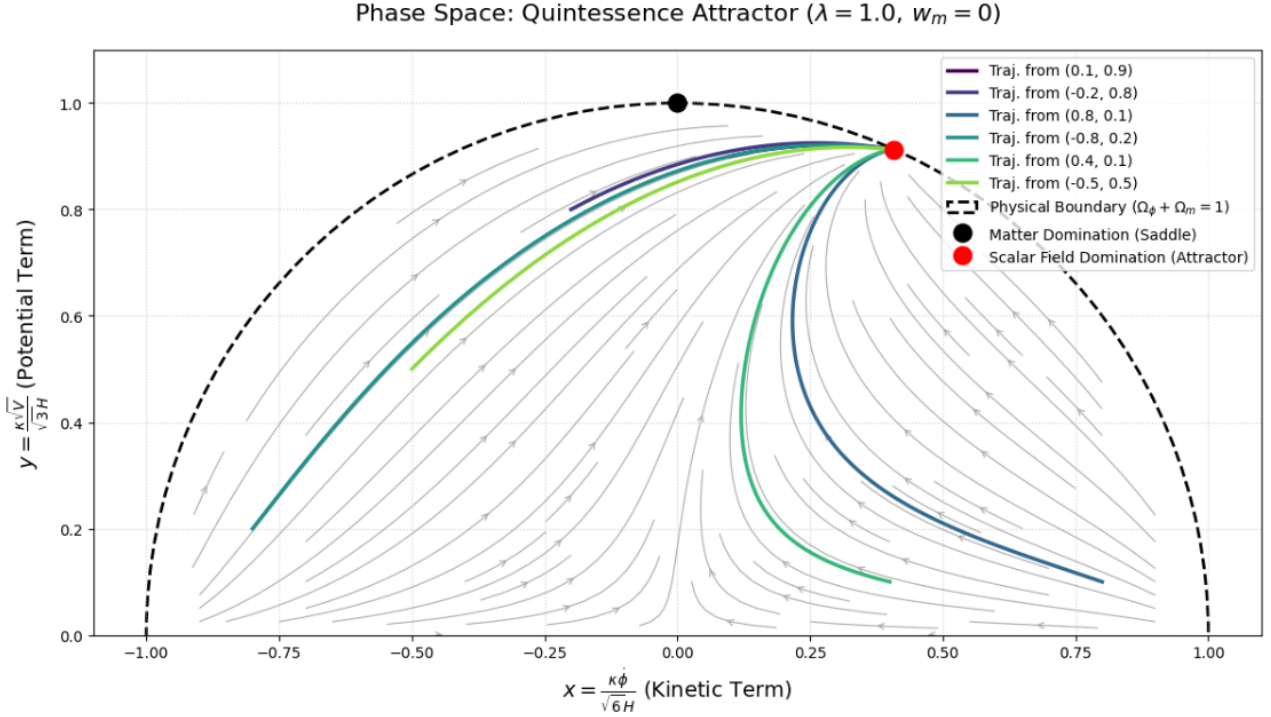


Figure 22: **Phase-space diagram for a quintessence attractor model with $\lambda = 1.0$ and $w_m = 0$.** The plot shows the flow of cosmological solutions in the (x, y) phase plane, where x represents the kinetic and y the potential energy of the scalar field. The gray streamplot and colored lines show that all trajectories, regardless of their initial conditions, are inevitably drawn to the stable **scalar-field dominated attractor** (red circle). This point represents a future universe completely dominated by dark energy ($\Omega_\phi = 1$) with an equation of state $w_\phi = -2/3$, driving cosmic acceleration. The matter-dominated epoch is represented by the unstable **saddle point** (black circle), from which all trajectories originate. This phase space illustrates the standard quintessence scenario where the universe transitions from a matter-dominated past to a dark-energy-dominated future.

- **Thawing Models:** In this class of models, the scalar field is initially "frozen" in place by Hubble friction at early times, typically because the potential is steep ($|V'/V|$ is large). At early times, $\dot{\phi} \approx 0$, and the field behaves like a cosmological constant with $w \approx -1$. As the universe expands and H decreases, the Hubble friction eventually becomes small enough that the field begins to roll down its potential. Consequently, its kinetic energy grows, and its equation of state begins to "thaw," evolving away from -1. These models are characterized by $w'(z) > 0$ at late times. The potential for these models is typically concave ($V'' < 0$).

5.5.2 Observational Signatures and Parametrization Bias

The two classes of models predict different evolutionary histories for $w(z)$. As shown by Pantazis et al. in [128], this has important implications for data analysis. A given two-parameter $w(z)$ parametrization is generically either a convex or a concave function. For example, the CPL parametrization (67) with $w_a > 0$ is a concave function of z , making it well-suited to describe thawing models. Conversely, a parametrization designed to fit freezing models will be convex.

The crucial point raised in [128] is that fitting data generated from a model of one class (e.g., freezing) with a parametrization designed for the other class (e.g., thawing, like CPL) can lead to significant and misleading biases. For instance, fitting freezing-model data with the CPL form can artificially induce a crossing of the phantom divide line ($w = -1$) in the best-fit curve, even when the underlying true model is a standard quintessence field that never violates the null energy condition (cf. Figure 25). This highlights the importance of using a diverse and physically motivated set of parametrizations when testing dark energy models.

5.6 The Statefinder Diagnostic

With a growing menagerie of dark energy models, it is useful to have a diagnostic tool that can distinguish between them in a model-independent way. The Statefinder diagnostic, proposed by Sahni et al. [129], provides such a tool based purely on the geometry of the universe's expansion history.

Phase Space: Scaling Solution Attractor ($\lambda = 4.0, w_m = 0$)

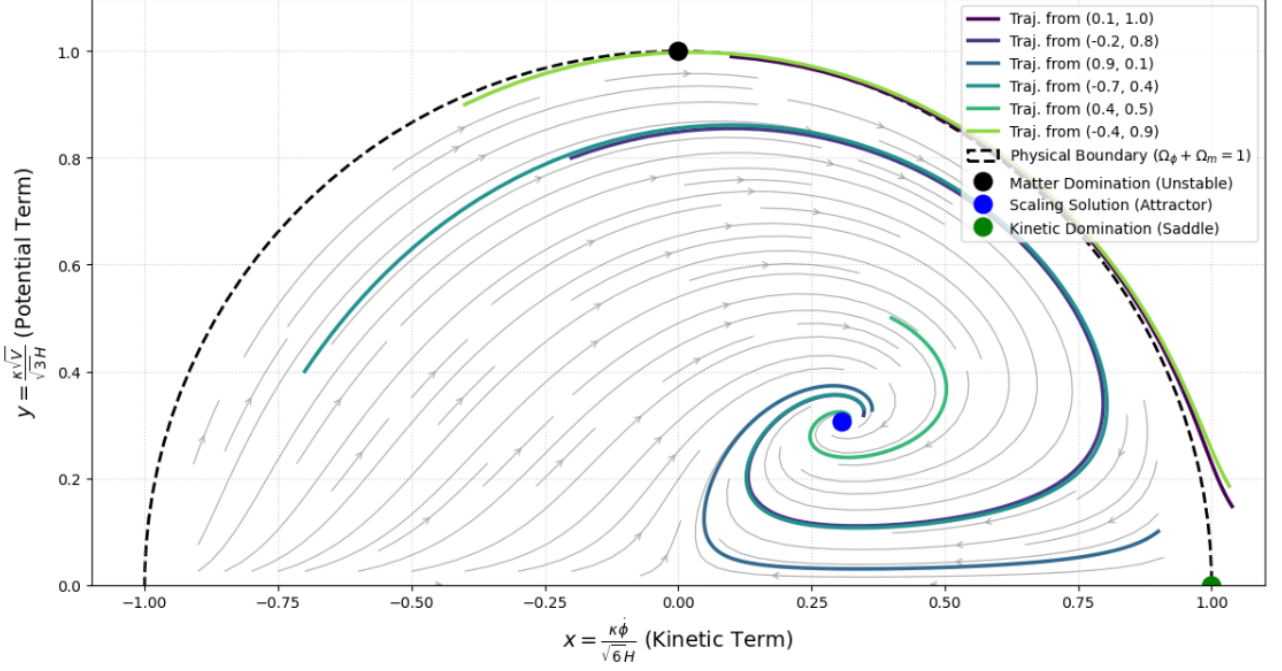


Figure 23: **Phase-space diagram for a scaling solution model with $\lambda = 4.0$ and $w_m = 0$.** For a steep potential, the final state of the universe is the stable **scaling solution attractor** (blue circle). All trajectories converge on this point, where the scalar field's energy density does not dominate but instead scales with the matter density, maintaining a constant fraction of the total energy ($\Omega_\phi = 3/16$) and having an equation of state $w_\phi = 0$. The matter-dominated point (black circle) is now an unstable node, and the kinetic-dominated solution (green circle) is a saddle point. This scenario is observationally ruled out as a model for dark energy because it cannot drive cosmic acceleration.

5.6.1 A Geometrical Probe of Dark Energy

The Statefinder extends the series of cosmological parameters derived from the scale factor. The first two are the Hubble parameter $H = \dot{a}/a$ and the deceleration parameter $q = -\ddot{a}/(aH^2)$. The Statefinder introduces a third parameter, r , constructed from the third time derivative of the scale factor:

$$r \equiv \frac{\ddot{a}}{aH^3} \quad (73)$$

A second parameter, s , is constructed from a linear combination of r and q :

$$s \equiv \frac{r - 1}{3(q - 1/2)} \quad (74)$$

The pair $\{r, s\}$ defines a point in the Statefinder plane. Different dark energy models trace out distinct trajectories in this plane as a function of redshift.

5.6.2 Statefinder Parameters in Terms of Cosmological Parameters

The power of the Statefinder diagnostic lies in its direct connection to the physical properties of dark energy, namely its equation of state w and its evolution. For a flat universe composed of matter and a dark energy component with a dynamical equation of state $w(z)$, the parameters q and r can be expressed as (cf. Derivation of the Statefinder Parameters in the Appendix for a full derivation):

$$q(z) = \frac{1}{2} + \frac{3}{2}w(z)\Omega_{DE}(z) \quad (75)$$

$$r(z) = 1 + \frac{9}{2}w(z)(1 + w(z))\Omega_{DE}(z) - \frac{3}{2H(z)}\frac{dw}{dt}\Omega_{DE}(z) \quad (76)$$

where $\Omega_{DE}(z)$ is the fractional energy density of dark energy at redshift z . The term dw/dt can be rewritten in terms of the redshift derivative as $dw/dt = -H(1 + z)dw/dz$. This makes it clear that r is sensitive to the time evolution of w , while s is constructed to depend primarily on the nature of the dark energy potential, being largely independent of Ω_{DE} .

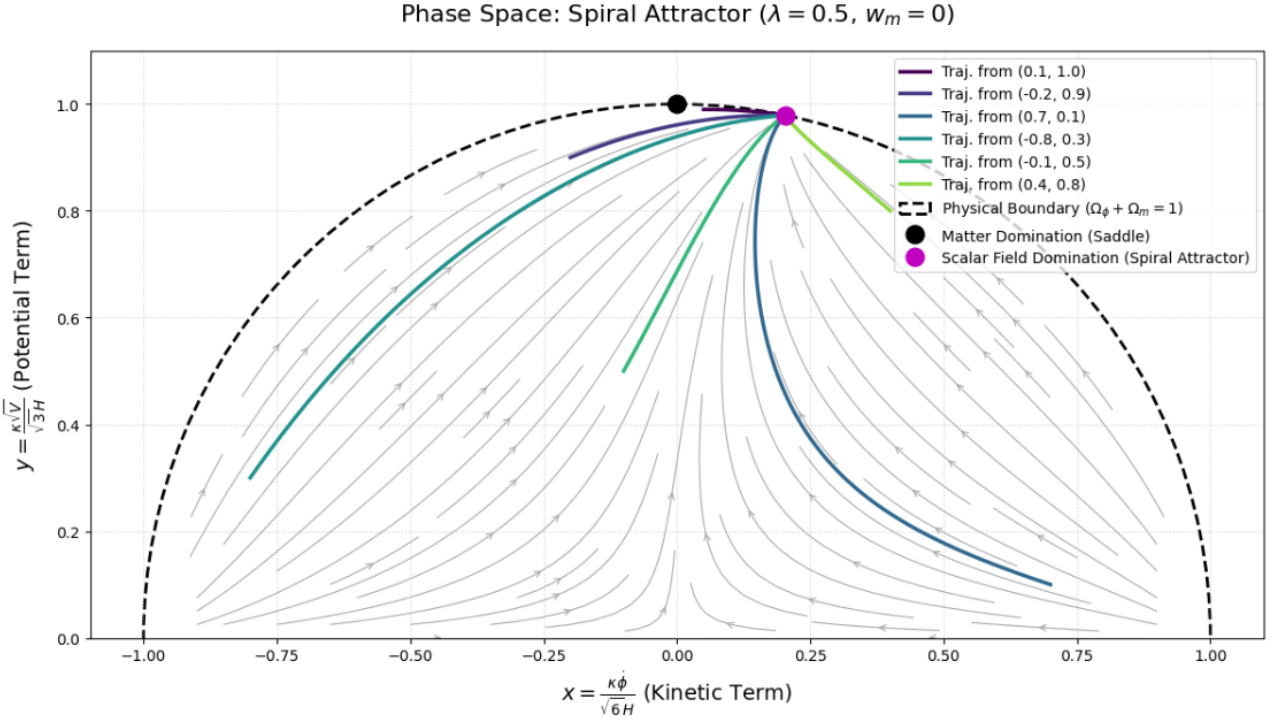


Figure 24: **Phase-space diagram for a spiral attractor model with $\lambda = 0.5$ and $w_m = 0$.** For a very flat potential, the scalar-field dominated attractor (magenta circle) becomes a **stable spiral**. The trajectories circle in towards the final fixed point, indicating that the universe approaches its ultimate state of acceleration via damped oscillations of the scalar field’s kinetic and potential energies. The final state is still one of complete dark energy domination with $w_\phi = -1 + (0.5)^2/3 \approx -0.92$, but the dynamical journey involves a spiral path in phase space rather than a direct approach.

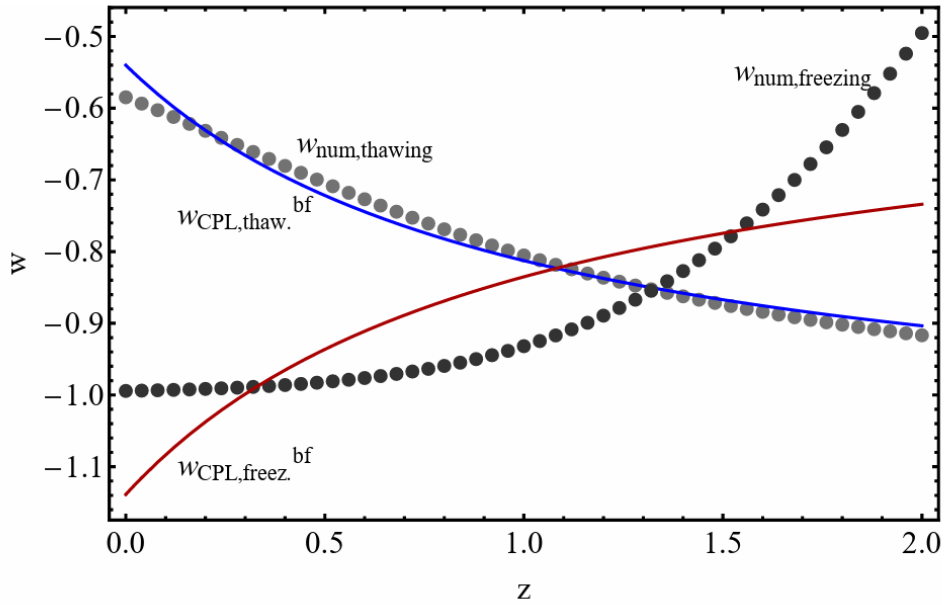


Figure 25: Illustration of parametrization bias in dark energy models. The true freezing model data (black dots, labeled **w_{num,freezing}**) remains above the phantom divide line ($w = -1$). However, when fit with the CPL parametrization (a model better suited for thawing data), the resulting best-fit curve (red line, labeled **wCPL,freez^{bf}**) is artificially induced to cross the phantom divide, creating a misleading signature of phantom energy. Adapted from [128]

5.6.3 The Statefinder Plane and Model Classification

The primary utility of the Statefinder diagnostic is its ability to map the often complex dynamics of different dark energy theories onto clear, visual trajectories in the $\{r, s\}$ plane. As demonstrated in the foundational analysis by [130], this provides a powerful method for classifying and distinguishing between competing cosmological models based on their geometric properties alone.

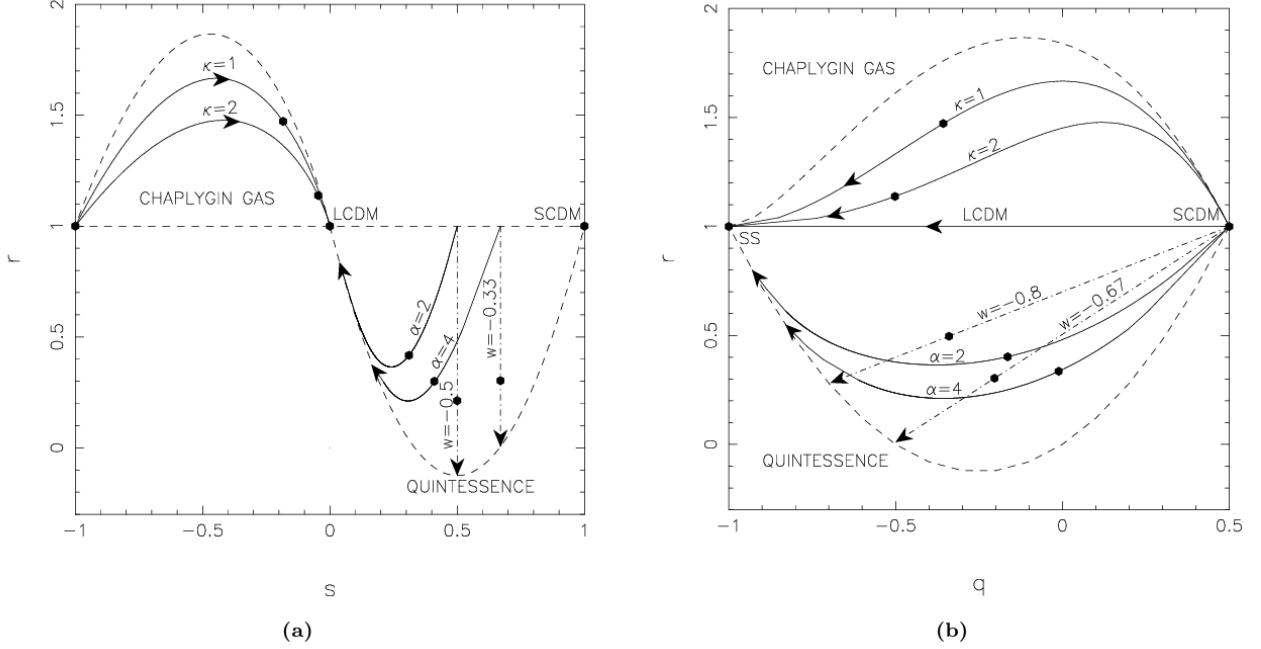


Figure 26: **Evolutionary trajectories of various dark energy models in the Statefinder planes.** **Panel (a)** shows the trajectories in the $\{r, s\}$ plane. The standard Λ CDM model is a fixed point at $\{1, 0\}$. All models begin at the matter-dominated (SCDM) fixed point $\{1, 1\}$ in the past and evolve towards the future (indicated by arrows). The plot clearly separates different physical theories: quintessence models with various potentials (‘SUGRA’, ‘ASUGRA’, ‘LQC’), the Chaplygin gas (‘CPL’), and a braneworld model (‘BRANE’) all trace unique and distinguishable paths. **Panel (b)** displays the same models in the $\{r, q\}$ plane, where q is the deceleration parameter. While models are still separated, their trajectories are more compressed and overlapping, particularly at late times as they approach the de Sitter state ($q \rightarrow -1$). This demonstrates the superior discriminatory power of the $\{r, s\}$ plane, which is specifically designed to stretch out these differences. Adapted from [130].

The analysis presented in Fig. 26 reveals the diagnostic’s key strengths. As shown in panel (a), each dark energy model leaves a unique “fingerprint” on the geometric history of the universe. For instance, different quintessence potentials, such as those motivated by supergravity (‘SUGRA’), trace distinct curves, demonstrating that the Statefinder is sensitive not just to the existence of a scalar field, but to the specific shape of its potential. This allows for a much finer level of model discrimination than is possible with just the equation of state w alone.

A crucial feature is that all models shown originate from the same point, $\{r, s\} = \{1, 1\}$, which represents the past matter-dominated era (the “SCDM” universe). As dark energy becomes dynamically important, the trajectories diverge. The goal of observational cosmology is to measure the values of $r(z)$ and $s(z)$ at different redshifts and plot these data points on the plane to determine which theoretical trajectory our universe is following.

The superiority of the $\{r, s\}$ plane over other combinations is highlighted in panel (b). When plotted against the deceleration parameter q , the model trajectories become crowded and degenerate, especially as they approach their future de Sitter state where $q \rightarrow -1$. The $\{s\}$ parameter is constructed specifically to resolve this degeneracy by effectively “zooming in” on the differences between models that are all approaching a similar expansion history. The clear separation of curves in panel (a) confirms that the $\{r, s\}$ pair provides the most effective geometrical framework for classifying dynamical dark energy.

5.6.4 Applications to Modern Cosmological Models

The Statefinder is a versatile tool that has been applied to a wide range of modern theories beyond simple quintessence.

Interacting Dark Energy. In models where dark energy and dark matter interact, exchanging energy at a rate Q , the standard conservation equations are modified. This interaction term directly affects the evolution of the Hubble parameter and its derivatives. As shown in recent analyses like [131], the presence of an interaction term Q significantly alters the trajectories in the $\{r, s\}$ plane. Different functional forms for Q (e.g., $Q \propto H\rho_c$, $Q \propto H\rho_x$) produce distinct, separable paths, offering a potential way to probe the nature of the dark sector coupling.

Modified Gravity. The Statefinder is particularly useful for testing modified gravity, as these theories alter the underlying gravitational dynamics and thus the geometric expansion history.

- **$f(T)$ and $f(Q)$ Gravity:** Theories based on modifying the teleparallel equivalent of GR, such as $f(T)$ and $f(Q)$ gravity, have been analyzed using the Statefinder diagnostic [132, 133]. These models can produce a wide range of trajectories, including some that mimic quintessence and others that evolve in the phantom region, depending on the chosen functional form for $f(T)$ or $f(Q)$.
- **Unique Trajectories:** Some models can produce unique and exotic paths in the $\{r, s\}$ plane. For instance, the analysis in [134] found that the "Abnormally Weighting Energy" (AWE) hypothesis can generate closed loops in the Statefinder plane, a feature not typically seen in standard quintessence models and which could serve as a smoking-gun signature.

The ability of the Statefinder to distinguish between these fundamentally different approaches to cosmic acceleration—whether it's a new fluid or new gravity—makes it a crucial diagnostic for future cosmological surveys.

A Appendix

A.1 Derivation of the Einstein Tensor from the Metric and conservation of energy/pomentum

To understand the full construction of the Einstein tensor $G_{\mu\nu}$, we begin with the metric tensor $g_{\mu\nu}$, which defines the geometry of spacetime. From the metric, we compute the Christoffel symbols of the second kind:

$$\Gamma_{\mu\nu}^{\rho} = \frac{1}{2}g^{\rho\sigma}(\partial_{\mu}g_{\nu\sigma} + \partial_{\nu}g_{\mu\sigma} - \partial_{\sigma}g_{\mu\nu}).$$

These symbols are not tensors but are used to define the covariant derivative, which preserves tensorial character under coordinate transformations. For a rank-1 covariant tensor V_{ν} , the covariant derivative is:

$$\nabla_{\mu}V_{\nu} = \partial_{\mu}V_{\nu} - \Gamma_{\mu\nu}^{\lambda}V_{\lambda}.$$

Using the Christoffel symbols, we construct the Riemann curvature tensor:

$$R^{\rho}_{\sigma\mu\nu} = \partial_{\mu}\Gamma_{\nu\sigma}^{\rho} - \partial_{\nu}\Gamma_{\mu\sigma}^{\rho} + \Gamma_{\mu\lambda}^{\rho}\Gamma_{\nu\sigma}^{\lambda} - \Gamma_{\nu\lambda}^{\rho}\Gamma_{\mu\sigma}^{\lambda}.$$

This rank-4 tensor encodes the full information about spacetime curvature. We obtain the Ricci tensor by contracting the Riemann tensor:

$$R_{\mu\nu} = R^{\lambda}_{\mu\lambda\nu},$$

and then the Ricci scalar by contracting the Ricci tensor with the inverse metric:

$$R = g^{\mu\nu}R_{\mu\nu}.$$

The Einstein tensor is finally defined as:

$$G_{\mu\nu} = R_{\mu\nu} - \frac{1}{2}Rg_{\mu\nu}.$$

The Bianchi's identity expresses a cyclic symmetry of the covariant derivative of the Riemann tensor:

$$\nabla_{\lambda}R^{\rho}_{\sigma\mu\nu} + \nabla_{\mu}R^{\rho}_{\sigma\nu\lambda} + \nabla_{\nu}R^{\rho}_{\sigma\lambda\mu} = 0.$$

Contracting indices ($\rho = \mu$) and using the definition of the Ricci and Einstein tensors, one obtains:

$$\nabla^{\mu}G_{\mu\nu} = 0,$$

which implies the local conservation of the energy-momentum tensor in general relativity:

$$\nabla^{\mu}T_{\mu\nu} = 0.$$

A.2 Derivation of the Einstein's Field Equation from the Einstein's - Hilbert's Action

A.2.1 Vacuum Einstein field equations

The Einstein-Hilbert action We adopt signature $(-+++)$ and units $c = 1$. From the Einstein's - Hilbert's Action, we have:

$$S_{\text{EH}}[g] = \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{-g} R \quad \text{with} \quad \kappa = 8\pi G$$

Variation of the determinant The determinant's mathematical⁶ identity $\delta g = g g^{\mu\nu} \delta g_{\mu\nu}$ gives:

$$\delta\sqrt{-g} = \frac{1}{2}\sqrt{-g} g^{\mu\nu} \delta g_{\mu\nu} = -\frac{1}{2}\sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \quad (77)$$

where indices are raised and lowered with the unvaried metric.

Variation of the Ricci scalar Write $R = g^{\mu\nu} R_{\mu\nu}$ so that

$$\delta R = \delta(g^{\mu\nu} R_{\mu\nu}) = R_{\mu\nu} \delta g^{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}.$$

⁶the identity $\delta g = g g^{\mu\nu} \delta g_{\mu\nu}$ is up to the reader to be proven and the analytic demonstration of similar mathematical expressions doesn't align with the purpose of this work.

Variation of the connection The Levi-Civita connection(Christoffel's Symbols) varies as:

$$\delta\Gamma_{\mu\nu}^{\rho} = \frac{1}{2}g^{\mu\nu}(\nabla_{\mu}\delta g_{\nu\sigma} + \nabla_{\nu}\delta g_{\mu\sigma} - \nabla_{\sigma}\delta g_{\mu\nu}).$$

Variation of the Ricci tensor With the variation of the connection:

$$\delta R_{\mu\nu} = \nabla_{\rho}\delta\Gamma_{\mu\nu}^{\rho} - \nabla_{\nu}\delta\Gamma_{\mu\rho}^{\rho} = \frac{1}{2}\left(\nabla_{\rho}\nabla_{\mu}\delta g^{\rho}_{\nu} + \nabla_{\rho}\nabla_{\nu}\delta g^{\rho}_{\mu} - \nabla_{\rho}\nabla^{\rho}\delta g_{\mu\nu} - \nabla_{\nu}\nabla_{\mu}\delta g^{\rho}_{\rho} - \nabla_{\nu}\nabla_{\rho}\delta g^{\rho}_{\mu} + \nabla_{\nu}\nabla_{\rho}\delta g_{\mu}^{\rho}\right).$$

By contracting, rearranging and using $\nabla_{\mu}g_{\mu\nu} = 0$ yields the compact identity:

$$g^{\mu\nu}\delta R_{\mu\nu} = (\nabla_{\mu}\nabla_{\nu} - g_{\mu\nu}g^{\rho\sigma}\nabla_{\rho}\nabla_{\sigma})\delta g^{\mu\nu}$$

Variation of the action By substituting all the previous variations, we get:

$$\delta S_{\text{EH}} = \frac{1}{2\kappa} \int d^4x \left[(\delta\sqrt{-g}) R + \sqrt{-g} \delta R \right] = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left[-\frac{1}{2} R g_{\mu\nu} \delta g^{\mu\nu} + R_{\mu\nu} \delta g^{\mu\nu} + (\nabla_{\mu}\nabla_{\nu} - g_{\mu\nu}g^{\rho\sigma}\nabla_{\rho}\nabla_{\sigma})\delta g^{\mu\nu} \right]$$

Isolating the total divergence To remove the second-derivative term in δS_{EH} , we invoke the *covariant divergence theorem*, the curved-space analogue of Gauss's theorem. For any smooth vector field V^{μ} on $(\mathcal{M}, g_{\mu\nu})$ with boundary $\partial\mathcal{M}$, we have:

$$\int_{\mathcal{M}} d^4x \sqrt{-g} \nabla_{\mu} V^{\mu} = \oint_{\partial\mathcal{M}} d^3\Sigma_{\mu} V^{\mu},$$

with

$$d^3\Sigma_{\mu} = n_{\mu} d^3\Sigma,$$

where n_{μ} is the outward-pointing unit normal and $d^3\Sigma$ is the induced volume element on the boundary. This identity follows from the relation: $\partial_{\mu}(\sqrt{-g}V^{\mu}) = \sqrt{-g}\nabla_{\mu}V^{\mu}$, which holds due to the metric compatibility condition $\nabla_{\mu}g = 0$.

Now by choosing:

$$V^{\mu} = \nabla_{\nu} \delta g^{\mu\nu}$$

Because the Levi-Civita connection is torsion-free, the covariant derivatives commute on the scalar function $\delta g^{\mu\nu}$, i.e.,

$$\nabla_{\mu}\nabla_{\nu}\delta g^{\mu\nu} = \nabla_{\nu}\nabla_{\mu}\delta g^{\mu\nu}$$

Substituting into the divergence theorem gives:

$$\int_{\mathcal{M}} d^4x \sqrt{-g} \nabla_{\mu}\nabla_{\nu}\delta g^{\mu\nu} = \oint_{\partial\mathcal{M}} d^3\Sigma_{\mu} \nabla_{\nu}\delta g^{\mu\nu}$$

Since the variation $\delta g^{\mu\nu}$ vanishes on the boundary $\delta g^{\mu\nu}|_{\partial\mathcal{M}} = 0$, its tangential derivatives also vanish on $\partial\mathcal{M}$ and the surface term is zero. Thus we integrate by parts again:

$$\int_{\mathcal{M}} d^4x \sqrt{-g} \nabla_{\mu}\nabla_{\nu}\delta g^{\mu\nu} = - \int_{\mathcal{M}} d^4x \sqrt{-g} \delta g^{\mu\nu} \nabla_{\mu}\nabla_{\nu}.$$

An identical argument applies to the term involving the Laplacian:

$$g_{\mu\nu}g^{\rho\sigma}\nabla_{\rho}\nabla_{\sigma}\delta g^{\mu\nu} \quad \Rightarrow \quad -g_{\mu\nu}g^{\rho\sigma} \int_{\mathcal{M}} d^4x \sqrt{-g} \delta g^{\mu\nu} \nabla_{\rho}\nabla_{\sigma}$$

Therefore, the total contribution of the second-derivative terms in the variation reduces to terms linear in $\delta g^{\mu\nu}$ and all boundary contributions vanish. Inserting these results back into the variation of the action, we get:

$$\delta S_{\text{EH}} = \frac{1}{2\kappa} \int_{\mathcal{M}} d^4x \sqrt{-g} (R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R) \delta g^{\mu\nu}.$$

Field equations From the principle of stationary action, it is necessary to have $\delta S_{\text{EH}} = 0$ for *arbitrary* $\delta g^{\mu\nu}$ implies:

$$R_{\mu\nu} - \frac{1}{2}g_{\mu\nu}R = 0 \quad \Longleftrightarrow \quad G_{\mu\nu} = 0$$

which are the **vacuum Einstein field equations**.

A.2.2 Including matter

Add a matter action $S_m[g, \Phi] = \int d^4x \sqrt{-g} \mathcal{L}_m$ and define $T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_m}{\delta g^{\mu\nu}}$. Repeating the same variation leads to:

$$G_{\mu\nu} = 8\pi G T_{\mu\nu}$$

completing the standard form of Einstein's equations.

A.3 Derivation of the Euler-Lagrange Equation

The dynamics of a physical system can be obtained by finding the path that extremizes the **action**:

$$S = \int_{t_1}^{t_2} L(q(t), \dot{q}(t), t) dt$$

where:

- L is the *Lagrangian*, typically $L = T - V$,
- $q(t)$ is the generalized coordinate,
- $\dot{q}(t) = \frac{dq}{dt}$ is the generalized velocity.

To find the path that makes the action stationary, we consider a small variation of the path:

$$q(t) \rightarrow q(t) + \epsilon \eta(t)$$

with fixed endpoints $\eta(t_1) = \eta(t_2) = 0$. The variation of the action is:

$$\delta S = \frac{d}{d\epsilon} \int_{t_1}^{t_2} L(q + \epsilon \eta, \dot{q} + \epsilon \dot{\eta}, t) dt \Big|_{\epsilon=0}$$

Expanding to first order in ϵ , we obtain:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} \eta(t) + \frac{\partial L}{\partial \dot{q}} \dot{\eta}(t) \right) dt$$

We integrate the second term by parts:

$$\int_{t_1}^{t_2} \frac{\partial L}{\partial \dot{q}} \dot{\eta}(t) dt = \left[\frac{\partial L}{\partial \dot{q}} \eta(t) \right]_{t_1}^{t_2} - \int_{t_1}^{t_2} \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \eta(t) dt$$

The boundary term vanishes due to $\eta(t_1) = \eta(t_2) = 0$, so:

$$\delta S = \int_{t_1}^{t_2} \left(\frac{\partial L}{\partial q} - \frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) \right) \eta(t) dt$$

Since $\eta(t)$ is arbitrary, the integrand must vanish:

$$\frac{d}{dt} \left(\frac{\partial L}{\partial \dot{q}} \right) - \frac{\partial L}{\partial q} = 0$$

A.4 Derivation of the Schwarzschild Metric

To derive the Schwarzschild solution, we solve Einstein's vacuum field equations, $R_{\mu\nu} = 0$, for a static, spherically symmetric spacetime.

The Metric Ansatz The assumptions of a static and spherically symmetric spacetime lead to a general form for the metric, known as the metric ansatz:

$$ds^2 = -B(r)c^2 dt^2 + A(r)dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

where $A(r)$ and $B(r)$ are unknown functions of the radial coordinate r that we must determine. The functions must approach 1 as $r \rightarrow \infty$ for the metric to be asymptotically flat.

Christoffel Symbols From this metric, we compute the non-zero Christoffel symbols. Using a prime (') to denote differentiation with respect to r , the key symbols are:

$$\begin{aligned}\Gamma_{tr}^t &= \frac{B'}{2B} & \Gamma_{tt}^r &= \frac{B'}{2A} \\ \Gamma_{rr}^r &= \frac{A'}{2A} & \Gamma_{\theta\theta}^r &= -\frac{r}{A} \\ \Gamma_{\phi\phi}^r &= -\frac{r \sin^2 \theta}{A} & \Gamma_{r\theta}^\theta &= \frac{1}{r} \\ \Gamma_{r\phi}^\phi &= \frac{1}{r} & \Gamma_{\phi\phi}^\theta &= -\sin \theta \cos \theta \\ \Gamma_{\theta\phi}^\phi &= \cot \theta\end{aligned}$$

Ricci Tensor Components Using the Christoffel symbols, we compute the components of the Ricci tensor $R_{\mu\nu}$. The vacuum field equations demand that all components are zero. The non-trivial diagonal components yield the following equations:

$$R_{tt} = \frac{B''}{2A} - \frac{B'}{4A} \left(\frac{A'}{A} + \frac{B'}{B} \right) + \frac{B'}{rA} = 0 \quad (78)$$

$$R_{rr} = -\frac{B''}{2B} + \frac{B'}{4B} \left(\frac{A'}{A} + \frac{B'}{B} \right) + \frac{A'}{rA} = 0 \quad (79)$$

$$R_{\theta\theta} = 1 - \frac{r}{2A} \left(\frac{A'}{A} - \frac{B'}{B} \right) - \frac{1}{A} = 0 \quad (80)$$

Solving the System of Equations We can simplify the system by combining equations. Adding $\frac{R_{tt}}{B}$ and $\frac{R_{rr}}{A}$ gives:

$$\frac{R_{tt}}{B} + \frac{R_{rr}}{A} = \frac{1}{rA} \left(\frac{A'}{A} + \frac{B'}{B} \right) = 0 \quad \Rightarrow \quad \frac{A'}{A} + \frac{B'}{B} = 0$$

Integrating this simple relation gives $\ln A + \ln B = \text{constant}$, which implies:

$$A(r)B(r) = K$$

where K is an integration constant. From the asymptotically flat condition ($A \rightarrow 1$ and $B \rightarrow 1$ as $r \rightarrow \infty$), we must have $K = 1$. Therefore, we find the crucial relation:

$$A(r) = \frac{1}{B(r)}$$

Now we substitute $A = 1/B$ into the $R_{\theta\theta}$ equation (80):

$$1 - \frac{r}{2(1/B)} \left(-\frac{B'}{B} - \frac{B'}{B} \right) - B = 0$$

$$1 + rB' - B = 0 \quad \Rightarrow \quad (rB)' = 1$$

Integrating this equation with respect to r yields:

$$rB(r) = r + C \quad \Rightarrow \quad B(r) = 1 + \frac{C}{r}$$

where C is another integration constant.

Physical Interpretation and Final Solution To determine the constant C , we compare our solution to the Newtonian limit. In the weak-field limit, as Einstein showed himself [1], the metric component g_{tt} is related to the Newtonian gravitational potential $\Phi = -GM/r$:

$$g_{tt} \approx -\left(1 + \frac{2\Phi}{c^2}\right) = -\left(1 - \frac{2GM}{c^2 r}\right)$$

Comparing this with our derived form $g_{tt} = -B(r) = -(1 + C/r)$, we identify the constant C as:

$$C = -\frac{2GM}{c^2}$$

This gives us the final forms for $B(r)$ and $A(r)$:

$$B(r) = 1 - \frac{2GM}{c^2 r} \quad \text{and} \quad A(r) = \left(1 - \frac{2GM}{c^2 r}\right)^{-1}$$

Substituting these back into our metric ansatz, we arrive at the Schwarzschild metric:

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2(d\theta^2 + \sin^2 \theta d\phi^2)$$

A.5 Derivation of the Relativistic Poisson Equation

Here, we derive the relativistic form of the Poisson equation (Equation ??) from the Einstein Field Equations (2) in the weak-field, static limit.

The Weak-Field and Static Approximations In the weak-field limit, the metric is $g_{\mu\nu} = \eta_{\mu\nu} + h_{\mu\nu}$ with $|h_{\mu\nu}| \ll 1$. The Ricci tensor can be linearized in $h_{\mu\nu}$. For a static field (all time derivatives are zero, $\partial_0 = 0$), the 00-component of the linearized Ricci tensor simplifies to:

$$R_{00} \approx -\frac{1}{2}\nabla^2 h_{00}$$

where $\nabla^2 = \delta^{ij}\partial_i\partial_j$ is the spatial Laplacian.

The Stress-Energy Tensor In the rest frame, the four-velocity is $u^\mu = (c, 0, 0, 0)$, so $u_0 \approx -c$. The components of the stress-energy tensor T (1) are:

$$\begin{aligned} T_{00} &= (\rho + p/c^2)(-c)^2 + p(g_{00}) \approx \rho c^2 + p - p = \rho c^2 \\ T_{ij} &= pg_{ij} \approx p\delta_{ij} \end{aligned}$$

The trace of the stress-energy tensor is $T = g^{\mu\nu}T_{\mu\nu} \approx \eta^{\mu\nu}T_{\mu\nu} = -T_{00} + \delta^{ij}T_{ij} = -\rho c^2 + 3p$.

Assembling the Equation We now focus on the 00-component of Equation 2:

$$R_{00} = \frac{8\pi G}{c^4} \left(T_{00} - \frac{1}{2}g_{00}T \right)$$

Substitute the approximations for each term:

$$\begin{aligned} R_{00} &\approx -\frac{1}{2}\nabla^2 h_{00} \\ T_{00} &\approx \rho c^2 \\ g_{00} &= \eta_{00} + h_{00} = -1 + h_{00} \approx -1 \\ T &\approx -\rho c^2 + 3p \end{aligned}$$

Plugging these in gives:

$$-\frac{1}{2}\nabla^2 h_{00} \approx \frac{8\pi G}{c^4} \left(\rho c^2 - \frac{1}{2}(-1)(-\rho c^2 + 3p) \right) \quad (81)$$

After calculations, we derive:

$$\nabla^2 h_{00} \approx -\frac{8\pi G}{c^4}(\rho c^2 + 3p) = -\frac{8\pi G}{c^2} \left(\rho + \frac{3p}{c^2} \right) \quad (82)$$

Identifying the Newtonian Potential To connect h_{00} to the Newtonian potential Φ , we examine the geodesic equation 4 for a slow-moving test particle in the static, weak field. The equation for the spatial components is $\frac{d^2 x^i}{dt^2} \approx -c^2 \Gamma_{00}^i$. For a static field, $\Gamma_{00}^i \approx -\frac{1}{2}\partial_i h_{00}$. Thus:

$$\frac{d^2 x^i}{dt^2} \approx \frac{c^2}{2}\partial_i h_{00}$$

Comparing this to the Newtonian equation of motion, $\frac{d^2 x^i}{dt^2} = -\partial_i \Phi$, we make the identification:

$$\Phi = -\frac{c^2}{2}h_{00} \quad \implies \quad h_{00} = -\frac{2\Phi}{c^2}$$

Note: The sign convention for Φ can vary; this one leads to the standard attractive force.

Final Result Substitute this expression for h_{00} back into Equation 82 and we consider $c=1$ for simplification:

$$\nabla^2 \Phi = 4\pi G (\rho + 3p)$$

A.6 Derivation of the Perihelion Precession Formula

We derive the formula for the advance of the perihelion of an orbit in the Schwarzschild metric.

1. Conserved Quantities from Killing Vectors The Schwarzschild metric is static and spherically symmetric, which implies it is independent of the time coordinate t and the azimuthal angle ϕ . These symmetries correspond to two Killing vectors, $\xi_t = \partial_t$ and $\xi_\phi = \partial_\phi$. For a particle moving along a geodesic, the quantities $p_\mu \xi^\mu$ are conserved, where $p^\mu = m \frac{dx^\mu}{d\tau}$ is the four-momentum. This gives us two conserved quantities: energy and angular momentum.

- **Energy Conservation:**

$$p_t = g_{t\mu} p^\mu = g_{tt} \frac{dt}{d\tau} = -mc^2 \left(1 - \frac{2GM}{c^2 r} \right) \frac{dt}{d\tau} \equiv -E$$

where E is the conserved energy per unit mass.

- **Angular Momentum Conservation:**

$$p_\phi = g_{\phi\mu} p^\mu = g_{\phi\phi} \frac{d\phi}{d\tau} = mr^2 \sin^2 \theta \frac{d\phi}{d\tau} \equiv L_z$$

Due to spherical symmetry, we can simplify the problem by considering motion in the equatorial plane, $\theta = \pi/2$, so $\sin \theta = 1$. Thus, the conserved angular momentum per unit mass is:

$$L = r^2 \frac{d\phi}{d\tau}$$

2. The Radial Geodesic Equation For a massive particle, the four-velocity is normalized such that $g_{\mu\nu} \frac{dx^\mu}{d\tau} \frac{dx^\nu}{d\tau} = -c^2$. Writing this out using the Schwarzschild metric (5) in the equatorial plane:

$$-\left(1 - \frac{2GM}{c^2 r}\right) c^2 \left(\frac{dt}{d\tau}\right)^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \left(\frac{dr}{d\tau}\right)^2 + r^2 \left(\frac{d\phi}{d\tau}\right)^2 = -c^2$$

We can substitute our conserved quantities into this equation. From the conservation equations, we have:

$$\frac{dt}{d\tau} = \frac{E/c^2}{1 - \frac{2GM}{c^2 r}}, \quad \frac{d\phi}{d\tau} = \frac{L}{r^2}$$

Substituting these into the normalization condition gives the radial equation of motion:

$$-\frac{E^2/c^2}{1 - \frac{2GM}{c^2 r}} + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} \left(\frac{dr}{d\tau}\right)^2 + \frac{L^2}{r^2} = -c^2$$

Rearranging to solve for $\left(\frac{dr}{d\tau}\right)^2$:

$$\left(\frac{dr}{d\tau}\right)^2 = E^2 - c^2 \left(1 - \frac{2GM}{c^2 r}\right) \left(1 + \frac{L^2}{c^2 r^2}\right)$$

3. Relativistic Binet's Equation To find the shape of the orbit, $r(\phi)$, we change the independent variable from proper time τ to the angle ϕ . Using the chain rule:

$$\frac{dr}{d\tau} = \frac{dr}{d\phi} \frac{d\phi}{d\tau} = \frac{L}{r^2} \frac{dr}{d\phi}$$

Now, we make the standard substitution $u = 1/r$, which implies $\frac{dr}{d\phi} = -\frac{1}{u^2} \frac{du}{d\phi}$. So:

$$\frac{dr}{d\tau} = -L \frac{du}{d\phi}$$

Substituting this into the radial equation and dividing by L^2 :

$$\left(\frac{du}{d\phi}\right)^2 = \frac{E^2 - c^2}{L^2} + \frac{2GM}{L^2}u + \frac{2GM}{c^2}u^3 - u^2$$

To get the Binet equation, we differentiate with respect to ϕ :

$$2\frac{du}{d\phi}\frac{d^2u}{d\phi^2} = \frac{2GM}{L^2}\frac{du}{d\phi} - 2u\frac{du}{d\phi} + \frac{6GM}{c^2}u^2\frac{du}{d\phi}$$

Dividing by $2\frac{du}{d\phi}$ (assuming non-circular orbit), we get the relativistic Binet's equation:

$$\frac{d^2u}{d\phi^2} + u = \frac{GM}{L^2} + \frac{3GM}{c^2}u^2$$

The term $\frac{GM}{L^2}$ is the familiar Newtonian term, which gives closed elliptical orbits. The new term, $\frac{3GM}{c^2}u^2$, is the General Relativistic correction that causes the precession.

4. Perturbative Solution The relativistic correction is very small. We can solve this equation by treating the u^2 term as a perturbation. The unperturbed (Newtonian) solution is an ellipse:

$$u_N(\phi) = \frac{GM}{L^2}(1 + e \cos \phi)$$

where e is the eccentricity. We substitute this approximate solution into the small correction term on the right-hand side:

$$\frac{d^2u}{d\phi^2} + u \approx \frac{GM}{L^2} + \frac{3GM}{c^2} \left[\frac{GM}{L^2}(1 + e \cos \phi) \right]^2$$

Expanding the squared term and keeping only the non-oscillatory (constant) part that contributes to the precession (averaging over an orbit, $\langle \cos \phi \rangle = 0$, $\langle \cos^2 \phi \rangle = 1/2$), the dominant constant perturbation is from the $(GM/L^2)^2$ and $(GM/L^2)^2 e^2 \cos^2 \phi$ terms. A more direct approach is to look for a solution of the form $u(\phi) = \frac{GM}{L^2}(1 + e \cos(k\phi))$. Substituting this into the equation and linearizing for small perturbations shows that $k \approx 1 - \frac{3G^2M^2}{c^2L^2}$.

The orbit returns to the same radius when the argument of the cosine changes by 2π . This occurs when $k\phi = 2\pi$, or $\phi = 2\pi/k$.

$$\phi \approx \frac{2\pi}{1 - \frac{3G^2M^2}{c^2L^2}} \approx 2\pi \left(1 + \frac{3G^2M^2}{c^2L^2} \right)$$

The precession angle $\delta\phi$ is the excess angle over one complete orbit:

$$\delta\phi = \phi - 2\pi = 2\pi \left(\frac{3G^2M^2}{c^2L^2} \right) = \frac{6\pi G^2M^2}{c^2L^2}$$

A.7 Derivation of the Radial Function $A(r)$ in a symmetric 3D Space

To determine the form of the radial function $A(r)$ in the spatial part of the FLRW metric, we begin with the general spherically symmetric 3-dimensional line element:

$$ds^2 = A(r) dr^2 + r^2 (d\theta^2 + \sin^2 \theta d\phi^2).$$

By computing the Riemann tensor components associated with this metric, we find:

$$R^{r\theta}_{\theta r} = R^{r\phi}_{\phi r} = \frac{1}{2r} \cdot \frac{A'}{A^2}, \quad R^{\theta\phi}_{\phi\theta} = \frac{1}{r^2 A} (A - 1).$$

By the homogeneity's hypothesis, we assume that the curvature is the same in all directions so we fix a constant K such that:

$$R^{r\theta}_{\theta r} = R^{r\phi}_{\phi r} = R^{\theta\phi}_{\phi\theta} = K$$

By solving for the $R^{\theta\phi}_{\phi\theta}$ component we get :

$$\frac{1}{r^2 A} (A - 1) = K \quad \Rightarrow \quad A(r) = \frac{1}{1 - Kr^2}.$$

A.8 Derivation of Friedmann and Fluid Equations from Einstein's Equations

After computing the Christoffel symbols that derive from the FLRW metric (9) and using them to compute the Ricci tensor components, we find:

$$R_0^0 = \frac{3}{c^2} \frac{\ddot{a}}{a}, \quad R_1^1 = R_2^2 = R_3^3 = \frac{1}{c^2} \left(\frac{\ddot{a}}{a} + \frac{2\dot{a}^2}{a^2} + \frac{2kc^2}{a^2} \right).$$

The Ricci scalar is the trace $R = R^\mu{}_\mu$ and evaluates to:

$$R = \frac{6}{c^2} \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2 + kc^2}{a^2} \right).$$

We now compute the 00 component of the Einstein tensor (with $g_{00} = -c^2$):

$$G_{00} = R_{00} - \frac{1}{2}g_{00}R = -\frac{3\ddot{a}}{a} + \frac{1}{2}c^2 \cdot \frac{6}{c^2} \left(\frac{\ddot{a}}{a} + \frac{\dot{a}^2 + kc^2}{a^2} \right) = 3 \left(\frac{\dot{a}^2 + kc^2}{a^2} \right).$$

On the other side of Einstein's equations, the energy-momentum tensor $T_{\mu\nu}$ for a perfect fluid with $u^\mu = (1, 0, 0, 0)$ in comoving coordinates, gives:

$$T_{00} = \rho$$

Plugging into Einstein's equation, we get:

$$G_{00} + \Lambda g_{00} = \frac{8\pi G}{c^4} T_{00} \quad \Rightarrow \quad 3 \left(\frac{\dot{a}^2 + kc^2}{a^2} \right) - \Lambda c^2 = \frac{8\pi G}{c^4} \rho.$$

The constant k then appears to have units = $time^{-2}$ and by setting $c = 1$ we make time and length units interchangeable:

$$\left(\frac{\dot{a}}{a} \right)^2 = \frac{8\pi G}{3} \rho - \frac{k}{a^2} + \frac{\Lambda}{3}$$

The fluid equation can be derived directly from the conservation of the energy-momentum tensor by focus on the time component $\nu = 0$ of the conservation law:

$$\nabla_\mu T^{\mu 0} = 0$$

We compute this term-by-term using the identity:

$$\nabla_\mu T^{\mu 0} = \partial_\mu T^{\mu 0} + \Gamma_{\mu\lambda}^\mu T^{\lambda 0} + \Gamma_{\mu\lambda}^0 T^{\mu\lambda}$$

By evaluating each term:

- $\partial_\mu T^{\mu 0} = \partial_0 T^{00} = \dot{\rho}$
- ⁷ $\Gamma_{\mu 0}^\mu = \partial_0(\ln \sqrt{-g}) = 3\frac{\dot{a}}{a}$ since $\sqrt{-g} = a^3$ for the spatial part of the FLRW metric
- So $\Gamma_{\mu 0}^\mu T^{00} = 3\frac{\dot{a}}{a}\rho$
- For the last term, note that $T^{ij} = pg^{ij}$ and $\Gamma_{ij}^0 = \frac{\dot{a}}{a}g_{ij}$, so:

$$\Gamma_{ij}^0 T^{ij} = \Gamma_{ij}^0 g^{ij} p = \frac{\dot{a}}{a} g_{ij} g^{ij} p = 3\frac{\dot{a}}{a}p$$

Putting it all together:

$$\nabla_\mu T^{\mu 0} = \dot{\rho} + 3\frac{\dot{a}}{a}\rho + 3\frac{\dot{a}}{a}p = 0$$

Which gives the fluid equation:

$$\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0$$

We now derive the second Friedmann equation by combining the first Friedmann equation with the fluid equation. We aim to obtain an equation for the cosmic acceleration $\ddot{a}$. To do so, we take the time derivative of both sides of equation (10):

$$\frac{d}{dt} \left[\left(\frac{\dot{a}}{a} \right)^2 \right] = \frac{d}{dt} \left[\frac{8\pi G}{3} \rho - \frac{k}{a^2} + \frac{\Lambda}{3} \right]$$

⁷the identity $\Gamma_{\mu 0}^\mu = \partial_0(\ln \sqrt{-g})$ is up to the reader to be proven.

The left-hand side:

$$\frac{d}{dt} \left[\left(\frac{\dot{a}}{a} \right)^2 \right] = 2 \frac{\dot{a}}{a} \left(\frac{\ddot{a}}{a} - \left(\frac{\dot{a}}{a} \right)^2 \right)$$

The right-hand side:

$$\frac{8\pi G}{3} \dot{\rho} + \frac{2k}{a^3} \dot{a}$$

Now substitute $\dot{\rho}$ using the fluid equation (11) and equating both sides:

$$\dot{\rho} = -3 \frac{\dot{a}}{a} (\rho + p) \Rightarrow 2 \frac{\dot{a}}{a} \left(\frac{\ddot{a}}{a} - \left(\frac{\dot{a}}{a} \right)^2 \right) = -8\pi G \frac{\dot{a}}{a} (\rho + p) + \frac{2k}{a^3} \dot{a}$$

By reforming and substituting the equation (10) one more time, we obtain the second Friedmann equation(12):

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3} (\rho + 3p) + \frac{\Lambda}{3}$$

A.9 Derivation of the Hubble's law

We consider a photon emitted by a nearby galaxy at time t , received today at $t = t_0$. When the galaxy is close enough such that $t_0 - t \ll 1$, we can expand the scale factor around t_0 :

$$a(t) = a(t_0) [1 - H_0(t_0 - t) + \dots]$$

Assuming a flat FLRW (9) universe and radial light travel (so $ds^2 = 0$), the comoving distance is given by:

$$\chi = \int_t^{t_0} \frac{c dt'}{a(t')}$$

Inserting the linear expansion of $a(t)$, we get:

$$\chi \approx \int_t^{t_0} \frac{c dt'}{a(t_0) [1 - H_0(t_0 - t')]}$$

This evaluates to:

$$\chi \approx \frac{c}{a(t_0)} \left[(t_0 - t) + \frac{1}{2} H_0 (t_0 - t)^2 \right]$$

Now we relate time to redshift using the first-order approximation valid for small z :

$$z \approx H_0(t_0 - t)$$

Substituting into the previous expression for χ and keeping only the first-order term in z :

$$\chi \approx \frac{cz}{a(t_0)H_0}$$

Thus, the proper distance today is:

$$d = a(t_0)\chi \approx \frac{cz}{H_0}$$

If we interpret redshift as a Doppler shift from a recession velocity v , we use:

$$v = cz$$

Combining the two results, we obtain Hubble's law in the limit $z \ll 1$:

$$v \approx H_0 \cdot d$$

A.10 Derivation of $\rho(a) \propto a^{-3(1+w)}$

Substitute the equation of state (14) in the fluid equation (11):

$$\dot{\rho} + 3\frac{\dot{a}}{a}\rho(1+w) = 0$$

This is a first-order linear differential equation. Rearranging:

$$\frac{\dot{\rho}}{\rho} = -3(1+w)\frac{\dot{a}}{a}$$

Integrate both sides:

$$\int \frac{d\rho}{\rho} = -3(1+w) \int \frac{da}{a}$$

$$\ln \rho = -3(1+w) \ln a + C \Rightarrow \rho(a) = \rho_0 \cdot a^{-3(1+w)}$$

where ρ_0 is the integration constant.

A.11 Derivation of $\rho(z) = \rho_0 \exp \left[3 \int_0^z \frac{1+w(z')}{1+z'} dz' \right]$

From the fluid equation (11) and using $a = \frac{1}{1+z}$, we obtain:

$$\frac{d\rho}{\rho} = -3(1+w(z))\frac{da}{a} = 3(1+w(z))\frac{dz}{1+z}$$

Integrating from $z = 0$ (today) to a given redshift z :

$$\int_{\rho_0}^{\rho(z)} \frac{d\rho'}{\rho'} = 3 \int_0^z \frac{1+w(z')}{1+z'} dz'$$

$$\ln \frac{\rho(z)}{\rho_0} = 3 \int_0^z \frac{1+w(z')}{1+z'} dz' \Rightarrow \rho(z) = \rho_0 \exp \left[3 \int_0^z \frac{1+w(z')}{1+z'} dz' \right]$$

A.12 Derivation of the Scaling Factor $a(t)$ for Flat Universe with Matter and Cosmological Constant

Let us define a dimensionless variable:

$$y := \frac{a(t)}{a_m}.$$

Replace $a = a_m y \Rightarrow a^{-3} = a_m^{-3} y^{-3}$ and by construction we have $\frac{\Omega_{m,0}}{\Omega_{\Lambda,0}} a_m^{-3} = 1$. Thus, the Friedmann equation becomes:

$$\left(\frac{\dot{a}}{a} \right)^2 = H_0^2 \Omega_{\Lambda,0} (y^{-3} + 1) \Rightarrow \frac{\dot{y}}{y} = H_0 \sqrt{\Omega_{\Lambda,0}} \sqrt{1 + y^{-3}}$$

Separating variables gives:

$$\frac{dy}{y\sqrt{1+y^{-3}}} = H_0 \sqrt{\Omega_{\Lambda,0}} dt$$

To solve the integral on the left, we first simplify the expression:

$$\int \frac{dy}{y\sqrt{1+y^{-3}}} = \int \frac{dy}{y\sqrt{\frac{y^3+1}{y^3}}} = \int \frac{y^{1/2} dy}{\sqrt{y^3+1}}.$$

Let $u = y^{3/2}$, which implies $du = \frac{3}{2} y^{1/2} dy$. The integral becomes:

$$\int \frac{\frac{2}{3} du}{\sqrt{u^2+1}} = \frac{2}{3} \sinh^{-1}(u) = \frac{2}{3} \ln(u + \sqrt{u^2+1}).$$

Substituting back $u = y^{3/2}$, we integrate both sides of the differential equation to get:

$$\frac{2}{3} \ln(y^{3/2} + \sqrt{1+y^3}) = H_0 \sqrt{\Omega_{\Lambda,0}} t + C.$$

By setting $a(0) = 0$ (so $y(0) = 0$), we find the integration constant $C = 0$. Solving for $H_0 t$ and using $\Omega_{\Lambda,0} = 1 - \Omega_{m,0}$ for a flat universe, we arrive at the final result:

$$H_0 t = \frac{2}{3\sqrt{1-\Omega_{m,0}}} \ln(y^{3/2} + \sqrt{1+y^3}) \quad \text{where} \quad y = \frac{a(t)}{a_m}$$

Early-Time Behavior $a \ll a_m \Rightarrow y \ll 1$: In this limit, $\sqrt{1+y^3} \approx 1 + \frac{1}{2}y^3$, so:

$$y^{3/2} + \sqrt{1+y^3} \approx y^{3/2} + 1 + \frac{1}{2}y^3 \approx 1 + y^{3/2}.$$

Using the approximation $\ln(1+x) \approx x$ for small x , we get:

$$\ln(1+y^{3/2}) \approx y^{3/2}.$$

So:

$$H_0 t \approx \frac{2}{3\sqrt{1-\Omega_{m,0}}} y^{3/2} = \frac{2}{3\sqrt{1-\Omega_{m,0}}} \left(\frac{a}{a_m} \right)^{3/2}.$$

Solving for $a(t)$:

$$a(t) \approx a_m \left(\frac{3}{2} \sqrt{1-\Omega_{m,0}} H_0 t \right)^{2/3}.$$

Late-Time Behavior $a \gg a_m \Rightarrow y \gg 1$: In this case, $\sqrt{1+y^3} \approx y^{3/2}$, so:

$$\ln(y^{3/2} + \sqrt{1+y^3}) \approx \ln(2y^{3/2}) = \ln 2 + \frac{3}{2} \ln y.$$

Then:

$$H_0 t \approx \frac{2}{3\sqrt{1-\Omega_{m,0}}} \left(\ln 2 + \frac{3}{2} \ln y \right).$$

Solving for $\ln y$ gives $\ln y \approx \sqrt{1-\Omega_{m,0}} H_0 t + \text{const}$, which implies the exponential behavior:

$$y = \frac{a}{a_m} \propto \exp\left(\sqrt{1-\Omega_{m,0}} H_0 t\right).$$

So the final behavior is:

$$a(t) \propto \exp\left(\sqrt{1-\Omega_{m,0}} H_0 t\right).$$

A.13 Derivation of the Scaling Factor's $a(t)$ - density parameter $\Omega(t)$ relationship

From the Friedmann equation (10) with curvature:

$$1 = \frac{8\pi G\rho}{3H^2} - \frac{k}{a^2 H^2} = \Omega - \frac{k}{a^2 H^2} \Rightarrow \Omega = 1 + \frac{k}{a^2 H^2}$$

Differentiate with respect to $\ln a$:

$$\frac{d\Omega}{d\ln a} = \frac{d}{d\ln a} \left(\frac{k}{a^2 H^2} \right) = \frac{d}{da} \left(\frac{k}{a^2 H^2} \right) \cdot a$$

Using:

$$\frac{d}{da} \left(\frac{1}{a^2 H^2} \right) = \frac{-2}{a^3 H^2} - \frac{2\dot{H}}{a^2 H^3} \cdot \frac{dt}{da} \quad \text{and} \quad \frac{dt}{da} = \frac{1}{\dot{a}} = \frac{1}{aH}$$

By using also the acceleration equation (12) and the fact that $\dot{H} = \frac{\ddot{a}}{a} - H^2$:

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3}(\rho + 3p) = -\frac{4\pi G}{3}\rho(1+3w) \Rightarrow \dot{H} = -4\pi G(\rho + p) = -\frac{3}{2}(1+w)H^2$$

Then:

$$\frac{d}{d\ln a} \left(\frac{k}{a^2 H^2} \right) = -2\frac{k}{a^2 H^2} - 2\frac{\dot{H}}{H^2} \cdot \frac{k}{a^2 H^2} = -2(\Omega - 1) + 3(1+w)(\Omega - 1)$$

Combining, we get:

$$\frac{d\Omega}{d\ln a} = (1+3w)\Omega(\Omega - 1)$$

A.14 Derivation of the Luminosity Distance in a Flat Λ CDM Universe

Here we derive the standard formula for the luminosity distance $d_L(z)$ in a spatially flat universe dominated by matter and a cosmological constant, which is essential for analyzing supernova data.

Comoving Distance and the Friedmann Equation The first Friedmann equation (10) for a spatially flat universe ($k = 0$) containing only matter and a cosmological constant (the contribution from radiation is negligible at the redshifts relevant to SNIa) becomes:

$$H(z')^2 = H_0^2 [\Omega_{m,0}(1+z')^3 + \Omega_{\Lambda,0}]$$

Solving for the Hubble parameter gives:

$$H(z') = H_0 \sqrt{\Omega_{m,0}(1+z')^3 + \Omega_{\Lambda,0}}$$

Assembling the Luminosity Distance Substituting this expression for $H(z')$ into the integral for the comoving distance (18) yields:

$$d_{co}(z) = c \int_0^z \frac{dz'}{H(z')} = \frac{c}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_{m,0}(1+z')^3 + \Omega_{\Lambda,0}}}$$

The final step is to convert this to the luminosity distance, $d_L(z)$. As established in Eq. (21), the luminosity distance is related to the metric distance by $d_L = d_m(1+z)$. In a flat universe, the metric distance is equal to the comoving distance ($d_m = d_{co}$). Therefore, we arrive at the final formula :

$$d_L(z; \Omega_{M,0}, \Omega_{\Lambda,0}, H_0) = \frac{c(1+z)}{H_0} \int_0^z \frac{dz'}{\sqrt{\Omega_{M,0}(1+z')^3 + \Omega_{\Lambda,0}}}$$

A.15 Derivation of $\rho_m \propto n_m$ and of $\rho_\gamma \propto n_\gamma^{4/3}$

Here we provide the derivations for the scaling relationship between energy density (ρ) and number density (n) for both non-relativistic matter and relativistic species.

Non-Relativistic Matter ($\rho_m \propto n_m$). For non-relativistic particles, such as baryons or cold dark matter, their kinetic energy is negligible compared to their rest mass energy. The energy of a single particle is therefore dominated by its rest mass, $E_{\text{particle}} = mc^2$. The total energy density ρ_m is the product of the number density of these particles, n_m , and the energy per particle:

$$\rho_m = n_m \cdot E_{\text{particle}} = n_m(mc^2)$$

Since the rest mass m and the speed of light c are constants for a given species, the energy density is directly proportional to the number density:

$$\rho_m \propto n_m$$

Relativistic Species ($\rho_\gamma \propto n_\gamma^{4/3}$). For relativistic species like photons, the energy of each particle is not constant but changes as the universe expands. This leads to a different scaling relation, which can be derived in two ways. We begin with the first law of thermodynamics for an adiabatically expanding fluid, which states that the change in internal energy U is due to the work done by pressure P :

$$dU = -PdV$$

where $U = \rho_\gamma V$ is the total energy in a volume V . For a relativistic fluid (radiation with $w = \frac{1}{3}$), the equation of state is $P = \frac{1}{3}\rho_\gamma$. Substituting these relations into the first law gives:

$$d(\rho_\gamma V) = -\frac{1}{3}\rho_\gamma dV$$

Applying the product rule to the left-hand side yields:

$$V d\rho_\gamma + \rho_\gamma dV = -\frac{1}{3}\rho_\gamma dV$$

Rearranging the terms to separate the variables ρ_γ and V :

$$V d\rho_\gamma = -\frac{4}{3}\rho_\gamma dV \implies \frac{d\rho_\gamma}{\rho_\gamma} = -\frac{4}{3} \frac{dV}{V}$$

Integrating this expression shows how energy density scales with volume:

$$\ln(\rho_\gamma) = -\frac{4}{3}\ln(V) + \text{const} \implies \rho_\gamma \propto V^{-4/3}$$

Since the number of particles N_γ is conserved, the number density is $n_\gamma = N_\gamma/V$, which implies $V \propto n_\gamma^{-1}$. Substituting this into our result gives the final relation:

$$\rho_\gamma \propto (n_\gamma^{-1})^{-4/3} = n_\gamma^{4/3}$$

A.16 Derivation of $S_b \propto n_\gamma/n_b$

In the early, hot, and dense universe, the cosmic plasma was dominated by relativistic particles (radiation), primarily photons (γ) and neutrinos. The contribution of non-relativistic matter (baryons, b) to the total *entropy* was negligible. Therefore, the total entropy density, s , can be well approximated by the entropy density of the photon gas, s_γ .

$$s \approx s_\gamma$$

Entropy and Number Density of a Photon Gas For a gas of relativistic bosons like photons, standard statistical mechanics and thermodynamics give us the following relations for energy density (ρ_γ) and entropy density (s_γ):

$$\rho_\gamma = \frac{\pi^2}{15} T^4 \quad (\text{in natural units where } \hbar = c = k_B = 1) \quad (83)$$

The entropy density is related to the energy density and pressure (P_γ) by the thermodynamic relation $s_\gamma = (\rho_\gamma + P_\gamma)/T$. For a photon gas, $P_\gamma = \rho_\gamma/3$. Substituting this in, we get:

$$s_\gamma = \frac{\rho_\gamma + \frac{1}{3}\rho_\gamma}{T} = \frac{4}{3} \frac{\rho_\gamma}{T}$$

Substituting the expression for ρ_γ (83):

$$s_\gamma = \frac{4}{3} \frac{1}{T} \left(\frac{\pi^2}{15} T^4 \right) = \frac{4\pi^2}{45} T^3 \quad (84)$$

Now, let's find the photon *number density*, n_γ . This is also a standard result from statistical mechanics for a black-body spectrum:

$$n_\gamma = \frac{2\zeta(3)}{\pi^2} T^3 \quad (85)$$

where $\zeta(3) \approx 1.202$ is the Riemann zeta function ⁸ evaluated at 3.

Establishing the Proportionality From Equation (84) and Equation (85), we can see that both the entropy density and the number density of photons are proportional to the cube of the temperature: $s_\gamma \propto T^3$, $n_\gamma \propto T^3$. Because both quantities share the same dependence on temperature, they must be directly proportional to each other: $s_\gamma \propto n_\gamma$. We can even find the constant of proportionality: $s_\gamma = \left(\frac{2\pi^4}{45\zeta(3)} \right) n_\gamma \approx 3.602 n_\gamma$.

Defining Entropy per Baryon The quantity "entropy per baryon" is a very useful concept in cosmology. It is defined as the total entropy density, s , divided by the baryon number density, n_b .

$$S_b \equiv \frac{s}{n_b}$$

Using our approximation $s \approx s_\gamma$ and the proportionality $s_\gamma \propto n_\gamma$, we arrive at the final relationship:

$$S_b \approx \frac{s_\gamma}{n_b} \implies \mathbf{S_b} \propto \frac{\mathbf{n_\gamma}}{\mathbf{n_b}}$$

A.17 Derivation of the Generalized Poisson Equation

Here, we provide a detailed derivation of the generalized Poisson equation, which governs the evolution of the metric potentials in response to density perturbations. The derivation begins with the Einstein field equations and applies them to the perturbed FLRW metric in the Conformal Newtonian Gauge.

The Perturbed Einstein and Energy-Momentum Tensors Our starting point is the time-time component of the Einstein field equations, expressed in mixed-tensor form, which remains valid for the perturbed spacetime:

$$G_0^0 = 8\pi G T_0^0$$

⁸The **Riemann zeta function**, $\zeta(s)$, is a function defined for a complex variable s with a real part greater than 1 as the infinite sum $\zeta(s) = \sum_{n=1}^{\infty} 1/n^s$. It appears in statistical physics because the integrals that describe the properties of a boson gas, like black-body radiation, evaluate to specific values of this function. The integral for number density involves $\zeta(3) \approx 1.202$, also known as Apéry's constant, while the integral for energy density involves $\zeta(4) = \pi^4/90$.

We expand both the Einstein tensor G_0^0 and the energy-momentum tensor T_0^0 into a background part (denoted by a superscript (0)) and a first-order perturbation:

$$G_0^{0(0)} + \delta G_0^0 = 8\pi G(T_0^{0(0)} + \delta T_0^0)$$

The background terms satisfy the unperturbed Friedmann equation, $G_0^{0(0)} = 8\pi G T_0^{0(0)}$, which corresponds to:

$$-\frac{3\mathcal{H}^2}{a^2} = 8\pi G(-\rho^{(0)})$$

This background relation cancels from both sides, leaving a linear equation that relates the perturbations:

$$\delta G_0^0 = 8\pi G \delta T_0^0 \quad (86)$$

The Perturbed Energy-Momentum Source In the Conformal Newtonian Gauge, the coordinates are chosen to be comoving with the total cosmic fluid. In this frame, the time-time component of the energy-momentum tensor for a perfect fluid is simply the negative of the total energy density, $T_0^0 = -\rho_{\text{tot}}$. The first-order perturbation is therefore the fluctuation in the total energy density:

$$\delta T_0^0 = -\delta \rho_{\text{tot}} \quad (87)$$

The Perturbed Geometric Term The most involved part of the derivation is calculating the perturbed component of the Einstein tensor, δG_0^0 , from the perturbed metric. This requires computing the Christoffel symbols for the perturbed metric, using them to find the perturbed Ricci tensor δR_ν^μ , and finally assembling $\delta G_0^0 = \delta R_0^0 - \frac{1}{2}\delta_0^\alpha \delta R_\alpha$. This is a standard but lengthy calculation in cosmological perturbation theory. For the Conformal Newtonian Gauge with potentials Φ (Newtonian) and Ψ (curvature), the result is:

$$\delta G_0^0 = \frac{2}{a^2} [\nabla^2 \Psi - 3\mathcal{H}(\Psi' + \mathcal{H}\Phi)] \quad (88)$$

where a prime denotes differentiation with respect to conformal time τ , and ∇^2 is the spatial Laplacian.

Assembling the Final Equation We now substitute the expressions for the source term, Eq. (87), and the geometric term, Eq. (88), back into the perturbed Einstein equation (86):

$$\frac{2}{a^2} [\nabla^2 \Psi - 3\mathcal{H}(\Psi' + \mathcal{H}\Phi)] = 8\pi G(-\delta \rho_{\text{tot}})$$

To analyze the spatial dependence, it is convenient to perform a spatial Fourier transform, which replaces the Laplacian operator with its wavenumber, $\nabla^2 \rightarrow -k^2$. Applying this transformation and simplifying gives:

$$-k^2 \Psi - 3\mathcal{H}(\Psi' + \mathcal{H}\Phi) = -4\pi G a^2 \delta \rho_{\text{tot}}$$

Finally, multiplying the entire equation by -1 yields the generalized Poisson equation in Fourier space:

$$k^2 \Psi + 3\mathcal{H}(\Psi' + \mathcal{H}\Phi) = 4\pi G a^2 \delta \rho_{\text{tot}}$$

A.18 Derivation of the Anisotropic Stress Equation

This section provides a detailed derivation of the algebraic relation between the gravitational potentials and the anisotropic stress (cf. equation (32)). This equation arises from the traceless part of the spatial ($i - j$) components of the perturbed Einstein field equations.

The Perturbed Einstein Equations We begin with the spatial components of the Einstein equations, $G_{ij} = 8\pi G T_{ij}$. At the linear perturbation level, this becomes an equation for the first-order perturbations:

$$\delta G_{ij} = 8\pi G \delta T_{ij}$$

To isolate the physics of anisotropic stress, we focus on the traceless part of this tensor equation. We will work with the mixed-index form, $\delta G_j^i = 8\pi G \delta T_j^i$, as it simplifies the decomposition.

The Geometric Term: Perturbed Einstein Tensor The traceless part of the perturbed Einstein tensor, δG_j^i , must be calculated from the perturbed metric in the Conformal Newtonian Gauge. This is a standard result in cosmological perturbation theory. For the metric convention used in this document, where Φ is the Newtonian potential and Ψ is the curvature perturbation, the traceless part of the spatial Einstein tensor is:

$$\delta G_j^i(\text{traceless}) = \frac{1}{a^2} \left(\partial^i \partial_j - \frac{1}{3} \delta_j^i \nabla^2 \right) (\Phi - \Psi) \quad (89)$$

The operator $(\partial^i \partial_j - \frac{1}{3} \delta_j^i \nabla^2)$ is a projection operator that explicitly extracts the traceless component of the quantity it acts upon.

The Source Term: Anisotropic Stress The perturbed energy-momentum tensor, δT_j^i , can be decomposed into an isotropic pressure perturbation and a traceless anisotropic stress tensor, Π_j^i :

$$\delta T_j^i = \delta p \delta_j^i + \Pi_j^i$$

By definition, Π_j^i is the traceless part of the spatial stress perturbation, so:

$$\delta T_j^i(\text{traceless}) = \Pi_j^i \quad (90)$$

The anisotropic stress tensor Π_j^i describes momentum fluxes that are not aligned with the direction of momentum, such as those generated by free-streaming relativistic particles like neutrinos.

Assembling the Equation in Fourier Space Equating the traceless parts of the geometric and source terms gives:

$$\frac{1}{a^2} \left(\partial^i \partial_j - \frac{1}{3} \delta_j^i \nabla^2 \right) (\Phi - \Psi) = 8\pi G \Pi_j^i$$

This equation is most conveniently analyzed in Fourier space. The differential operator becomes an algebraic operator involving the wavevector $\mathbf{k}$:

$$\mathcal{F} \left[\partial^i \partial_j - \frac{1}{3} \delta_j^i \nabla^2 \right] \rightarrow (-k^i k_j + \frac{1}{3} \delta_j^i k^2) = -k^2 \left(\hat{k}^i \hat{k}_j - \frac{1}{3} \delta_j^i \right)$$

where $\hat{k}^i = k^i/k$ is the unit wavevector. The Fourier-transformed equation is:

$$-\frac{k^2}{a^2} \left(\hat{k}^i \hat{k}_j - \frac{1}{3} \delta_j^i \right) (\tilde{\Phi} - \tilde{\Psi}) = 8\pi G \tilde{\Pi}_j^i$$

The anisotropic stress tensor $\tilde{\Pi}_j^i$ can also be decomposed using the same projection operator, which defines the scalar anisotropic stress, σ :

$$\tilde{\Pi}_j^i \equiv (\rho_{\text{tot}} + p_{\text{tot}}) \left(\hat{k}^i \hat{k}_j - \frac{1}{3} \delta_j^i \right) \sigma$$

Substituting this definition into our Fourier-space equation, we have:

$$-\frac{k^2}{a^2} \left(\hat{k}^i \hat{k}_j - \frac{1}{3} \delta_j^i \right) (\tilde{\Phi} - \tilde{\Psi}) = 8\pi G (\rho_{\text{tot}} + p_{\text{tot}}) \left(\hat{k}^i \hat{k}_j - \frac{1}{3} \delta_j^i \right) \sigma$$

The tensor projection operator $(\hat{k}^i \hat{k}_j - \frac{1}{3} \delta_j^i)$ is common to both sides and can be cancelled. This leaves a simple algebraic relation between the scalar quantities. After multiplying by $-a^2/k^2$, we arrive at the final result:

$$\Phi - \Psi = \frac{8\pi G a^2}{k^2} (\rho_{\text{tot}} + p_{\text{tot}}) \sigma$$

A.19 Derivation of the Linearized Boltzmann Equation for Photons

Here, we provide the derivation for the linearized Boltzmann equation for photons, which describes the evolution of the temperature fluctuation Θ in the presence of gravitational perturbations.

The Boltzmann Equation and the Metric The fundamental starting point is the relativistic Boltzmann equation, which states that the total derivative of the photon distribution function f with respect to conformal time τ is equal to the collision term $C[f]$:

$$\frac{df}{d\tau} = \frac{\partial f}{\partial \tau} + \frac{dx^i}{d\tau} \frac{\partial f}{\partial x^i} + \frac{dq}{d\tau} \frac{\partial f}{\partial q} = C[f]$$

We neglect the term involving $dn_i/d\tau$ as its contribution is of second order. We work in the **Conformal Newtonian Gauge**, where the line element for a spatially flat universe is given by (30)

The Geodesic Equation and Photon Momentum Evolution The crucial term in the Boltzmann equation is $\frac{dq}{d\tau}$, which describes how a photon's comoving momentum q changes as it travels through the perturbed spacetime. The change in the physical energy p is the sum of the zeroth-order cosmological redshift and first-order effects from the gravitational potentials.

1. Cosmological Redshift from Expansion This is a zeroth-order effect present even in a perfectly homogeneous and isotropic universe. It arises directly from the expansion of space.

- A photon's physical energy is inversely proportional to its wavelength: $p \propto 1/\lambda$.
- As the universe expands, the wavelength is stretched in direct proportion to the scale factor: $\lambda(\tau) \propto a(\tau)$.
- Therefore, the photon's energy must decrease as the universe expands: $p(\tau) \propto a(\tau)^{-1}$.

Taking the logarithmic derivative with respect to conformal time τ (where a prime denotes $d/d\tau$) gives the energy loss rate from expansion:

$$\left(\frac{d \ln p}{d\tau}\right)_{\text{Expansion}} = \frac{d}{d\tau}(-\ln a) = -\frac{1}{a} \frac{da}{d\tau} = -\frac{a'}{a} = -\mathcal{H}$$

This is the standard cosmological redshift, where $\mathcal{H}$ is the conformal Hubble parameter.

2. Gravitational Potential Effects In an inhomogeneous universe, a photon's energy is also affected by its passage through gravitational potentials. The full derivation for this contribution comes from a detailed analysis of the geodesic equation in the perturbed metric.

A **comoving observer** is at rest with respect to the spatial coordinates, so their four-velocity U^μ has only a time component. From the normalization condition $g_{\mu\nu}U^\mu U^\nu = -1$, we find to first order:

$$-a^2(1 + 2\Psi)(U^0)^2 = -1 \implies U^0 = \frac{1}{a\sqrt{1 + 2\Psi}} \approx \frac{1 - \Psi}{a}$$

A photon travels along a null geodesic, $ds^2 = 0$, with four-momentum P^μ . The **physical energy** of the photon as measured by our comoving observer is $p \equiv -P_\mu U^\mu$. A detailed calculation involving the geodesic equation for the photon's four-momentum in the perturbed metric shows that the total rate of change of this physical energy is given by:

$$\frac{d \ln p}{d\tau} = -\mathcal{H} + n^i \frac{\partial \Phi}{\partial x^i} - \frac{\partial \Psi}{\partial \tau} \quad (91)$$

where n^i is a unit vector in the direction of photon propagation. The term $-\mathcal{H}$ is the standard redshift from expansion. The other two terms arise from the perturbations: $n^i \partial_i \Phi$ from the gravitational redshift (Sachs-Wolfe effect) as the photon moves through the Newtonian potential Φ , and $-\Psi'$ from the effect of a time-varying spatial curvature Ψ (a contribution to the Integrated Sachs-Wolfe effect).

Evolution of Comoving Momentum For analyzing cosmological perturbations, the quantity of interest is the photon's **comoving momentum**, q , which is related to its physical energy p by the scale factor: $q = ap$. The comoving momentum is constant in a perfectly homogeneous universe. Its change in a perturbed universe isolates the effect of the potentials. We find the evolution of q using the product rule and our previous result:

$$\frac{d \ln q}{d\tau} = \frac{d}{d\tau}(\ln(ap)) = \frac{a'}{a} + \frac{d \ln p}{d\tau} = \mathcal{H} + \left(-\mathcal{H} + n^i \frac{\partial \Phi}{\partial x^i} - \frac{\partial \Psi}{\partial \tau}\right)$$

The background expansion terms cancel, leaving the change due to the potentials:

$$\frac{d \ln q}{d\tau} = n^i \frac{\partial \Phi}{\partial x^i} - \frac{\partial \Psi}{\partial \tau}$$

This gives us the momentum change term needed for the full Boltzmann equation:

$$\frac{dq}{d\tau} = q \left(n^i \frac{\partial \Phi}{\partial x^i} - \frac{\partial \Psi}{\partial \tau} \right)$$

Linearization We now linearize the Boltzmann equation. We write the distribution function as a small perturbation Θ_p around the homogeneous and isotropic background distribution $f_0(q)$:

$$f(\mathbf{x}, q, \mathbf{n}, \tau) = f_0(q) [1 + \Theta_p(\mathbf{x}, q, \mathbf{n}, \tau)]$$

Substituting this into the Boltzmann equation, keeping only first-order terms, and using the geodesic result for $dq/d\tau$, we obtain:

$$\frac{\partial \Theta_p}{\partial \tau} + n^i \frac{\partial \Theta_p}{\partial x^i} + \left(n^i \frac{\partial \Phi}{\partial x^i} - \frac{\partial \Psi}{\partial \tau} \right) \frac{d \ln f_0}{d \ln q} = \frac{C[f]}{f_0}$$

It is convenient to work in Fourier space, where $\partial/\partial x^i \rightarrow ik_i$. Defining the cosine of the angle between the wavevector and the photon direction as $\mu \equiv \hat{k} \cdot \hat{n}$, the equation becomes:

$$\frac{\partial \Theta_p}{\partial \tau} + ik\mu\Theta_p + \left(ik\mu\Phi - \frac{\partial \Psi}{\partial \tau} \right) \frac{d \ln f_0}{d \ln q} = \frac{C[f]}{f_0}$$

Relating to Temperature Fluctuation The perturbation to the distribution function, Θ_p , is directly related to the observable photon temperature fluctuation, $\Theta \equiv \delta T/T$. A local temperature fluctuation modifies the argument of the background Bose-Einstein distribution: $f = f_0(q/(1 + \Theta))$. Expanding this to first order in Θ gives:

$$f \approx f_0(q) - \Theta q \frac{\partial f_0}{\partial q} = f_0(q) \left(1 - \Theta \frac{d \ln f_0}{d \ln q} \right)$$

Comparing this with our definition of Θ_p , we find the relation:

$$\Theta_p = -\Theta \left(\frac{d \ln f_0}{d \ln q} \right)$$

We substitute this expression for Θ_p back into the linearized Boltzmann equation:

$$-\left(\frac{d \ln f_0}{d \ln q} \right) \frac{\partial \Theta}{\partial \tau} - ik\mu\Theta \left(\frac{d \ln f_0}{d \ln q} \right) + \left(ik\mu\Phi - \frac{\partial \Psi}{\partial \tau} \right) \frac{d \ln f_0}{d \ln q} = \frac{C[\Theta]}{f_0}$$

The term $(d \ln f_0/d \ln q)$ is a non-zero constant for the relativistic photon gas and can be divided out from the entire equation. Multiplying by -1 and rearranging the terms, we arrive at the final form:

$$\frac{\partial \Theta}{\partial \tau} + ik\mu\Theta + \frac{\partial \Psi}{\partial \tau} - ik\mu\Phi = C[\Theta]$$

A.20 Derivation of the Klein-Gordon Equation for a Homogeneous Scalar Field

The dynamics of a homogeneous scalar field in a cosmological context can be derived from the principle of energy conservation. For a perfect fluid in an FLRW universe, this is expressed by the fluid equation (11): $\dot{\rho} + 3\frac{\dot{a}}{a}(\rho + p) = 0$. A homogeneous scalar field, $\phi(t)$, can be treated as such a perfect fluid, with an effective energy density ρ_ϕ and pressure p_ϕ given by Eq. (35), (36). To find the equation of motion for ϕ , we substitute these definitions into the

First, we calculate the time derivative of the energy density, $\dot{\rho}_\phi$. Using the chain rule for the potential term:

$$\begin{aligned} \dot{\rho}_\phi &= \frac{d}{dt} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right) \\ &= \frac{1}{2} (2\dot{\phi}\ddot{\phi}) + \frac{dV}{d\phi} \frac{d\phi}{dt} \\ &= \dot{\phi}\ddot{\phi} + V'(\phi)\dot{\phi} \end{aligned}$$

where $V'(\phi) \equiv dV/d\phi$.

Next, we find the term $(\rho_\phi + p_\phi)$:

$$\begin{aligned} \rho_\phi + p_\phi &= \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right) + \left(\frac{1}{2} \dot{\phi}^2 - V(\phi) \right) \\ &= \dot{\phi}^2 \end{aligned}$$

Now, we substitute these two results back into the fluid equation:

$$(\dot{\phi}\ddot{\phi} + V'(\phi)\dot{\phi}) + 3H(\dot{\phi}^2) = 0$$

Assuming the field is dynamical (i.e., $\dot{\phi} \neq 0$), we can divide the entire equation by $\dot{\phi}$:

$$\ddot{\phi} + V'(\phi) + 3H\dot{\phi} = 0$$

Rearranging the terms yields the Klein-Gordon equation for a homogeneous scalar field in an expanding universe:

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0$$

A.21 Derivation of the Slow-Roll Equations of Motion

We start with the equations governing a homogeneous scalar field $\phi(t)$ in a flat FRW universe which are the Klein-Gordon equation (37) combined with the Friedmann equation :

$$H^2 = \frac{1}{3m_{Pl}^2} \left(\frac{1}{2}\dot{\phi}^2 + V(\phi) \right) \quad (92)$$

where $m_{Pl}^2 = (8\pi G)^{-1}$ and a prime on V denotes $d/d\phi$.

The slow-roll approximation consists of two conditions. First, the kinetic energy of the field is assumed to be much smaller than its potential energy:

$$\frac{1}{2}\dot{\phi}^2 \ll V(\phi) \quad (93)$$

Applying this condition to the Friedmann equation (92), we immediately obtain the first slow-roll equation of motion:

$$H^2 \approx \frac{V(\phi)}{3m_{Pl}^2} \quad (94)$$

The second slow-roll condition is that the acceleration of the field is negligible compared to the Hubble friction term:

$$|\ddot{\phi}| \ll |3H\dot{\phi}| \quad (95)$$

Applying this condition to the Klein-Gordon equation (37), we obtain the second slow-roll equation of motion:

$$3H\dot{\phi} \approx -V'(\phi) \quad (96)$$

These two equations, (94) and (96), form the basis of the slow-roll dynamics. We can self-consistently check the conditions by defining the slow-roll parameters. Differentiating Eq. (94) with respect to time gives $2H\dot{H} \approx V'\dot{\phi}/(3m_{Pl}^2)$. Using Eq. (96) to eliminate V' , we find $\dot{H} \approx -\dot{\phi}^2/(2m_{Pl}^2)$. The first slow-roll parameter is defined as:

$$\epsilon \equiv -\frac{\dot{H}}{H^2} \approx \frac{\dot{\phi}^2}{2m_{Pl}^2 H^2} = \frac{3m_{Pl}^2 \dot{\phi}^2}{2V} = \frac{m_{Pl}^2}{2} \left(\frac{V'}{V} \right)^2$$

The condition $\epsilon \ll 1$ is equivalent to the first slow-roll condition (93). The second parameter is:

$$\eta \equiv \epsilon - \frac{1}{2H} \frac{d \ln \epsilon}{dt} \approx m_{Pl}^2 \frac{V''}{V}$$

The condition $|\eta| \ll 1$ ensures that the acceleration $\ddot{\phi}$ is small, satisfying the second condition (95).

A.22 Derivation of Quantum Fluctuations in de Sitter Space

We wish to calculate the variance of a massless scalar field in a pure de Sitter background, where $a(t) = e^{Ht}$ with H being constant. It is most convenient to work in conformal time, $\eta = \int dt/a = -1/(aH)$. The line element is $ds^2 = a^2(\eta)(d\eta^2 - d\mathbf{x}^2) = (-H\eta)^{-2}(d\eta^2 - d\mathbf{x}^2)$.

The action for a massless scalar field ϕ is $S = \int d^4x \sqrt{-g} (-\frac{1}{2}g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi)$. It is convenient to define a rescaled field $\chi \equiv a\phi$. The action for χ becomes:

$$S = \frac{1}{2} \int d\eta d^3x \left((\chi')^2 - (\nabla \chi)^2 + \frac{a''}{a} \chi^2 \right)$$

where a prime denotes $d/d\eta$. In de Sitter space, $a''/a = 2/\eta^2$. The equation of motion for a Fourier mode χ_k is:

$$\chi_k'' + \left(k^2 - \frac{2}{\eta^2} \right) \chi_k = 0$$

This is the equation for a quantum harmonic oscillator with a time-dependent frequency. To quantize the field, we promote χ_k to an operator $\hat{\chi}_k(\eta) = v_k(\eta)\hat{a}_k + v_k^*(\eta)\hat{a}_{-k}^\dagger$, where $\hat{a}_k$ and $\hat{a}_k^\dagger$ are annihilation and creation operators. The mode function $v_k(\eta)$ must satisfy the equation of motion.

In the early universe, at large negative η , the modes are deep inside the Hubble radius ($k|\eta| \gg 1$). In this limit, the k^2 term dominates and the equation becomes $\chi_k'' + k^2 \chi_k \approx 0$. The solution should match the standard flat-space vacuum mode functions. The correctly normalized positive frequency solution is:

$$v_k(\eta) = \frac{e^{-ik\eta}}{\sqrt{2k}} \quad (k|\eta| \rightarrow \infty)$$

The exact solution to the full equation that matches this asymptotic form is:

$$v_k(\eta) = \frac{e^{-ik\eta}}{\sqrt{2k}} \left(1 - \frac{i}{k\eta} \right)$$

We are interested in the variance of the original field $\phi = \chi/a$. The two-point correlation function in Fourier space is defined by $\langle \hat{\phi}_{\mathbf{k}} \hat{\phi}_{\mathbf{k}'} \rangle = (2\pi)^3 \delta^{(3)}(\mathbf{k} + \mathbf{k}') P_\phi(k)$. The power spectrum is given by $P_\phi(k) = |v_k|^2/a^2$.

$$P_\phi(k) = \frac{1}{a^2} \frac{1}{2k} \left(1 + \frac{1}{k^2 \eta^2} \right) = \frac{H^2 \eta^2}{2k} \left(1 + \frac{1}{k^2 \eta^2} \right)$$

We evaluate this spectrum on super-horizon scales, i.e., in the limit $k|\eta| \ll 1$. In this limit, the $1/(k^2 \eta^2)$ term dominates:

$$P_\phi(k) \approx \frac{H^2}{2k^3} \quad (k|\eta| \rightarrow 0)$$

The dimensionless power spectrum is defined as $\Delta_\phi^2(k) = \frac{k^3}{2\pi^2} P_\phi(k)$. Substituting our result, we find the famous scale-invariant spectrum for a massless scalar field in de Sitter space:

$$\Delta_\phi^2(k) = \frac{H^2}{4\pi^2}$$

The root-mean-square fluctuation on a given scale is therefore $\langle (\delta\phi)^2 \rangle^{1/2} \approx H/(2\pi)$.

A.23 Derivation of the Scalar Perturbation Power Spectrum

A rigorous derivation requires the gauge-invariant formalism for cosmological perturbations. The action for scalar perturbations can be reduced to the action of a single gauge-invariant field, the Mukhanov-Sasaki variable v , which is related to the comoving curvature perturbation ζ by $v = z\zeta$ where $z \equiv a\sqrt{2\epsilon_V}/(c_s H)$. In the slow-roll limit, $c_s \approx 1$ and $z \approx a\dot{\phi}/H$. The action for v is:

$$S = \frac{1}{2} \int d\eta d^3x \left((v')^2 - c_s^2 (\nabla v)^2 + \frac{z''}{z} v^2 \right)$$

The equation of motion for a Fourier mode v_k is:

$$v_k'' + \left(c_s^2 k^2 - \frac{z''}{z} \right) v_k = 0$$

This equation is analogous to the one for the rescaled inflaton fluctuation χ_k in the previous section. The term z''/z can be evaluated in the slow-roll limit. To lowest order in slow-roll parameters, $a \approx -1/(H\eta)$ and $z \propto a \propto 1/|\eta|$. This gives $z''/z \approx 2/\eta^2$. The equation becomes identical to the one for a massless field in de Sitter space.

The quantization proceeds identically. The mode functions are $v_k(\eta) = \frac{e^{-ik\eta}}{\sqrt{2k}} (1 - \frac{i}{k\eta})$, and the power spectrum for v on super-horizon scales is $P_v(k) = |v_k|^2 \approx 1/(2k^3 \eta^2)$. The power spectrum for the curvature perturbation ζ is then:

$$P_\zeta(k) = \frac{P_v(k)}{z^2} \approx \frac{1}{z^2} \frac{1}{2k^3 \eta^2}$$

We evaluate this at horizon crossing, $k = aH = 1/|\eta|$. At this time, $z^2 \approx (a\dot{\phi}/H)^2 = 2\epsilon_V m_{Pl}^2 a^2$.

$$P_\zeta(k) \approx \frac{1}{2\epsilon_V m_{Pl}^2 a^2} \frac{1}{2k^3} (aH)^2 = \frac{H^2}{4k^3 \epsilon_V m_{Pl}^2}$$

The dimensionless power spectrum is:

$$\Delta_\zeta^2(k) = \frac{k^3}{2\pi^2} P_\zeta(k) = \frac{H^2}{8\pi^2 \epsilon_V m_{Pl}^2} = \frac{1}{24\pi^2 m_{Pl}^4} \frac{V}{\epsilon_V} \Big|_{k=aH}$$

This gives the amplitude of the scalar perturbations.

A.24 Derivation of the Tensor Perturbation Power Spectrum

Tensor perturbations are described by the transverse-traceless part of the metric perturbation, h_{ij} . The action for tensor perturbations, expanded to second order, is:

$$S_h = \frac{m_{Pl}^2}{8} \int d\eta d^3x a^2(\eta) ((h'_{ij})^2 - (\nabla h_{ij})^2)$$

We can decompose h_{ij} into two polarization states, h_+ and $h_\times$. For each polarization, we can define a canonically normalized field, for example $\tilde{h} \equiv am_{Pl}h_+/\sqrt{2}$. The action for $\tilde{h}$ becomes:

$$S_{\tilde{h}} = \frac{1}{2} \int d\eta d^3x \left((\tilde{h}')^2 - (\nabla \tilde{h})^2 + \frac{a''}{a} \tilde{h}^2 \right)$$

This action is identical in form to that for the rescaled scalar field fluctuation χ in a de Sitter background. The quantization and calculation of the power spectrum are therefore identical. The dimensionless power spectrum for one polarization state $\tilde{h}$ is $\Delta_{\tilde{h}}^2 = (H/2\pi)^2$. To get the total tensor power spectrum, we sum over the two polarizations and convert back to the original metric perturbation h .

$$\Delta_h^2(k) = 2 \times \frac{2}{a^2 m_{Pl}^2} \Delta_{\tilde{h}}^2 = \frac{4}{a^2 m_{Pl}^2} \left(\frac{H}{2\pi} \right)^2 = \frac{H^2}{\pi^2 m_{Pl}^2}$$

In the slow-roll approximation, $H^2 \approx V/(3m_{Pl}^2)$, so the amplitude is:

$$\Delta_h^2(k) = \frac{2V}{3\pi^2 m_{Pl}^4} \Big|_{k=aH}$$

A.25 Derivation of the Brans-Dicke Field Equations

We start from the Brans-Dicke action (40), presented here in a form consistent with the final field equations, where the matter Lagrangian $\mathcal{L}_m$ is appropriately scaled by 16π :

$$S = \int d^4x \sqrt{-g} \left[\phi R - \frac{\omega}{\phi} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + 16\pi \mathcal{L}_m \right]$$

The principle of stationary action states implies that $\delta S = 0$. We will vary the action with respect to the inverse metric $g^{\mu\nu}$, as it simplifies the calculations. The total variation can be split into three parts:

$$\delta S = \delta S_g + \delta S_\phi + \delta S_m = 0$$

where S_g is the gravitational part (ϕR), S_ϕ is the scalar kinetic part, and S_m is the matter part.

1. Variation of the Matter Term (δS_m) The energy-momentum tensor is defined by the variation of the matter action with respect to the metric:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta(\sqrt{-g} \mathcal{L}_m)}{\delta g^{\mu\nu}}$$

By using the definition of the variation of the matter action and the mathematical identity (77), we find that:

$$\delta S_m = \delta \int d^4x \sqrt{-g} (16\pi \mathcal{L}_m) = \int d^4x (16\pi) \delta(\sqrt{-g} \mathcal{L}_m) = \int d^4x (16\pi) \left(-\frac{1}{2} \sqrt{-g} T_{\mu\nu} \right) \delta g^{\mu\nu}$$

$$\delta S_m = \int d^4x \sqrt{-g} (-8\pi T_{\mu\nu}) \delta g^{\mu\nu}$$

2. Variation of the Scalar Field Kinetic Term (δS_ϕ) This term does not involve derivatives of the metric, so its variation is straightforward. We vary $g^{\mu\nu}$ while keeping ϕ and its derivatives fixed:

$$\delta S_\phi = \delta \int d^4x \sqrt{-g} \left(-\frac{\omega}{\phi} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi \right)$$

We need to vary both $\sqrt{-g}$ and $g^{\mu\nu}$ and we get:

$$\begin{aligned}\delta S_\phi &= \int d^4x \left[\delta(\sqrt{-g}) \left(-\frac{\omega}{\phi} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi \right) + \sqrt{-g} \left(-\frac{\omega}{\phi} \partial_\mu \phi \partial_\nu \phi \right) \delta g^{\mu\nu} \right] \\ &= \int d^4x \sqrt{-g} \left[\left(-\frac{1}{2} g_{\mu\nu} \right) \left(-\frac{\omega}{\phi} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi \right) - \frac{\omega}{\phi} \partial_\mu \phi \partial_\nu \phi \right] \delta g^{\mu\nu} \\ &= \int d^4x \sqrt{-g} \left[-\frac{\omega}{\phi} \left(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} \partial_\lambda \phi \partial^\lambda \phi \right) \right] \delta g^{\mu\nu}\end{aligned}$$

Note that the term in the square brackets is precisely the energy-momentum tensor of the scalar field, $T_{\mu\nu}^{(\phi)}$, scaled by $-\omega/\phi$.

3. Variation of the Gravitational Term (δS_g) This is the most involved part. We start with $S_g = \int d^4x \sqrt{-g} \phi R$.

$$\delta S_g = \int d^4x \delta(\sqrt{-g} \phi R) = \int d^4x [\delta(\sqrt{-g}) \phi R + \sqrt{-g} \phi \delta R]$$

Since we are varying with respect to $g^{\mu\nu}$, ϕ is held constant, so $\delta(\phi R) = \phi \delta R$.

$$\delta S_g = \int d^4x \left[\left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) \phi R + \sqrt{-g} \phi \delta R \right]$$

We now use the identity $\delta R = \delta(g^{\mu\nu} R_{\mu\nu}) = R_{\mu\nu} \delta g^{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}$:

$$\delta S_g = \int d^4x \sqrt{-g} \left[\phi \left(R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R \right) \delta g^{\mu\nu} + \phi g^{\mu\nu} \delta R_{\mu\nu} \right] \quad (97)$$

The first term contains the Einstein tensor $G_{\mu\nu}$. The second term, involving $\delta R_{\mu\nu}$, contains second derivatives of the metric variation $\delta g_{\mu\nu}$. It is a standard result from the derivation of the Einstein-Hilbert action A.2 that this term can be expressed as a total divergence. We handle this term by integrating by parts twice.

$$\begin{aligned}\int d^4x \sqrt{-g} \phi (g^{\mu\nu} \delta R_{\mu\nu}) &= \int d^4x \sqrt{-g} \phi (\nabla_\mu \nabla_\nu \delta g^{\mu\nu} - g^{\alpha\beta} \nabla_\alpha \nabla_\beta (\delta g)) \\ (\text{Integrate by parts once}) &= - \int d^4x \sqrt{-g} (\nabla_\mu \phi) (\nabla_\nu \delta g^{\mu\nu} - \nabla^\mu \delta g) \\ (\text{Integrate by parts again}) &= \int d^4x \sqrt{-g} (\nabla_\nu \nabla_\mu \phi) \delta g^{\mu\nu} - \int d^4x \sqrt{-g} (\nabla_\mu \nabla^\mu \phi) \delta g \\ &= \int d^4x \sqrt{-g} [(\nabla_\mu \nabla_\nu \phi) \delta g^{\mu\nu} - (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) g_{\mu\nu} \delta g^{\mu\nu}] \\ &= \int d^4x \sqrt{-g} (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)) \delta g^{\mu\nu}\end{aligned}$$

Here, boundary terms are discarded, as is standard. Substituting this back into Eq. (97), we get the full variation of the gravitational part:

$$\delta S_g = \int d^4x \sqrt{-g} [\phi G_{\mu\nu} + (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi))] \delta g^{\mu\nu}$$

4. Assembling the Field Equation Now we combine all the variations and set the total variation to zero:

$$\delta S = \int d^4x \sqrt{-g} \left[\phi G_{\mu\nu} + (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)) - \frac{\omega}{\phi} \left(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} \partial_\lambda \phi \partial^\lambda \phi \right) - 8\pi T_{\mu\nu} \right] \delta g^{\mu\nu} = 0$$

Since this must hold for any arbitrary variation $\delta g^{\mu\nu}$, the entire expression in the square brackets must be zero.

$$\phi G_{\mu\nu} + (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)) - \frac{\omega}{\phi} \left(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} \partial_\lambda \phi \partial^\lambda \phi \right) - 8\pi T_{\mu\nu} = 0$$

Rearranging to solve for $G_{\mu\nu}$ and dividing by ϕ , we arrive at the Brans-Dicke gravitational field equation (41):

$$G_{\mu\nu} = \frac{8\pi}{\phi} T_{\mu\nu} + \frac{\omega}{\phi^2} \left(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} \partial_\lambda \phi \partial^\lambda \phi \right) + \frac{1}{\phi} (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi))$$

A.26 Derivation of the Brans-Dicke Scalar Field Equation

The equation of motion for the scalar field ϕ is obtained by varying the Brans-Dicke action with respect to ϕ and applying the principle of stationary action, $\delta S/\delta\phi = 0$.

We begin again with the action:

$$S = \int d^4x \sqrt{-g} \left[\phi R - \frac{\omega}{\phi} g^{\mu\nu} \partial_\mu \phi \partial_\nu \phi + 16\pi \mathcal{L}_m \right]$$

The Lagrangian density is $\mathcal{L} = \sqrt{-g} \left(\phi R - \frac{\omega}{\phi} (\partial\phi)^2 + 16\pi \mathcal{L}_m \right)$. The Euler-Lagrange equation (3) for the field ϕ is:

$$\frac{\partial \mathcal{L}}{\partial \phi} - \partial_\mu \left(\frac{\partial \mathcal{L}}{\partial (\partial_\mu \phi)} \right) = 0$$

In curved spacetime, this correctly gives rise to the covariant equation by using the definition $\nabla_\mu \mathcal{L} = \frac{1}{\sqrt{-g}} \partial_\mu (\sqrt{-g} \mathcal{L})$:

$$\frac{1}{\sqrt{-g}} \frac{\partial (\sqrt{-g} L)}{\partial \phi} - \nabla_\mu \left(\frac{1}{\sqrt{-g}} \frac{\partial (\sqrt{-g} L)}{\partial (\partial_\mu \phi)} \right) = 0$$

where $L = \frac{\mathcal{L}}{\sqrt{-g}}$ is the Lagrangian part inside the integral.

Deriving the Equation of Motion for ϕ via Euler-Lagrange The matter Lagrangian $\mathcal{L}_m$ is assumed not to depend on ϕ , so its variation is zero. We only need to consider the gravitational and scalar kinetic terms.

1. Variation of the ϕR term:

$$\frac{\partial (\phi R)}{\partial \phi} = R$$

2. Variation of the kinetic term, $L_\phi = -\frac{\omega}{\phi} (\partial\phi)^2$:

First, the partial derivative with respect to ϕ :

$$\frac{\partial L_\phi}{\partial \phi} = \frac{\partial}{\partial \phi} (-\omega \phi^{-1} g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi) = -\omega (-1) \phi^{-2} (\partial\phi)^2 = \frac{\omega}{\phi^2} (\partial\phi)^2$$

Next, the partial derivative with respect to the field gradient $\partial_\mu \phi$:

$$\frac{\partial L_\phi}{\partial (\partial_\mu \phi)} = -\frac{\omega}{\phi} \frac{\partial}{\partial (\partial_\mu \phi)} (g^{\alpha\beta} \partial_\alpha \phi \partial_\beta \phi) = -\frac{\omega}{\phi} (2g^{\mu\nu} \partial_\nu \phi) = -\frac{2\omega}{\phi} \partial^\mu \phi$$

Now we take the covariant divergence of this term:

$$\begin{aligned} \nabla_\mu \left(\frac{\partial L}{\partial (\partial_\mu \phi)} \right) &= \nabla_\mu \left(-\frac{2\omega}{\phi} \partial^\mu \phi \right) \\ &= -2\omega \left[\left(\nabla_\mu \frac{1}{\phi} \right) \partial^\mu \phi + \frac{1}{\phi} (\nabla_\mu \partial^\mu \phi) \right] \\ &= -2\omega \left[\left(-\frac{1}{\phi^2} \partial_\mu \phi \right) \partial^\mu \phi + \frac{1}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) \right] \\ &= \frac{2\omega}{\phi^2} (\partial\phi)^2 - \frac{2\omega}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) \end{aligned}$$

Combining all the pieces into the Euler-Lagrange equation:

$$\left(R + \frac{\omega}{\phi^2} (\partial\phi)^2 \right) - \left(\frac{2\omega}{\phi^2} (\partial\phi)^2 - \frac{2\omega}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) \right) = 0$$

$$R - \frac{\omega}{\phi^2} (\partial\phi)^2 + \frac{2\omega}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) = 0$$

Solving for R , we find our first expression:

$$R = \frac{\omega}{\phi^2} (\partial_\lambda \phi \partial^\lambda \phi) - \frac{2\omega}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) \quad (98)$$

Taking the Trace of the Gravitational Field Equation Next, we take the trace of the gravitational field equation (41) derived previously. The equation is:

$$G_{\mu\nu} = \frac{8\pi}{\phi} T_{\mu\nu} + \frac{\omega}{\phi^2} \left(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (\partial\phi)^2 \right) + \frac{1}{\phi} (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi))$$

We contract both sides with $g^{\mu\nu}$. The trace of the Einstein tensor is $g^{\mu\nu} G_{\mu\nu} = g^{\mu\nu} (R_{\mu\nu} - \frac{1}{2} g_{\mu\nu} R) = R - \frac{1}{2} (4) R = -R$. The traces of the terms on the right-hand side are:

- $g^{\mu\nu} \left(\frac{8\pi}{\phi} T_{\mu\nu} \right) = \frac{8\pi}{\phi} T$, where $T = g^{\mu\nu} T_{\mu\nu}$
- $g^{\mu\nu} \left[\frac{\omega}{\phi^2} (\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (\partial\phi)^2) \right] = \frac{\omega}{\phi^2} ((\partial\phi)^2 - \frac{1}{2} (4) (\partial\phi)^2) = -\frac{\omega}{\phi^2} (\partial\phi)^2$
- $g^{\mu\nu} \left[\frac{1}{\phi} (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)) \right] = \frac{1}{\phi} (g^{\mu\nu} \nabla_\mu \nabla_\nu \phi - 4 (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)) = -\frac{3}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)$

Assembling the traced equation:

$$-R = \frac{8\pi}{\phi} T - \frac{\omega}{\phi^2} (\partial\phi)^2 - \frac{3}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)$$

Solving for R gives our second expression:

$$R = -\frac{8\pi}{\phi} T + \frac{\omega}{\phi^2} (\partial\phi)^2 + \frac{3}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) \quad (99)$$

Combining the Results We now have two independent expressions for the Ricci scalar R . By equating Eq. (98) and Eq. (99), we can eliminate R and find the desired wave equation for ϕ .

$$\frac{\omega}{\phi^2} (\partial\phi)^2 - \frac{2\omega}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) = -\frac{8\pi}{\phi} T + \frac{\omega}{\phi^2} (\partial\phi)^2 + \frac{3}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)$$

The kinetic term $\frac{\omega}{\phi^2} (\partial\phi)^2$ conveniently cancels from both sides:

$$-\frac{2\omega}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) = -\frac{8\pi}{\phi} T + \frac{3}{\phi} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)$$

Multiplying the entire equation by ϕ to simplify:

$$-2\omega (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) = -8\pi T + 3(g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)$$

Finally, we group the terms and solve:

$$8\pi T = 3(g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) + 2\omega (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi) = (3 + 2\omega) (g^{\alpha\beta} \nabla_\alpha \nabla_\beta \phi)$$

This yields the final form of the Brans-Dicke scalar field equation (42):

$$g^{\mu\nu} \nabla_\mu \nabla_\nu \phi = \frac{8\pi}{3 + 2\omega} T$$

This equation shows explicitly how the trace of the energy-momentum tensor of matter acts as the source for the scalar field ϕ .

A.27 Derivation of the Scalar Field Solution in the Weak-Field Limit

Here we derive the solution for the scalar field perturbation φ sourced by a static, localized mass M in the weak-field approximation of the Brans-Dicke theory.

The starting point is the linearized wave equation for the scalar field perturbation φ , as given in the main text:

$$\eta^{\mu\nu} \partial_\mu \partial_\nu \varphi = \frac{8\pi}{3 + 2\omega} T \quad (100)$$

where $\eta^{\mu\nu}$ is the Minkowski metric, T is the trace of the energy-momentum tensor, and ω is the Brans-Dicke parameter.

1. Static Field Approximation We are looking for a solution for a static gravitational source. In this case, the scalar field perturbation φ is time-independent, meaning all time derivatives of φ are zero:

$$\frac{\partial \varphi}{\partial t} = \partial_0 \varphi = 0$$

The operator $\eta^{\mu\nu} \partial_\mu \partial_\nu$ with the $(-+++)$ metric signature expands to:

$$\eta^{\mu\nu} \partial_\mu \partial_\nu \varphi = \eta^{00} \partial_0 \partial_0 \varphi + \eta^{ij} \partial_i \partial_j \varphi = -\frac{\partial^2 \varphi}{\partial t^2} + \nabla^2 \varphi$$

where ∇^2 is the standard Laplacian operator. Since the field is static, the time derivative term vanishes, and the operator simplifies to the Laplacian:

$$\eta^{\mu\nu} \partial_\mu \partial_\nu \varphi \rightarrow \nabla^2 \varphi$$

Thus, the wave equation (100) reduces to a Poisson-like equation:

$$\nabla^2 \varphi(\mathbf{r}) = \frac{8\pi}{3+2\omega} T(\mathbf{r}) \quad (101)$$

2. The Source Term The source for the field is a static, localized object of mass M . In the non-relativistic, weak-field limit, the energy-momentum tensor is dominated by the mass-energy component. The trace of the energy-momentum tensor, T , can be effectively identified with the mass density ρ . For a point mass M located at the origin ($\mathbf{r} = 0$), the mass density is described by a Dirac delta function:

$$T(\mathbf{r}) \approx \rho(\mathbf{r}) = M \delta^{(3)}(\mathbf{r})$$

Substituting this source term into our Poisson equation (101), we get:

$$\nabla^2 \varphi(\mathbf{r}) = \frac{8\pi M}{3+2\omega} \delta^{(3)}(\mathbf{r}) \quad (102)$$

3. Solving the Poisson Equation This is a standard Poisson's equation of the form $\nabla^2 u = f(\mathbf{r})$. The solution can be found using a Green's function. The Green's function for the Laplacian operator, which satisfies $\nabla^2 G(\mathbf{r}) = \delta^{(3)}(\mathbf{r})$, is:

$$G(\mathbf{r}) = -\frac{1}{4\pi r}$$

where $r = |\mathbf{r}|$. The solution $\varphi(\mathbf{r})$ is given by the convolution of the source term with the Green's function:

$$\varphi(\mathbf{r}) = \int G(\mathbf{r} - \mathbf{r}') f(\mathbf{r}') d^3 r'$$

In our case, the source term is $f(\mathbf{r}') = \frac{8\pi M}{3+2\omega} \delta^{(3)}(\mathbf{r}')$. Plugging this in, we obtain:

$$\begin{aligned} \varphi(r) &= \int \left(-\frac{1}{4\pi |\mathbf{r} - \mathbf{r}'|} \right) \left(\frac{8\pi M}{3+2\omega} \delta^{(3)}(\mathbf{r}') \right) d^3 r' \\ &= -\frac{8\pi M}{4\pi(3+2\omega)} \int \frac{\delta^{(3)}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 r' \end{aligned}$$

The integral is evaluated using the sifting property of the Dirac delta function, which sets $\mathbf{r}' = 0$:

$$\int \frac{\delta^{(3)}(\mathbf{r}')}{|\mathbf{r} - \mathbf{r}'|} d^3 r' = \frac{1}{|\mathbf{r}|} = \frac{1}{r}$$

Substituting this result back, we arrive at the final solution for the scalar field perturbation:

$$\varphi(r) = -\frac{2M}{3+2\omega} \frac{1}{r} = -\frac{2M}{r(3+2\omega)}$$

A.28 Derivation of the Static Spherically Symmetric Vacuum Solution in Brans-Dicke Theory

The solution, first found by Brans [34], describes the spacetime geometry and the scalar field outside a non-rotating, uncharged body.

Field Equations and Metric Ansatz We begin with the Brans-Dicke field equations (42), (41) in vacuum ($T_{\mu\nu} = 0$), which implies the trace $T = 0$. The equations are:

$$G_{\mu\nu} = \frac{\omega}{\phi^2} \left(\partial_\mu \phi \partial_\nu \phi - \frac{1}{2} g_{\mu\nu} (\partial\phi)^2 \right) + \frac{1}{\phi} (\nabla_\mu \nabla_\nu \phi - g_{\mu\nu} \phi) \quad (103)$$

$$\phi = g^{\mu\nu} \nabla_\mu \nabla_\nu \phi = 0 \quad (104)$$

We assume a static, spherically symmetric spacetime, for which the metric can be written in isotropic coordinates as:

$$ds^2 = -e^{\nu(r)} dt^2 + e^{\lambda(r)} (dr^2 + r^2 d\Omega^2)$$

Due to symmetry, the scalar field ϕ is also a function of the radial coordinate r only, i.e., $\phi = \phi(r)$.

The System of Differential Equations Substituting the metric ansatz into the field equations (103) and (104) yields a system of coupled, non-linear ordinary differential equations for the functions $\nu(r)$, $\lambda(r)$, and $\phi(r)$. The scalar wave equation (104) becomes:

$$\phi'' + \left(\frac{\nu' + \lambda'}{2} + \frac{2}{r} \right) \phi' = 0 \quad (105)$$

The gravitational field equations provide two other independent equations. After significant algebraic manipulation, a convenient combination of the Einstein tensor components yields:

$$\frac{1}{2}(\nu'' - \lambda'') + \frac{1}{4}(\nu' - \lambda')(\nu' + \lambda') - \frac{1}{r}(\nu' + \lambda') = 0 \quad (106)$$

$$\lambda'' + \frac{(\lambda')^2}{2} + \frac{\lambda'}{r} - \frac{\nu' \lambda'}{2} = \frac{\omega}{2\phi^2} e^\lambda (\phi')^2 + \frac{e^\lambda}{\phi} \left(\phi'' + (\lambda' + \frac{1}{r}) \phi' \right) \quad (107)$$

Method of Solution The structure of the equations suggests a solution of the following form, parameterized by constants A, C, K :

$$e^{\nu(r)} = \left(\frac{1 - K/r}{1 + K/r} \right)^A \quad (108)$$

$$e^{\lambda(r)} = (1 - K^2/r^2)^2 \left(\frac{1 - K/r}{1 + K/r} \right)^C \quad (109)$$

$$\phi(r) = \phi_0 \left(\frac{1 - K/r}{1 + K/r} \right)^{-(A+C)/2} \quad (110)$$

The form for $\phi(r)$ is chosen to identically satisfy the scalar wave equation (105), which can be integrated once to give $\phi' \propto r^{-2} e^{-(\nu+\lambda)/2}$. Substituting the ansatz forms (108)-(110) into this integrated scalar equation confirms that it is satisfied.

The remaining task is to substitute these functional forms into the gravitational equations to determine the constants A, C, K . After a lengthy calculation, substituting the ansatz into equations (106) and (107) reduces them to a set of algebraic equations for the exponents A and C . The primary constraint equation that emerges is:

$$A(A - C - 2) = C(C + 2) + \omega(A + C)^2$$

This equation relates the geometric properties of the solution (A, C) to the fundamental parameter of the theory (ω) .

The Physical Solution To connect this general mathematical solution to a physical source of mass M , we must match the solution to the weak-field post-Newtonian limit. This procedure identifies the constants as follows:

- The constant K is related to the mass M of the source. In the weak-field limit, $K \approx \frac{GM}{2c^2}$. For simplicity in theoretical expressions, it is often denoted as B .
- The exponents are determined by requiring consistency with observations, particularly the deflection of light. This leads to a specific choice from the family of solutions.

The standard solution (known as the Brans Class I solution) that is asymptotically flat and corresponds to a point mass is found by setting:

$$A = \frac{2}{2 + \gamma}, \quad C = \frac{2(\gamma - 1)}{2 + \gamma}$$

where $\gamma = \frac{1+\omega}{2+\omega}$ is the PPN parameter. Substituting these into the constraint equation confirms their validity.

With this identification, the exponent for the scalar field in equation (110) becomes:

$$-\frac{A+C}{2} = -\frac{1}{2} \left(\frac{2}{2+\gamma} + \frac{2(\gamma-1)}{2+\gamma} \right) = -\frac{1}{2} \left(\frac{2\gamma}{2+\gamma} \right) = -\frac{\gamma}{2+\gamma}$$

Now, we express this in terms of ω :

$$-\frac{\gamma}{2+\gamma} = -\frac{(1+\omega)/(2+\omega)}{(2+\omega)/(2+\omega) + (1+\omega)/(2+\omega)} = -\frac{1+\omega}{3+2\omega}$$

A.29 Derivation of the Field Equations for Metric f(R) Gravity

We begin with the generalized action for $f(R)$ gravity in the metric formalism, which includes a standard matter action S_M . The total action is given by:

$$S = S_g + S_M = \frac{1}{2\kappa} \int d^4x \sqrt{-g} f(R) + S_M(g_{\mu\nu}, \psi)$$

where $\kappa = 8\pi G$, g is the determinant of the metric tensor $g_{\mu\nu}$, R is the Ricci scalar constructed from the Levi-Civita connection of $g_{\mu\nu}$, and ψ represents the matter fields.

The principle of stationary action dictates that the variation of the action with respect to the metric must vanish, i.e., $\delta S = 0$. We will vary the gravitational and matter parts separately.

Variation of the Matter Action The variation of the matter action defines the energy-momentum tensor $T_{\mu\nu}$. By definition:

$$T_{\mu\nu} = -\frac{2}{\sqrt{-g}} \frac{\delta S_M}{\delta g^{\mu\nu}}$$

This can be rewritten to express the variation of the action as:

$$\delta S_M = -\frac{1}{2} \int d^4x \sqrt{-g} T_{\mu\nu} \delta g^{\mu\nu}$$

Variation of the Gravitational Action The variation of the gravitational part, S_g , is more involved.

$$\delta S_g = \frac{1}{2\kappa} \delta \int d^4x \sqrt{-g} f(R) = \frac{1}{2\kappa} \int d^4x \delta(\sqrt{-g} f(R))$$

We apply the product rule to the term inside the integral:

$$\delta(\sqrt{-g} f(R)) = (\delta\sqrt{-g}) f(R) + \sqrt{-g} (\delta f(R))$$

Using the identity for the variation of the determinant, $\delta\sqrt{-g} = -\frac{1}{2}\sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu}$, and the chain rule, $\delta f(R) = f'(R) \delta R$, we get:

$$\delta(\sqrt{-g} f(R)) = -\frac{1}{2} \sqrt{-g} g_{\mu\nu} f(R) \delta g^{\mu\nu} + \sqrt{-g} f'(R) \delta R$$

The variation of the Ricci scalar, $\delta R = \delta(g^{\mu\nu} R_{\mu\nu})$, is:

$$\delta R = R_{\mu\nu} \delta g^{\mu\nu} + g^{\mu\nu} \delta R_{\mu\nu}$$

Substituting this back, the variation of the action term becomes:

$$\delta(\sqrt{-g} f(R)) = \sqrt{-g} \left[\left(f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} \right) \delta g^{\mu\nu} + f'(R) g^{\mu\nu} \delta R_{\mu\nu} \right]$$

The full variation of the gravitational action is therefore:

$$\delta S_g = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left[\left(f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} \right) \delta g^{\mu\nu} + f'(R) g^{\mu\nu} \delta R_{\mu\nu} \right] \quad (111)$$

The term $g^{\mu\nu} \delta R_{\mu\nu}$ contains second derivatives of the metric variation $\delta g^{\mu\nu}$. It is a standard result A.2 from the derivation of the Einstein-Hilbert action that this term can be written as a total divergence, which under an integral can be manipulated via integration by parts. The identity is:

$$g^{\mu\nu} \delta R_{\mu\nu} = (g_{\alpha\beta} g^{\lambda\sigma} \nabla_\lambda \nabla_\sigma - \nabla_\alpha \nabla_\beta) \delta g^{\alpha\beta}$$

where boundary terms have been neglected. We substitute this into the last term of eq. (111):

$$\int d^4x \sqrt{-g} f'(R) g^{\mu\nu} \delta R_{\mu\nu} = \int d^4x \sqrt{-g} f'(R) (g_{\mu\nu} g^{\alpha\beta} \nabla_\alpha \nabla_\beta - \nabla_\mu \nabla_\nu) \delta g^{\mu\nu}$$

We now integrate by parts twice to move the derivative operators from $\delta g^{\mu\nu}$ onto $f'(R)$. This procedure yields:

$$\int d^4x \sqrt{-g} f'(R) g^{\mu\nu} \delta R_{\mu\nu} = \int d^4x \sqrt{-g} [(g^{\alpha\beta} \nabla_\alpha \nabla_\beta f'(R)) g_{\mu\nu} - \nabla_\mu \nabla_\nu f'(R)] \delta g^{\mu\nu}$$

Substituting this result back into the expression for δS_g (eq. (111)), we obtain:

$$\delta S_g = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \left[f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} + g_{\mu\nu} (g^{\alpha\beta} \nabla_\alpha \nabla_\beta f'(R)) - \nabla_\mu \nabla_\nu f'(R) \right] \delta g^{\mu\nu}$$

Assembling the Field Equations Now we set the total variation of the action to zero:

$$\delta S = \delta S_g + \delta S_M = 0$$

$$\int d^4x \sqrt{-g} \left\{ \frac{1}{2\kappa} \left[f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} - (\nabla_\mu \nabla_\nu - g_{\mu\nu} g^{\alpha\beta} \nabla_\alpha \nabla_\beta) f'(R) \right] - \frac{1}{2} T_{\mu\nu} \right\} \delta g^{\mu\nu} = 0$$

Since this equation must hold for an arbitrary variation $\delta g^{\mu\nu}$, the entire expression within the curly braces must be zero. This gives the field equations for metric $f(R)$ gravity:

$$f'(R) R_{\mu\nu} - \frac{1}{2} f(R) g_{\mu\nu} - [\nabla_\mu \nabla_\nu - g_{\mu\nu} g^{\alpha\beta} \nabla_\alpha \nabla_\beta] f'(R) = \kappa T_{\mu\nu}$$

A.30 Derivation of the Field Equations for Palatini $f(R)$ Gravity

In the Palatini formalism of $f(R)$ gravity, the action is varied by treating the metric tensor $g_{\mu\nu}$ and the connection coefficients $\Gamma_{\mu\nu}^\lambda$ as independent fields. The action is given by:

$$S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} f(\mathcal{R}) + S_M(g_{\mu\nu}, \psi)$$

Here, the Ricci scalar $\mathcal{R} = g^{\mu\nu} \mathcal{R}_{\mu\nu}$ is constructed using the independent connection Γ , where $\mathcal{R}_{\mu\nu} = \mathcal{R}_{\mu\nu}(\Gamma)$. A fundamental assumption is that the matter action S_M depends only on the metric and the matter fields ψ , not on the connection Γ .

The principle of stationary action, $\delta S = 0$, requires that the variations with respect to both independent fields vanish separately.

1. Variation with respect to the metric $g_{\mu\nu}$ We first vary the action with respect to the metric $g_{\mu\nu}$, keeping the connection Γ fixed. This means that the Ricci tensor $\mathcal{R}_{\mu\nu}$ is treated as a constant during this variation.

$$\delta_g S = \delta_g \left(\frac{1}{2\kappa} \int d^4x \sqrt{-g} f(\mathcal{R}) \right) + \delta_g S_M = 0$$

The variation of the matter action defines the energy-momentum tensor:

$$\delta_g S_M = \int d^4x \frac{\delta S_M}{\delta g^{\mu\nu}} \delta g^{\mu\nu} = \int d^4x \left(-\frac{1}{2} \sqrt{-g} T_{\mu\nu} \right) \delta g^{\mu\nu}$$

For the gravitational part, we vary $\sqrt{-g}$ and $g^{\mu\nu}$ inside $\mathcal{R}$:

$$\begin{aligned} \delta_g (\sqrt{-g} f(\mathcal{R})) &= (\delta_g \sqrt{-g}) f(\mathcal{R}) + \sqrt{-g} \delta_g (f(\mathcal{R})) \\ &= \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} \delta g^{\mu\nu} \right) f(\mathcal{R}) + \sqrt{-g} f'(\mathcal{R}) \delta_g (g^{\alpha\beta} \mathcal{R}_{\alpha\beta}) \\ &= \left(-\frac{1}{2} \sqrt{-g} g_{\mu\nu} f(\mathcal{R}) \right) \delta g^{\mu\nu} + \sqrt{-g} f'(\mathcal{R}) (\mathcal{R}_{\alpha\beta} \delta g^{\alpha\beta}) \\ &= \sqrt{-g} \left(f'(\mathcal{R}) \mathcal{R}_{\mu\nu} - \frac{1}{2} f(\mathcal{R}) g_{\mu\nu} \right) \delta g^{\mu\nu} \end{aligned}$$

Combining the variations and setting the total variation to zero:

$$\delta_g S = \int d^4x \sqrt{-g} \left[\frac{1}{2\kappa} \left(f'(\mathcal{R}) \mathcal{R}_{\mu\nu} - \frac{1}{2} f(\mathcal{R}) g_{\mu\nu} \right) - \frac{1}{2} T_{\mu\nu} \right] \delta g^{\mu\nu} = 0$$

Since this must hold for an arbitrary variation $\delta g^{\mu\nu}$, the term in the brackets must vanish. This yields the first field equation. Note that while $\mathcal{R}_{\mu\nu}(\Gamma)$ is not necessarily symmetric for a general connection, the left-hand side must be symmetric because the energy-momentum tensor $T_{\mu\nu}$ is symmetric. Thus, we obtain:

$$f'(\mathcal{R})\mathcal{R}_{(\mu\nu)} - \frac{1}{2}f(\mathcal{R})g_{\mu\nu} = \kappa T_{\mu\nu} \quad (112)$$

2. Variation with respect to the connection $\Gamma_{\mu\nu}^\lambda$ Next, we vary the action with respect to the connection Γ , keeping the metric $g_{\mu\nu}$ fixed. The matter action S_M is independent of Γ , so its variation is zero.

$$\delta_\Gamma S = \delta_\Gamma \left(\frac{1}{2\kappa} \int d^4x \sqrt{-g} f(\mathcal{R}) \right) = 0$$

Since $g_{\mu\nu}$ and thus $\sqrt{-g}$ are held constant, the variation acts only on $f(\mathcal{R})$:

$$\delta_\Gamma S = \frac{1}{2\kappa} \int d^4x \sqrt{-g} \delta_\Gamma (f(\mathcal{R})) = \frac{1}{2\kappa} \int d^4x \sqrt{-g} f'(\mathcal{R}) \delta_\Gamma \mathcal{R} = 0$$

With $g^{\mu\nu}$ constant, the variation of the Ricci scalar is $\delta_\Gamma \mathcal{R} = g^{\mu\nu} \delta_\Gamma \mathcal{R}_{\mu\nu}$. The variation becomes:

$$\int d^4x \sqrt{-g} f'(\mathcal{R}) g^{\mu\nu} \delta_\Gamma \mathcal{R}_{\mu\nu} = 0$$

To proceed, we use the Palatini identity for the variation of the Ricci tensor with respect to the connection:

$$\delta_\Gamma \mathcal{R}_{\mu\nu} = \bar{\nabla}_\lambda (\delta \Gamma_{\nu\mu}^\lambda) - \bar{\nabla}_\nu (\delta \Gamma_{\lambda\mu}^\lambda)$$

where $\bar{\nabla}$ denotes the covariant derivative associated with the independent connection Γ . Substituting this into the integral gives:

$$\int d^4x \sqrt{-g} f'(\mathcal{R}) g^{\mu\nu} (\bar{\nabla}_\lambda (\delta \Gamma_{\nu\mu}^\lambda) - \bar{\nabla}_\nu (\delta \Gamma_{\lambda\mu}^\lambda)) = 0$$

We can now integrate by parts. The boundary terms are assumed to vanish. Let us define the tensor density $\mathcal{P}^{\mu\nu} = \sqrt{-g} f'(\mathcal{R}) g^{\mu\nu}$. The integral becomes:

$$\int d^4x [- (\bar{\nabla}_\lambda \mathcal{P}^{\mu\nu}) \delta \Gamma_{\nu\mu}^\lambda + (\bar{\nabla}_\nu \mathcal{P}^{\mu\nu}) \delta \Gamma_{\lambda\mu}^\lambda] = 0$$

We can relabel the dummy indices in the second term ($\nu \rightarrow \lambda, \mu \rightarrow \sigma, \lambda \rightarrow \nu$):

$$\int d^4x [- (\bar{\nabla}_\lambda \mathcal{P}^{\mu\nu}) \delta \Gamma_{\nu\mu}^\lambda + (\bar{\nabla}_\lambda \mathcal{P}^{\sigma\lambda}) \delta \Gamma_{\nu\sigma}^\nu] = 0$$

This can be rewritten as:

$$\int d^4x [- (\bar{\nabla}_\lambda \mathcal{P}^{\mu\nu}) + \delta_\lambda^\mu (\bar{\nabla}_\sigma \mathcal{P}^{\nu\sigma})] \delta \Gamma_{\nu\mu}^\lambda = 0$$

Since the variation $\delta \Gamma_{\nu\mu}^\lambda$ is arbitrary, the coefficient in the square brackets must be zero:

$$- (\bar{\nabla}_\lambda \mathcal{P}^{\mu\nu}) + \delta_\lambda^\mu (\bar{\nabla}_\sigma \mathcal{P}^{\nu\sigma}) = 0$$

Contracting this equation by setting $\mu = \lambda$, we find:

$$- (\bar{\nabla}_\mu \mathcal{P}^{\mu\nu}) + \delta_\mu^\mu (\bar{\nabla}_\sigma \mathcal{P}^{\nu\sigma}) = 0 \quad \Rightarrow \quad - (\bar{\nabla}_\mu \mathcal{P}^{\mu\nu}) + 4 (\bar{\nabla}_\sigma \mathcal{P}^{\nu\sigma}) = 0$$

This implies $3 (\bar{\nabla}_\sigma \mathcal{P}^{\nu\sigma}) = 0$, and therefore $\bar{\nabla}_\sigma \mathcal{P}^{\nu\sigma} = 0$. Substituting this back into the coefficient equation forces the first term to be zero as well, leading to the simpler condition:

$$\bar{\nabla}_\lambda \mathcal{P}^{\mu\nu} = 0$$

Substituting the definition of $\mathcal{P}^{\mu\nu}$ gives the second field equation:

$$\bar{\nabla}_\lambda (\sqrt{-g} f'(\mathcal{R}) g^{\mu\nu}) = 0 \quad (113)$$

This equation states that the tensor density $f'(\mathcal{R}) g^{\mu\nu}$ is covariantly constant with respect to the connection Γ .

A.31 Derivation of the Tensor Perturbation Action in Horndeski Theory

Here we sketch the derivation of the quadratic action for tensor perturbations in the Horndeski theory, which leads to the expression for the speed of gravitational waves. The derivation follows the methodology detailed in [66].

We start with a spatially flat FLRW background metric (9) and consider tensor perturbations h_{ij} , which are transverse ($\partial^i h_{ij} = 0$) and traceless ($h_i^i = 0$). The perturbed line element is:

$$ds^2 = -dt^2 + a(t)^2(\delta_{ij} + h_{ij})dx^i dx^j$$

We expand the full Horndeski action (52) to second order in h_{ij} . The terms G_2 and G_3 do not contribute at this order as they do not contain curvature tensors. The contributions arise from the G_4 and G_5 terms.

Contribution from the G_4 term The part of the Lagrangian containing G_4 is:

$$\mathcal{L}_4 = G_4(\phi, X)R + G_{4X} [(g^{\mu\nu}\nabla_\mu\nabla_\nu\phi)^2 - \phi_{\mu\nu}\phi^{\mu\nu}]$$

We need to compute the perturbations of the Ricci scalar R , the D'Alembertian of the scalar field $g^{\mu\nu}\nabla_\mu\nabla_\nu\phi$, and the second covariant derivative $\phi_{\mu\nu}$ to second order in h_{ij} . The perturbed Ricci scalar is $\delta^{(2)}R = \frac{1}{4}\dot{h}_{ij}\dot{h}^{ij} - \frac{1}{4a^2}(\partial_k h_{ij})(\partial^k h^{ij})$. The other terms require more algebra. After a lengthy calculation, the quadratic action from the $\mathcal{L}_4$ part is found to be:

$$S_4^{(2)} = \frac{1}{2} \int dt d^3x a^3 \left[(G_4 - 2XG_{4X})\dot{h}_{ij}^2 - \frac{G_4}{a^2}(\partial_k h_{ij})^2 \right]$$

Contribution from the G_5 term The part of the Lagrangian containing G_5 is:

$$\mathcal{L}_5 = G_5(\phi, X)G_{\mu\nu}\phi^{\mu\nu} - \frac{G_{5X}}{6} [(g^{\mu\nu}\nabla_\mu\nabla_\nu\phi)^3 - \dots]$$

This term involves the Einstein tensor $G_{\mu\nu}$. Its perturbation to first order, $\delta^{(1)}G_{ij}$, couples to the first-order perturbation of ϕ_{ij} . The calculation is involved, but the key result is that the time derivatives from δG_{ij} combine with those from other terms. The final contribution to the quadratic action from the $\mathcal{L}_5$ part is:

$$S_5^{(2)} = \frac{1}{2} \int dt d^3x a^3 \left[-X(\dot{\phi}HG_{5X} - G_{5\phi})\dot{h}_{ij}^2 + \frac{X}{a^2}(\ddot{\phi}G_{5X} + G_{5\phi})(\partial_k h_{ij})^2 \right]$$

Here, we have used the background equations of motion to simplify the expression.

Total Action and Propagation Speed Combining the contributions from $S_4^{(2)}$ and $S_5^{(2)}$, the total quadratic action for tensor modes is:

$$S_{\text{tensor}}^{(2)} = \frac{1}{8} \int dt d^3x a^3 \left[G_T \dot{h}_{ij}^2 - \frac{F_T}{a^2}(\partial_k h_{ij})^2 \right]$$

where the coefficients G_T and F_T are defined as:

$$\begin{aligned} G_T &:= 2 \left[G_4 - 2XG_{4X} - X(\dot{\phi}HG_{5X} - G_{5\phi}) \right] \\ F_T &:= 2 \left[G_4 - X(\ddot{\phi}G_{5X} + G_{5\phi}) \right] \end{aligned}$$

From this action, the propagation speed squared of gravitational waves is read off as $c_T^2 = F_T/G_T$.

A.32 Derivation of the Tracker Solution for an Inverse Power-Law Potential

Here we derive the tracker solution for a scalar field with an inverse power-law potential $V(\phi) = A\phi^{-\alpha}$ (where $A = M^{4+\alpha}$) in a flat FRW universe dominated by a background fluid with a constant equation of state w_m . The derivation follows the logic of [122, 111].

The Tracker Condition The key to the tracker solution is the assumption that the scalar field's energy density, ρ_ϕ , scales with the background fluid's energy density, ρ_m . In a power-law expanding universe, where $a(t) \propto t^p$, we have $H = p/t$ and $\rho_m \propto a^{-3(1+w_m)} \propto t^{-3p(1+w_m)}$. Since $H^2 \propto \rho_m$, we must have $2 = 3p(1+w_m)$, which gives the standard result $p = 2/(3(1+w_m))$. The tracker condition is that the potential energy of the field is of the same order as the total energy density, so $V(\phi) \propto H^2$. This implies ρ_ϕ is a nearly constant fraction of ρ_m .

Solving the Equations of Motion We look for a power-law solution for the scalar field of the form $\phi(t) = ct^q$. The Klein-Gordon equation (37) is:

$$\ddot{\phi} + 3H\dot{\phi} + V'(\phi) = 0$$

Substituting $V' = -A\alpha\phi^{-\alpha-1}$, $H = p/t$, and our ansatz for $\phi(t)$:

$$cq(q-1)t^{q-2} + 3\frac{p}{t}cqt^{q-1} - A\alpha(ct^q)^{-\alpha-1} = 0$$

$$cq(q-1)t^{q-2} + 3pcqt^{q-2} - A\alpha c^{-\alpha-1}t^{-q(\alpha+1)} = 0$$

For this equation to hold at all times, the powers of t must be equal:

$$q-2 = -q(\alpha+1) \implies q-2 = -q\alpha - q \implies 2q\alpha = 2 \implies q = \frac{2}{\alpha+2}$$

Now we solve for the coefficient c by setting the sum of coefficients to zero:

$$cq(q-1) + 3pcq - A\alpha c^{-\alpha-1} = 0$$

Substituting q and solving for $c^{\alpha+2}$:

$$c^{\alpha+2} [q(q-1) + 3pq] = A\alpha \implies c^{\alpha+2} = \frac{A\alpha}{q(q-1+3p)}$$

This confirms that a self-consistent power-law solution exists.

The Effective Equation of State Now we can calculate the kinetic and potential energies of the field for this solution.

$$\text{Kinetic Energy: } \frac{1}{2}\dot{\phi}^2 = \frac{1}{2}(cqt^{q-1})^2 = \frac{c^2q^2}{2}t^{2(q-1)}$$

$$\text{Potential Energy: } V(\phi) = A(ct^q)^{-\alpha} = Ac^{-\alpha}t^{-q\alpha}$$

We check the ratio of kinetic to potential energy. Note that $2(q-1) = 2(\frac{2}{\alpha+2} - 1) = \frac{4-2\alpha-4}{\alpha+2} = \frac{-2\alpha}{\alpha+2}$ and $-q\alpha = \frac{-2\alpha}{\alpha+2}$. The time dependence is the same, as required for a constant w_ϕ . The ratio is:

$$\frac{\frac{1}{2}\dot{\phi}^2}{V(\phi)} = \frac{\frac{1}{2}c^2q^2t^{2(q-1)}}{Ac^{-\alpha}t^{-q\alpha}} = \frac{c^{\alpha+2}q^2}{2A}$$

Substitute the expression for $c^{\alpha+2}$:

$$\frac{\frac{1}{2}\dot{\phi}^2}{V(\phi)} = \frac{A\alpha}{q(q-1+3p)} \frac{q^2}{2A} = \frac{\alpha q}{2(q-1+3p)}$$

Now substitute $q = 2/(\alpha+2)$ and $p = 2/(3(1+w_m))$:

$$q-1+3p = \frac{2}{\alpha+2} - 1 + \frac{6}{3(1+w_m)} = \frac{2-(\alpha+2)}{\alpha+2} + \frac{2}{1+w_m} = \frac{-\alpha}{\alpha+2} + \frac{2}{1+w_m} = \frac{-\alpha(1+w_m) + 2(\alpha+2)}{(\alpha+2)(1+w_m)} = \frac{\alpha - \alpha w_m + 4}{(\alpha+2)(1+w_m)}$$

So the ratio becomes:

$$\frac{\frac{1}{2}\dot{\phi}^2}{V(\phi)} = \frac{\alpha \frac{2}{\alpha+2}}{2 \frac{\alpha - \alpha w_m + 4}{(\alpha+2)(1+w_m)}} = \frac{\alpha(1+w_m)}{\alpha - \alpha w_m + 4}$$

This ratio is a constant, which means the equation of state w_ϕ is also a constant.

$$w_\phi = \frac{\frac{1}{2}\dot{\phi}^2 - V(\phi)}{\frac{1}{2}\dot{\phi}^2 + V(\phi)} = \frac{\frac{\frac{1}{2}\dot{\phi}^2}{V(\phi)} - 1}{\frac{\frac{1}{2}\dot{\phi}^2}{V(\phi)} + 1}$$

A simpler way is to use the energy conservation equation for the scalar field: $\dot{\rho}_\phi + 3H(1+w_\phi)\rho_\phi = 0$. Since $\rho_\phi \propto t^{-2\alpha/(\alpha+2)}$ and $H \propto 1/t$, we get:

$$-\frac{2\alpha}{\alpha+2} \frac{\rho_\phi}{t} + 3\frac{p}{t}(1+w_\phi)\rho_\phi = 0 \implies -\frac{2\alpha}{\alpha+2} + 3p(1+w_\phi) = 0$$

$$3\frac{2}{3(1+w_m)}(1+w_\phi) = \frac{2\alpha}{\alpha+2} \implies 1+w_\phi = \frac{\alpha(1+w_m)}{\alpha+2}$$

$$w_\phi = \frac{\alpha(1+w_m)}{\alpha+2} - 1 = \frac{\alpha + \alpha w_m - (\alpha+2)}{\alpha+2} = \frac{\alpha w_m - 2}{\alpha+2}$$

This completes the derivation of the effective equation of state for the tracker solution.

A.33 Derivation of the Autonomous System Equations for a Scalar Field

Here we derive the autonomous system of equations that governs the cosmological evolution of a spatially flat FRW universe containing a canonical scalar field ϕ and a single barotropic perfect fluid (e.g., matter or radiation), as discussed in [111, 126]. The goal is to transform the second-order differential equations of cosmology into a set of first-order autonomous equations.

Dimensionless Variables To create an autonomous system, we define a set of dimensionless variables that represent the ratios of different energy components. The standard choice is:

$$x \equiv \frac{\kappa\dot{\phi}}{\sqrt{6}H}, \quad y \equiv \frac{\kappa\sqrt{V(\phi)}}{\sqrt{3}H} \quad (114)$$

We also define a new time variable as the number of e-folds, $N \equiv \ln a$. The derivative with respect to N is related to the time derivative by $\frac{d}{dN} = \frac{1}{H} \frac{d}{dt}$.

From these definitions, we can express the density parameters for the scalar field and the fluid:

$$\begin{aligned} \Omega_\phi &= \frac{\kappa^2 \rho_\phi}{3H^2} = \frac{\kappa^2}{3H^2} \left(\frac{1}{2} \dot{\phi}^2 + V(\phi) \right) = \frac{\kappa^2}{3H^2} \left(\frac{3H^2 x^2}{\kappa^2} + \frac{3H^2 y^2}{\kappa^2} \right) = x^2 + y^2 \\ \Omega_m &= \frac{\kappa^2 \rho_m}{3H^2} = 1 - \Omega_\phi = 1 - x^2 - y^2 \end{aligned}$$

The equation of state for the scalar field can also be expressed in terms of x and y :

$$w_\phi = \frac{p_\phi}{\rho_\phi} = \frac{\frac{1}{2} \dot{\phi}^2 - V(\phi)}{\frac{1}{2} \dot{\phi}^2 + V(\phi)} = \frac{x^2 - y^2}{x^2 + y^2} \quad (115)$$

For the system to be closed, we need to express the slope of the potential in a dimensionless way. We define:

$$\lambda \equiv -\frac{V'(\phi)}{\kappa V(\phi)} \quad (116)$$

Deriving the Evolution Equations We now derive the evolution equations for x and y with respect to N .

First, we need an expression for $\dot{H}$. Differentiating the Friedmann equation (10):

$$2H\dot{H} = \frac{\kappa^2}{3}(\dot{\rho}_\phi + \dot{\rho}_m)$$

Using the conservation equations for ϕ and the fluid ($\dot{\rho}_\phi = -3H(1 + w_\phi)\rho_\phi = -3H\dot{\phi}^2$), we get:

$$\begin{aligned} 2H\dot{H} &= \frac{\kappa^2}{3}(-3H\dot{\phi}^2 - 3H(1 + w_m)\rho_m) \\ \frac{\dot{H}}{H^2} &= -\frac{\kappa^2}{2H^2}(\dot{\phi}^2 + (1 + w_m)\rho_m) = -\frac{\kappa^2}{2}\left(\frac{6x^2}{\kappa^2} + (1 + w_m)\frac{3\Omega_m}{\kappa^2}\right) \\ \frac{\dot{H}}{H^2} &= -3x^2 - \frac{3}{2}(1 + w_m)(1 - x^2 - y^2) \end{aligned}$$

Now we differentiate x with respect to N :

$$\frac{dx}{dN} = \frac{1}{H} \frac{d}{dt} \left(\frac{\kappa\dot{\phi}}{\sqrt{6}H} \right) = \frac{\kappa}{\sqrt{6}H^2} \left(\ddot{\phi} - \dot{\phi} \frac{\dot{H}}{H} \right)$$

Substitute $\ddot{\phi}$ from the Klein-Gordon Eq. (37) and the expression for $\dot{H}/H^2$:

$$\begin{aligned} \frac{dx}{dN} &= \frac{\kappa}{\sqrt{6}H^2} \left((-3H\dot{\phi} - V') - \dot{\phi} \frac{\dot{H}}{H} \right) \\ &= -\frac{3\kappa\dot{\phi}}{\sqrt{6}H} - \frac{\kappa V'}{\sqrt{6}H^2} - \frac{\kappa\dot{\phi}}{\sqrt{6}H} \frac{\dot{H}}{H^2} \\ &= -3x - \frac{\kappa V'}{\sqrt{6}H^2} - x \left(-3x^2 - \frac{3}{2}(1 + w_m)(1 - x^2 - y^2) \right) \end{aligned}$$

Using $V' = -\lambda\kappa V = -\lambda\kappa \frac{3H^2 y^2}{\kappa^2}$, we get $\frac{\kappa V'}{\sqrt{6}H^2} = -\frac{\sqrt{3}\lambda y^2}{\sqrt{2}} = -\sqrt{\frac{3}{2}}\lambda y^2$. Substituting this gives the final equation for x :

$$\frac{dx}{dN} = -3x + \sqrt{\frac{3}{2}}\lambda y^2 + x \left(3x^2 + \frac{3}{2}(1+w_m)(1-x^2-y^2) \right) \quad (117)$$

Next, we differentiate y with respect to N :

$$\frac{dy}{dN} = \frac{1}{H} \frac{d}{dt} \left(\frac{\kappa\sqrt{V}}{\sqrt{3}H} \right) = \frac{\kappa}{\sqrt{3}H^2} \left(\frac{\dot{V}}{2\sqrt{V}} - \sqrt{V} \frac{\dot{H}}{H} \right)$$

Using $\dot{V} = V'\dot{\phi} = (-\lambda\kappa V)\dot{\phi}$:

$$\begin{aligned} \frac{dy}{dN} &= \frac{\kappa}{\sqrt{3}H^2} \left(\frac{-\lambda\kappa V\dot{\phi}}{2\sqrt{V}} \right) - y \frac{\dot{H}}{H^2} \\ &= -\frac{\lambda\kappa^2\sqrt{V}\dot{\phi}}{2\sqrt{3}H^2} - y \frac{\dot{H}}{H^2} \\ &= -\frac{\lambda}{2\sqrt{3}} \left(\frac{\kappa\sqrt{V}}{H} \right) \left(\frac{\kappa\dot{\phi}}{H} \right) - y \frac{\dot{H}}{H^2} \\ &= -\frac{\lambda}{2\sqrt{3}} (\sqrt{3}y)(\sqrt{6}x) - y \left(-3x^2 - \frac{3}{2}(1+w_m)(1-x^2-y^2) \right) \\ &= -\sqrt{\frac{3}{2}}\lambda xy + y \left(3x^2 + \frac{3}{2}(1+w_m)(1-x^2-y^2) \right) \end{aligned}$$

This is the final equation for y . Rearranging the terms in both equations to match the conventional form gives the autonomous system presented in equations (71) and (72).

A.34 Derivation of the Statefinder Parameters

Here we derive the expressions for the Statefinder parameters $\{r, s\}$ in terms of the equation of state w and the dark energy density parameter Ω_{DE} for a spatially flat universe. The derivation follows the original methodology of [129, 130].

Basic Definitions and Relations We begin with the definitions of the Hubble parameter H , the deceleration parameter q , and the Statefinder parameter r :

$$q = -\frac{\ddot{a}}{aH^2}, \quad r = \frac{\ddot{a}}{aH^3}$$

From the definition of q , we can express $\ddot{a}$ as $\ddot{a} = -qaH^2$. We also need the time derivative of the Hubble parameter, $\dot{H}$. Differentiating $H = \dot{a}/a$ gives $\dot{H} = \frac{\ddot{a}}{a} - H^2$. Using the definition of q :

$$\dot{H} = -qH^2 - H^2 = -(1+q)H^2 \quad (118)$$

The Friedmann equation for a flat universe with matter ($w_m = 0$) and dark energy (w) is $H^2 = \frac{8\pi G}{3}(\rho_m + \rho_{DE})$. Taking its time derivative gives:

$$2H\dot{H} = \frac{8\pi G}{3}(\dot{\rho}_m + \dot{\rho}_{DE}) = \frac{8\pi G}{3}(-3H\rho_m - 3H(1+w)\rho_{DE})$$

Dividing by $2H^2$ gives:

$$\frac{\dot{H}}{H^2} = -\frac{4\pi G}{H^2}(\rho_m + (1+w)\rho_{DE}) = -\frac{3}{2}(\Omega_m + (1+w)\Omega_{DE})$$

Equating this with $\dot{H}/H^2 = -(1+q)$ and using $\Omega_m = 1 - \Omega_{DE}$, we find the expression for q :

$$1+q = \frac{3}{2}(1 - \Omega_{DE} + (1+w)\Omega_{DE}) = \frac{3}{2}(1 + w\Omega_{DE}) \implies q = \frac{1}{2} + \frac{3}{2}w\Omega_{DE}$$

Deriving the Parameter r To find r , we need the third derivative of the scale factor, $\ddot{a}$. We start by differentiating $\ddot{a} = -qaH^2$:

$$\begin{aligned}\ddot{a} &= -\dot{q}aH^2 - q\dot{a}H^2 - 2qaH\dot{H} \\ \frac{\ddot{a}}{aH^3} &= r = -\frac{\dot{q}}{H} - q - \frac{2q\dot{H}}{H^2}\end{aligned}$$

Substituting $\dot{H}/H^2 = -(1+q)$, we get:

$$r = -\frac{\dot{q}}{H} - q - 2q(-1-q) = -\frac{\dot{q}}{H} - q + 2q + 2q^2 = 2q^2 + q - \frac{\dot{q}}{H}$$

Now we need $\dot{q}$. Differentiating $q = \frac{1}{2} + \frac{3}{2}w\Omega_{DE}$:

$$\dot{q} = \frac{3}{2}(\dot{w}\Omega_{DE} + w\dot{\Omega}_{DE})$$

We also need $\dot{\Omega}_{DE}$. $\Omega_{DE} = \frac{8\pi G\rho_{DE}}{3H^2}$.

$$\dot{\Omega}_{DE} = \frac{8\pi G}{3} \left(\frac{\dot{\rho}_{DE}}{H^2} - \frac{2\rho_{DE}\dot{H}}{H^3} \right) = \Omega_{DE} \left(\frac{\dot{\rho}_{DE}}{\rho_{DE}} - \frac{2\dot{H}}{H^2} \right)$$

Using $\dot{\rho}_{DE} = -3H(1+w)\rho_{DE}$ and $\dot{H}/H^2 = -\frac{3}{2}(1+w\Omega_{DE})$:

$$\dot{\Omega}_{DE} = \Omega_{DE} (-3H(1+w) + 3H(1+w\Omega_{DE})) = 3H\Omega_{DE}(w\Omega_{DE} - 1 - w) = -3H\Omega_{DE}(1 - \Omega_{DE})(1 + w)$$

Using $\Omega_m = 1 - \Omega_{DE}$, this is $\dot{\Omega}_{DE} = -3H\Omega_{DE}\Omega_m(1 + w)$. Now, substitute back into $\dot{q}$:

$$\frac{\dot{q}}{H} = \frac{3}{2H}(\dot{w}\Omega_{DE} - 3Hw\Omega_{DE}\Omega_m(1 + w)) = \frac{3}{2} \frac{\dot{w}}{H} \Omega_{DE} - \frac{9}{2}w(1 + w)\Omega_{DE}(1 - \Omega_{DE})$$

Finally, substitute this into the expression for r :

$$\begin{aligned}r &= 2q^2 + q - \left[\frac{3}{2} \frac{\dot{w}}{H} \Omega_{DE} - \frac{9}{2}w(1 + w)\Omega_{DE}(1 - \Omega_{DE}) \right] \\ &= 2 \left(\frac{1}{2} + \frac{3}{2}w\Omega_{DE} \right)^2 + \left(\frac{1}{2} + \frac{3}{2}w\Omega_{DE} \right) - \frac{3}{2H}\dot{w}\Omega_{DE} + \frac{9}{2}w(1 + w)\Omega_{DE}(1 - \Omega_{DE}) \\ &= \frac{1}{2} + 3w\Omega_{DE} + \frac{9}{2}w^2\Omega_{DE}^2 + \frac{1}{2} + \frac{3}{2}w\Omega_{DE} - \frac{3}{2H}\dot{w}\Omega_{DE} + \frac{9}{2}w(1 + w)\Omega_{DE} - \frac{9}{2}w(1 + w)\Omega_{DE}^2 \\ &= 1 + \frac{9}{2}w\Omega_{DE} + \frac{9}{2}w^2\Omega_{DE} - \frac{3}{2H}\dot{w}\Omega_{DE} \\ &= 1 + \frac{9}{2}w(1 + w)\Omega_{DE} - \frac{3}{2H}\dot{w}\Omega_{DE}\end{aligned}$$

This is the final expression for r . For the Λ CDM model, $w = -1$ and $\dot{w} = 0$, which immediately gives $r = 1$.

Deriving the Parameter s With the expressions for q and r , the parameter s is found by direct substitution into its definition:

$$s = \frac{r - 1}{3(q - 1/2)} = \frac{\frac{9}{2}w(1 + w)\Omega_{DE} - \frac{3}{2H}\dot{w}\Omega_{DE}}{3(\frac{3}{2}w\Omega_{DE})} = \frac{\frac{3}{2}w\Omega_{DE}(3(1 + w) - \dot{w}/(Hw))}{\frac{9}{2}w\Omega_{DE}} = 1 + w - \frac{\dot{w}}{3Hw}$$

For the Λ CDM model, $w = -1$ and $\dot{w} = 0$, which gives $s = 0$.

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